

Disorders of the Auditory and Vestibular System

Matthew Ng, MD
Clinical Associate Professor
Department of Otolaryngology-HNS
UNLV School of Medicine

Goals & Objectives

- Review clinical applications of knowledge base on auditory and vestibular neurophysiology
- Review common auditory and vestibular disorders
  - Auditory disorders: types of hearing loss (conductive vs sensorineural vs mixed hearing loss), age-related, noise-induced, sudden idiopathic, ototoxicity, asymmetric) and tinnitus
  - Vestibular disorders: BPPV, vestibular neuritis and labyrinthitis, Meniere's disease and vestibulotoxicity

Case #1

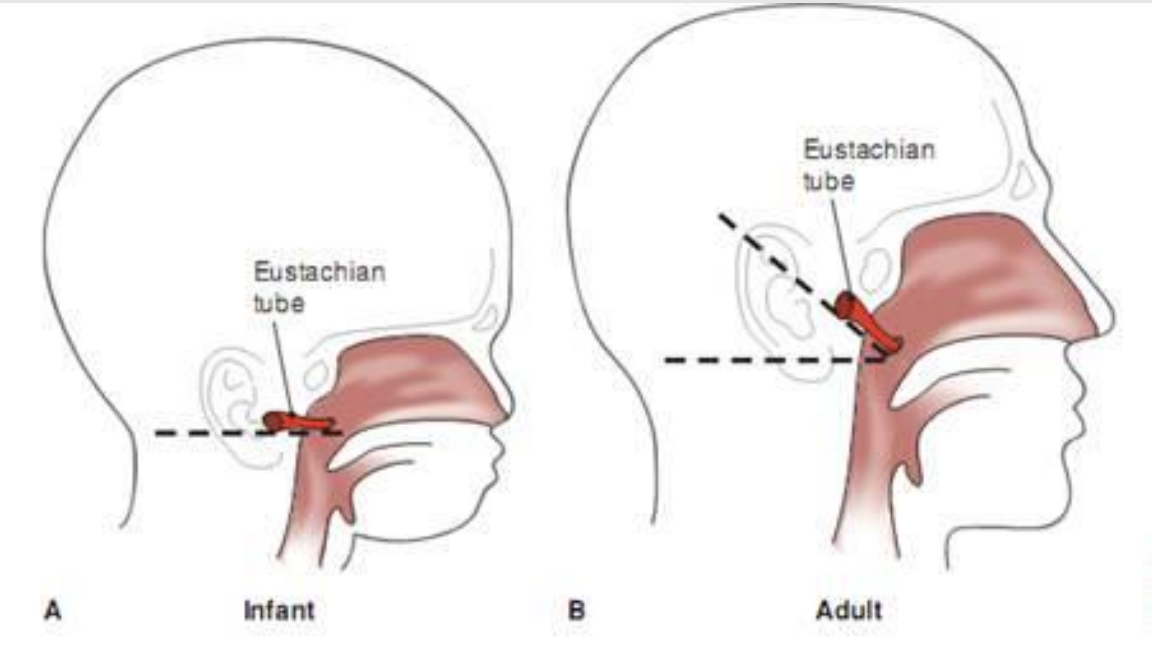
3 yo M brought in by mom with concerns for recurrent ear infections. Pt has had an ear infection, sometimes bilateral, about 6 times this past one year. In between ear infections, the pediatrician notices amber-colored discoloration of the eardrums.







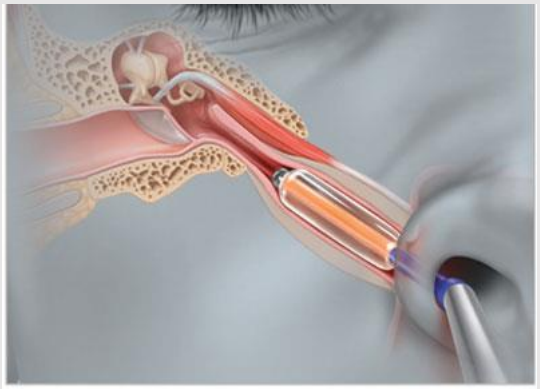
Tubes were placed and mom reports pt is happier and hears better.



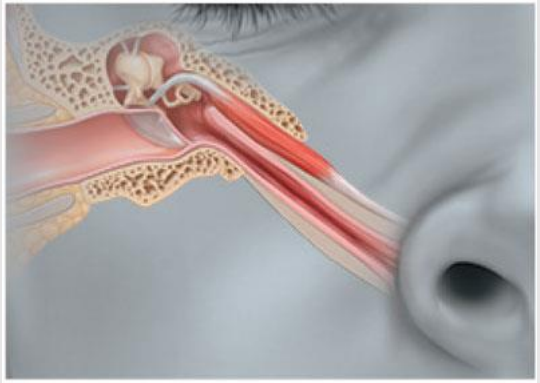


- Newer treatment for Eustachian tube dysfunction





1 Balloon is placed in the Eustachian tube through the nose and inflated to compress soft tissue lining the Eustachian tube.



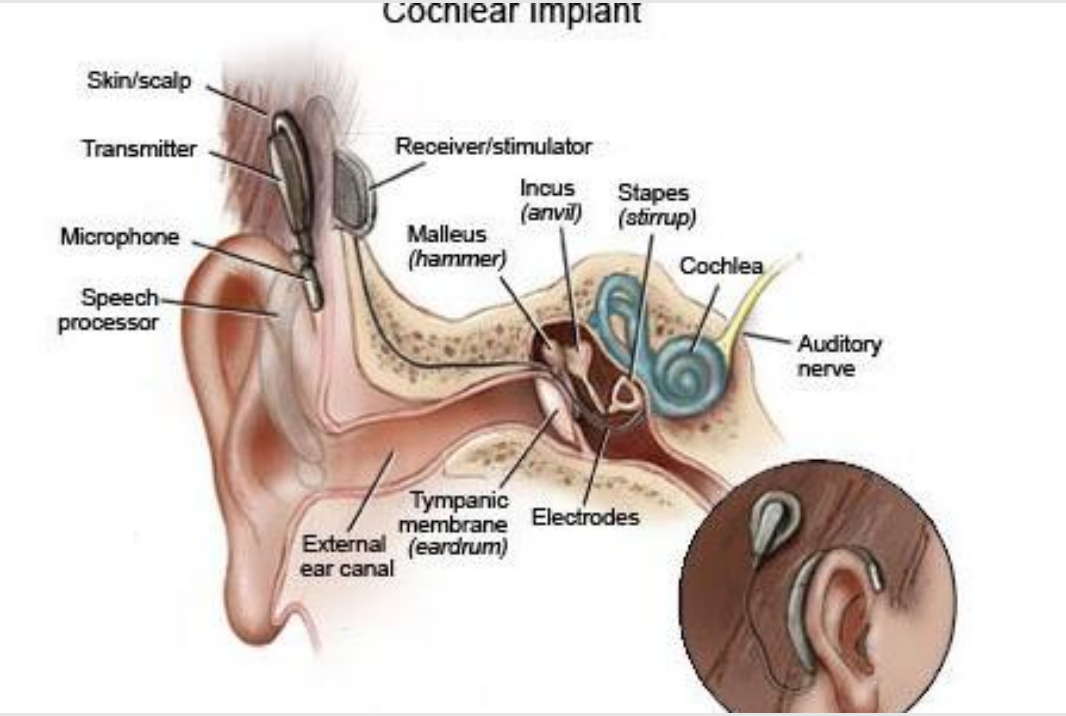
2 Once treatment is complete, the balloon is deflated and removed.

Case #2

6 month-old infant girl (no family hx of hearing loss) failed her state-mandated newborn hearing screen at birth. Audiologic testing at 3 mos of age with auditory brainstem response testing revealed bilateral profound hearing loss. Genetic testing revealed a mutation in the GBJ2 gene (connexin gene). Hearing aid trial at 6 mos of age did not make any difference in the patient's response to sound. Parents desire patient to hear and develop speech normally.



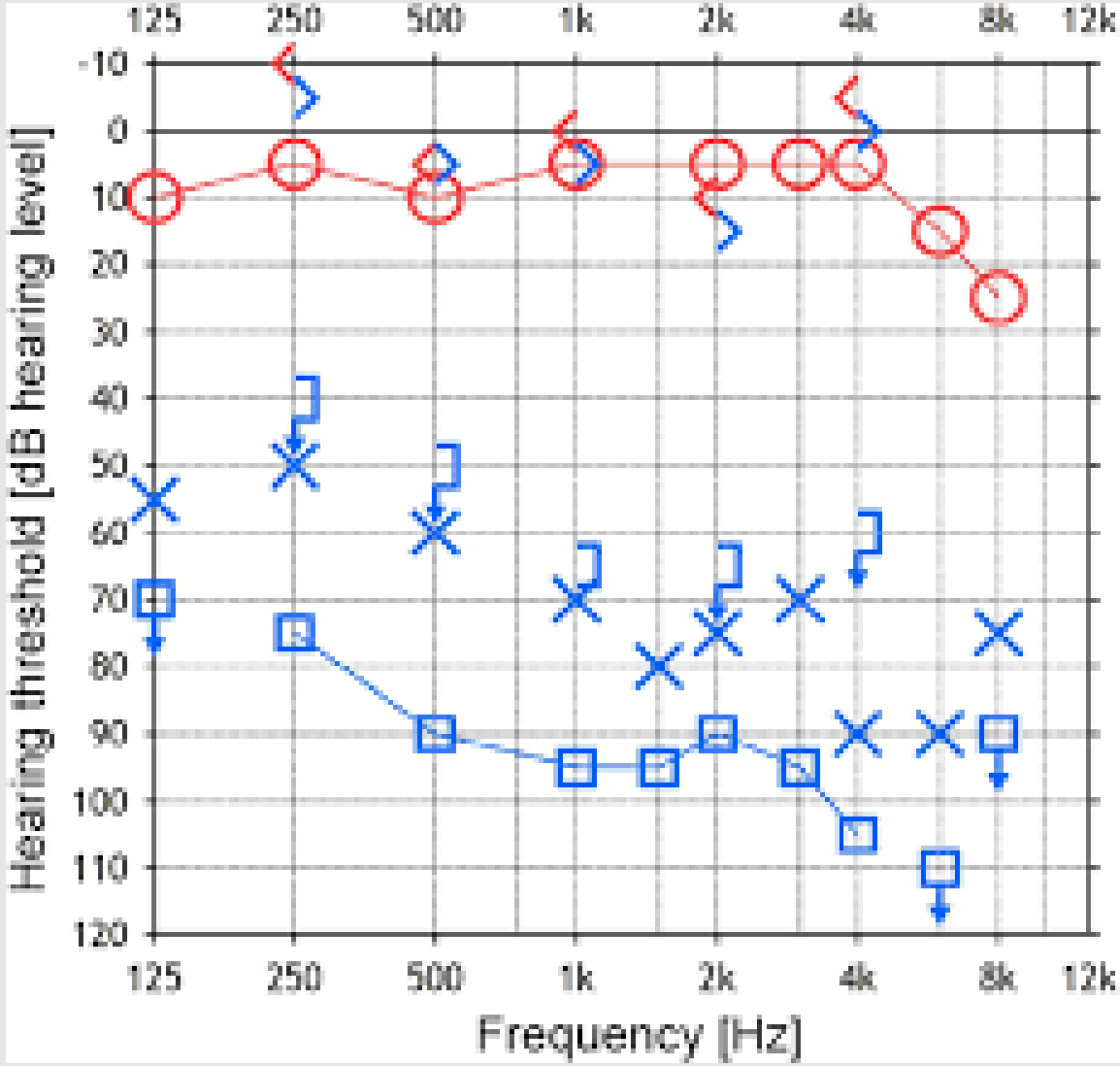
As early as 9 mos of age, bilateral cochlear implants were surgically placed. By age 5 years, patient is attending mainstream school and is academically on par with her peers in the classroom.





Case #3

55 yo M presents with gradual left hearing loss out of proportion to the right. This has been ongoing for at least 5 years. In addition to the near-total hearing loss, he has left unremitting, high-pitch tinnitus. He notices that when he walks, he veers to the left. Audiometric evaluation reveals an asymmetric left sensorineural hearing loss.

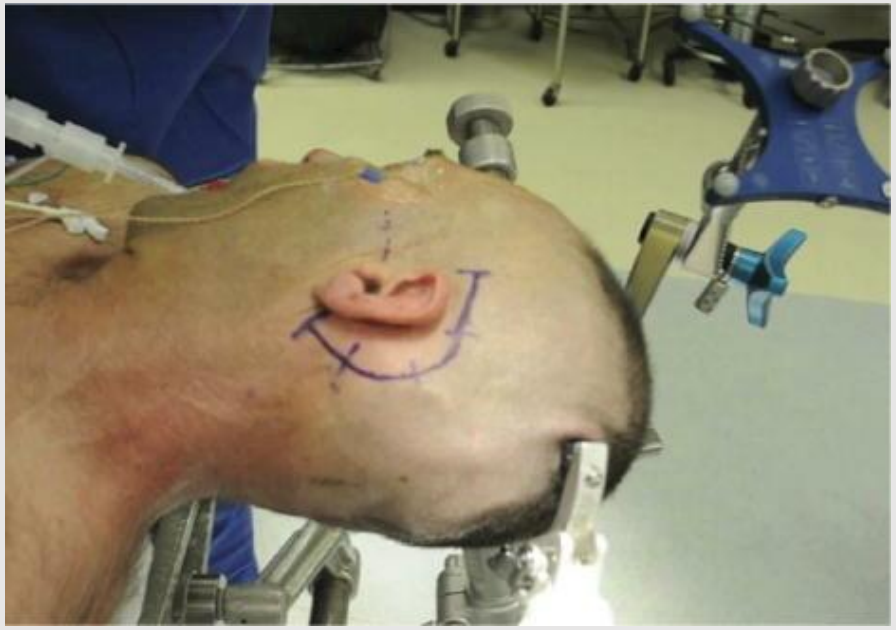


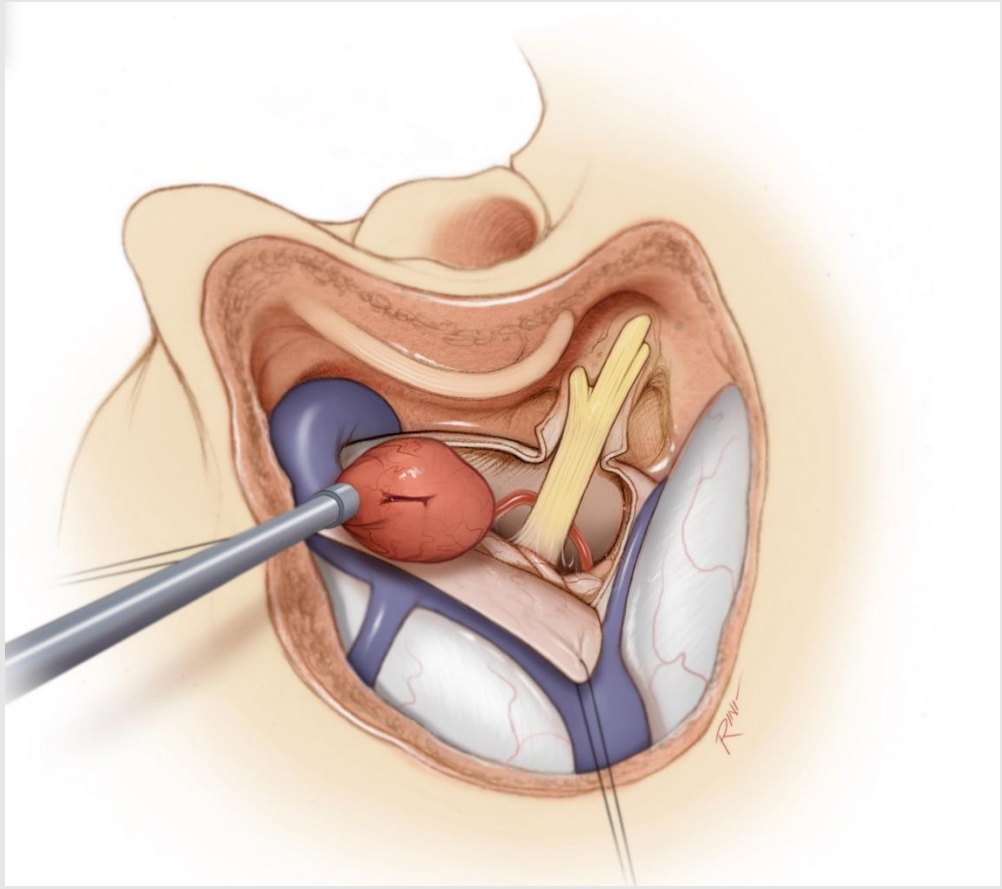
This asymmetry in hearing prompts an MRI of the internal auditory canal with and without gadolinium.

Left vestibular schwannoma.



Surgery is performed to remove the vestibular schwannoma tumor in conjunction with neurosurgery via translabyrinthine approach. The vestibular end-organ is drilled away to gain access to the cerebellopontine angle to permit tumor removal and facial nerve preservation.







Case #4

82 yo female c/o “dizziness”. Upon further elicitation of her history, she has gait imbalance when she is on her feet. She loses her orientation with quick head movements or rapid changes in her position. She feels like she is going to fall. She will be having cataract surgery because her vision is blurred. She recently had bilateral knee replacement surgery. Why is she so imbalanced?

Balance Mechanisms

(multiple inputs to brain)

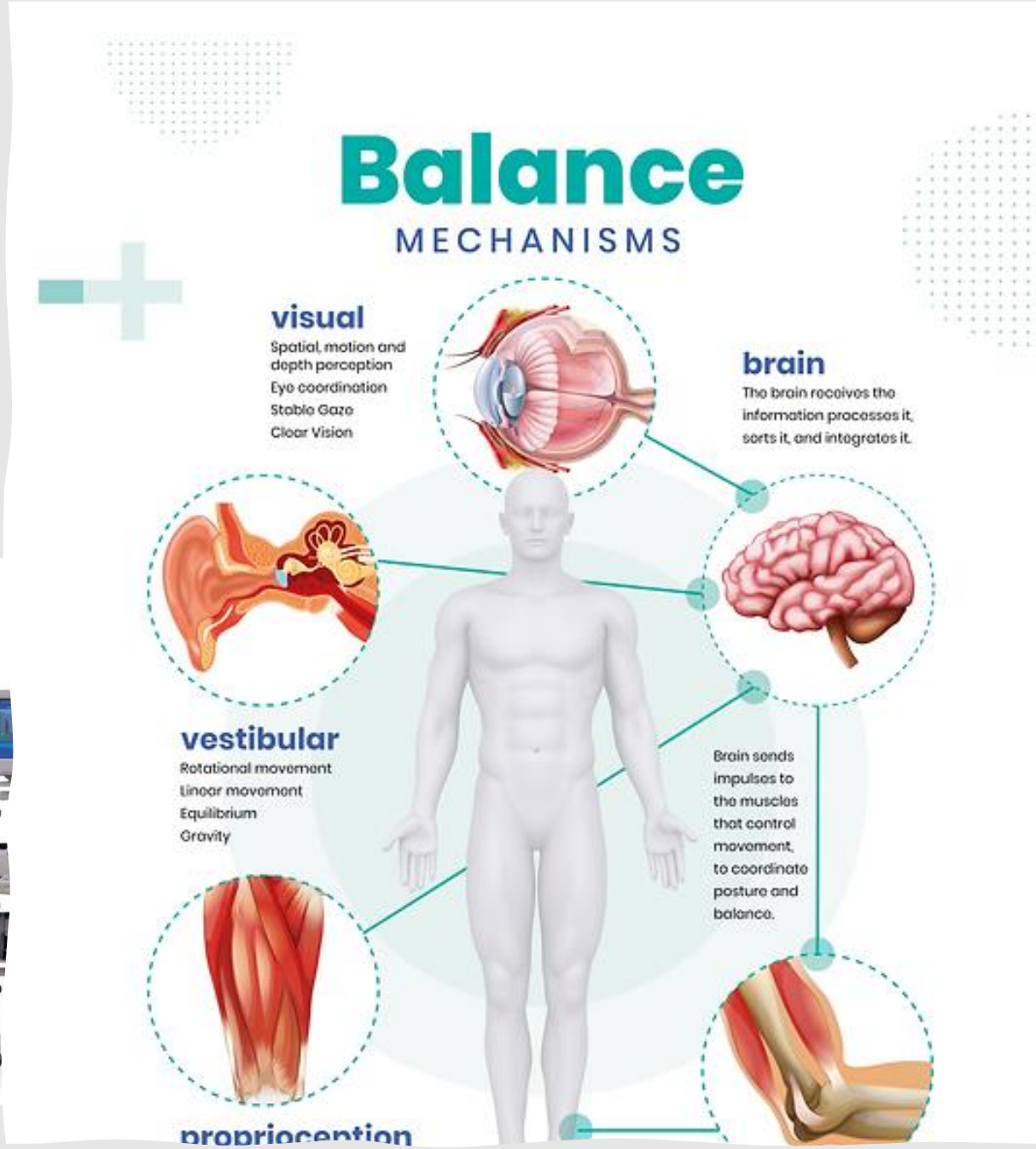
Balance treatment: vestibular rehabilitation/ balance-retraining









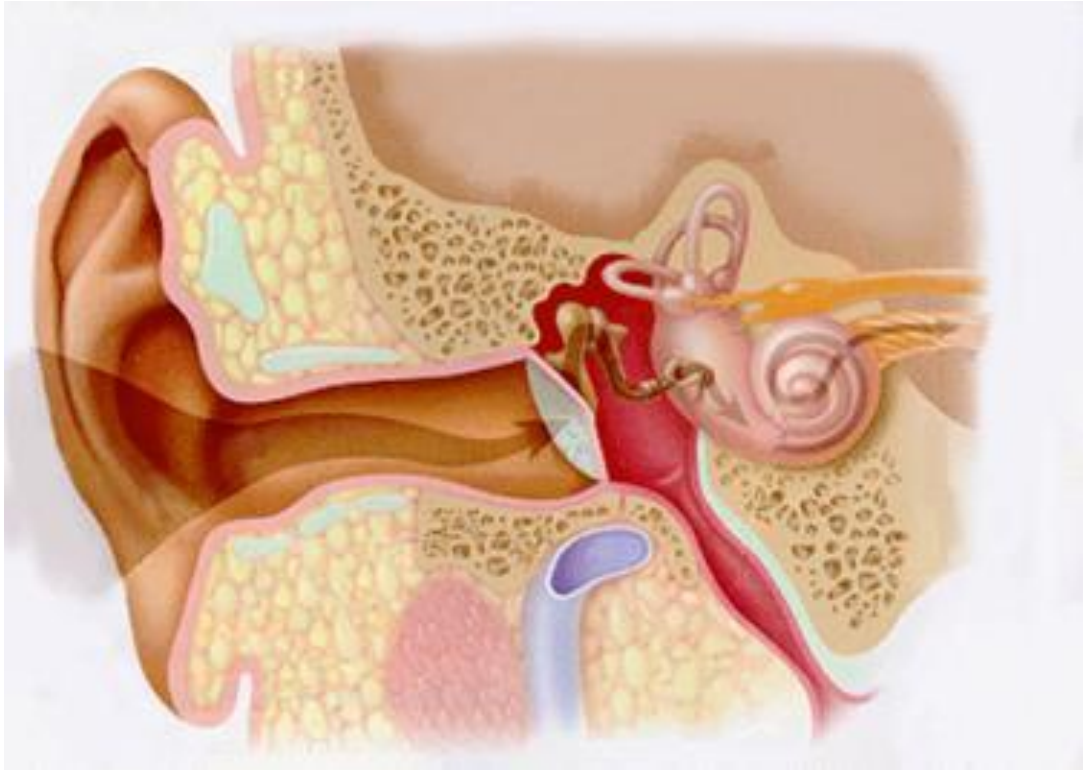


HEARING LOSS

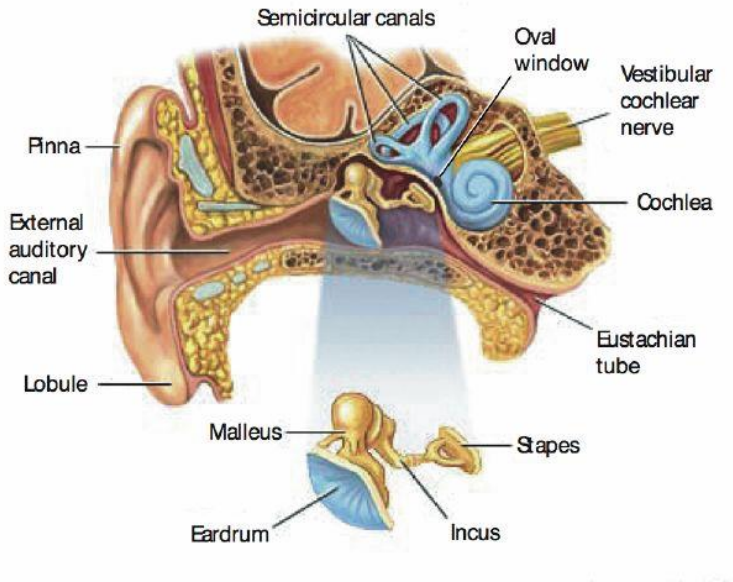
Hearing…what happens?

- Phenomenon of capturing sound energy (acoustic energy) traveling via air medium by outer ear
- Transforming the sound energy by the middle ear (impedance matching system)
- Tranducing the sound energy by the inner ear to neural activity…
- Decoded by central auditory nervous system into complex auditory perceptions (speech and music)

External Ear

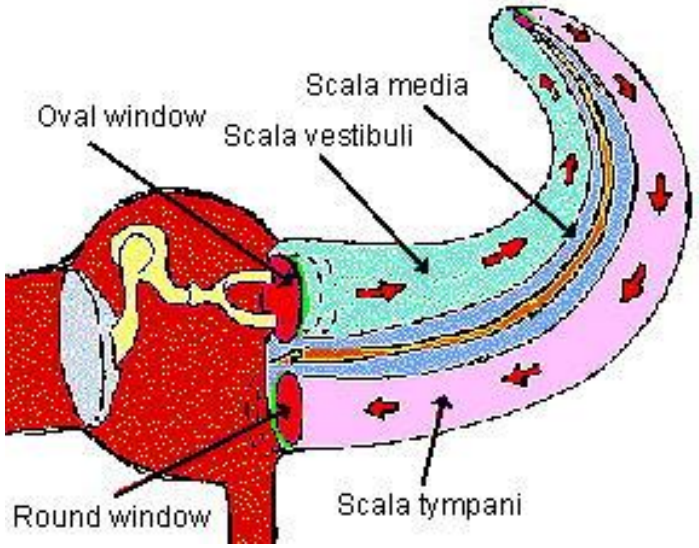


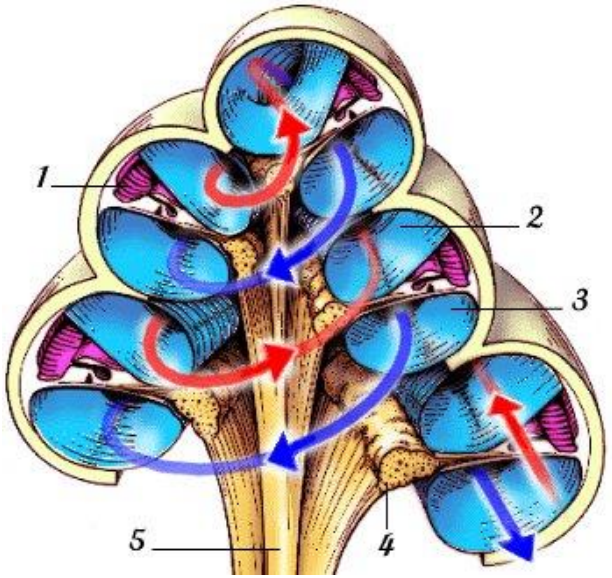
Middle Ear



INNER EAR

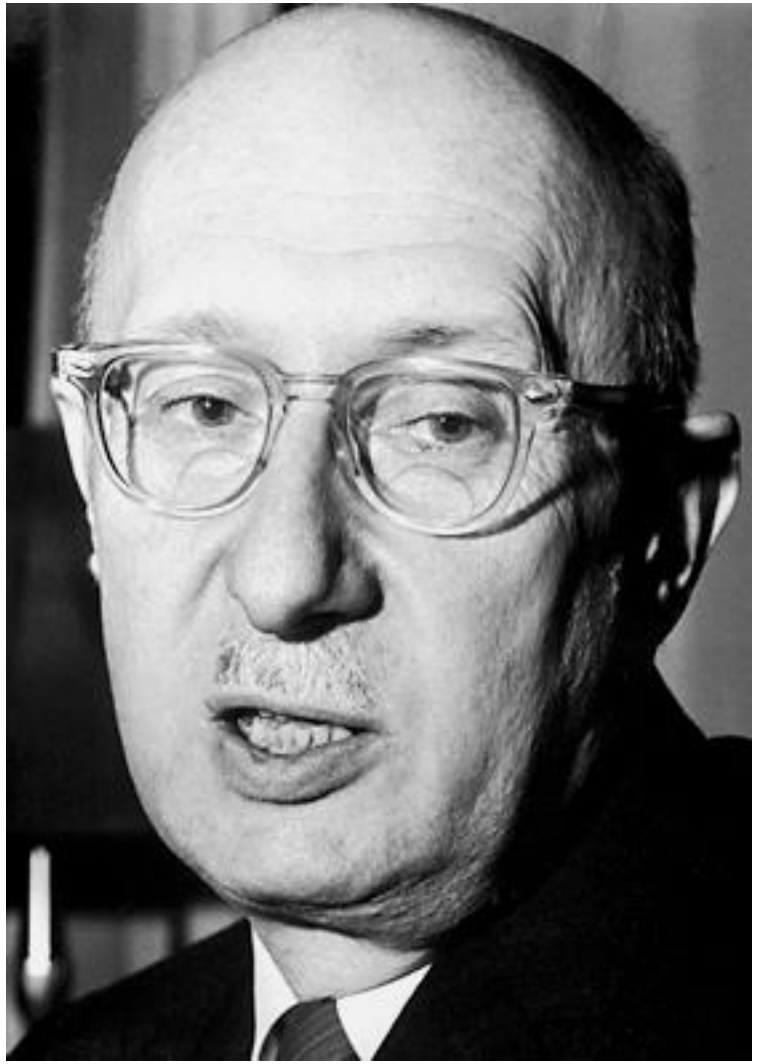
- Acoustic energy from middle ear transferred to inner ear fluids (hydromechanical energy)
- Inner ear fluids: endolymph and perilymph
- Cochlear fluid movement results in basilar membrane movement (Traveling Wave Theory)



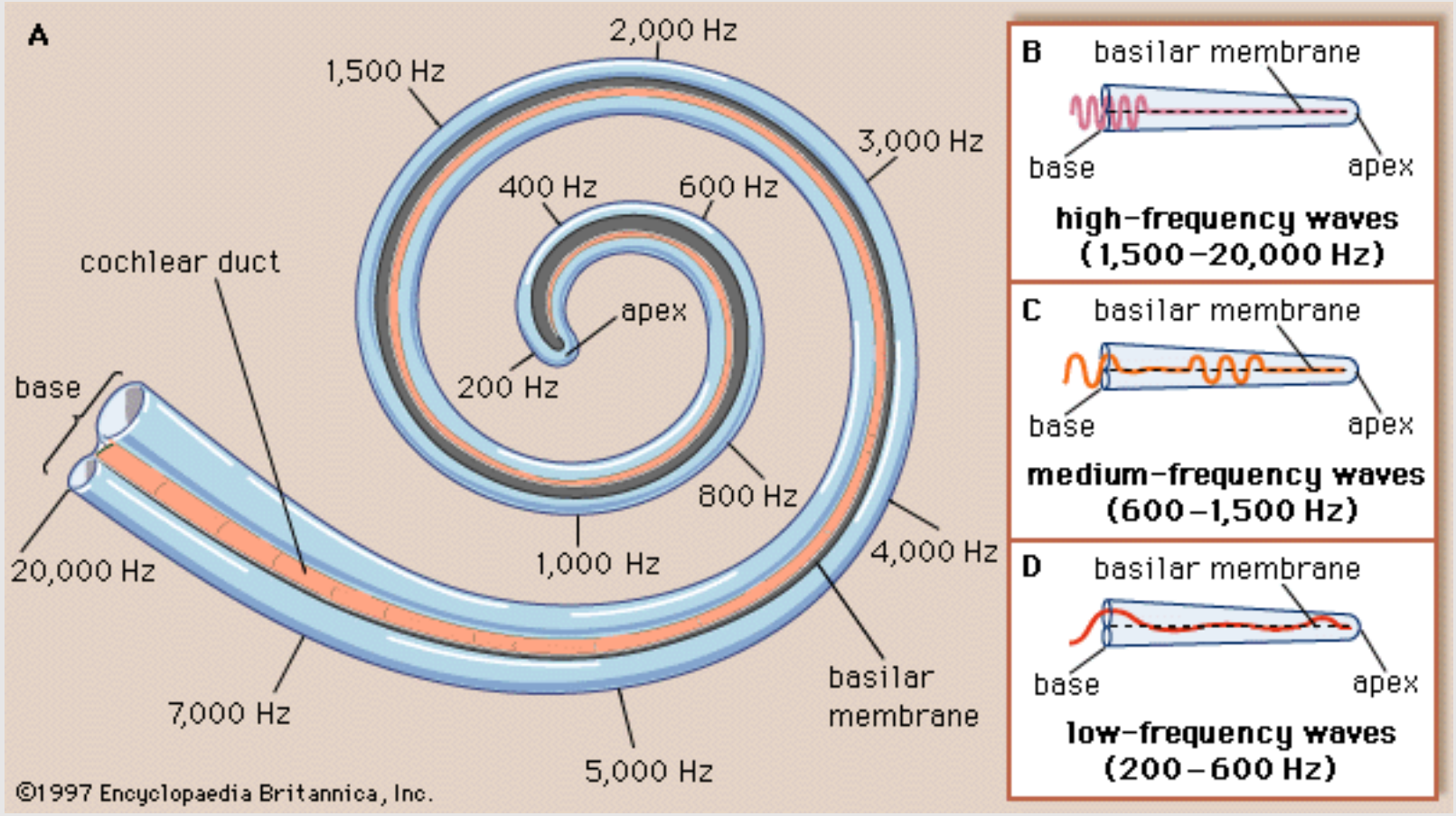


Traveling Wave Theory

- Georg von Bekesy
- 1961 Nobel Prize in Physiology/Medicine
- Harvard University

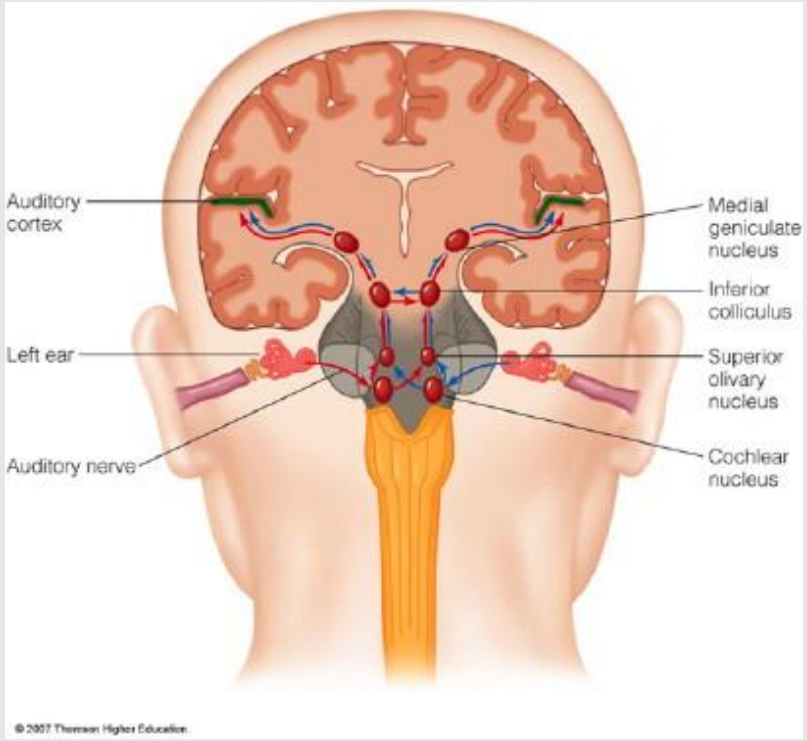


Cochlear Tonotopicity



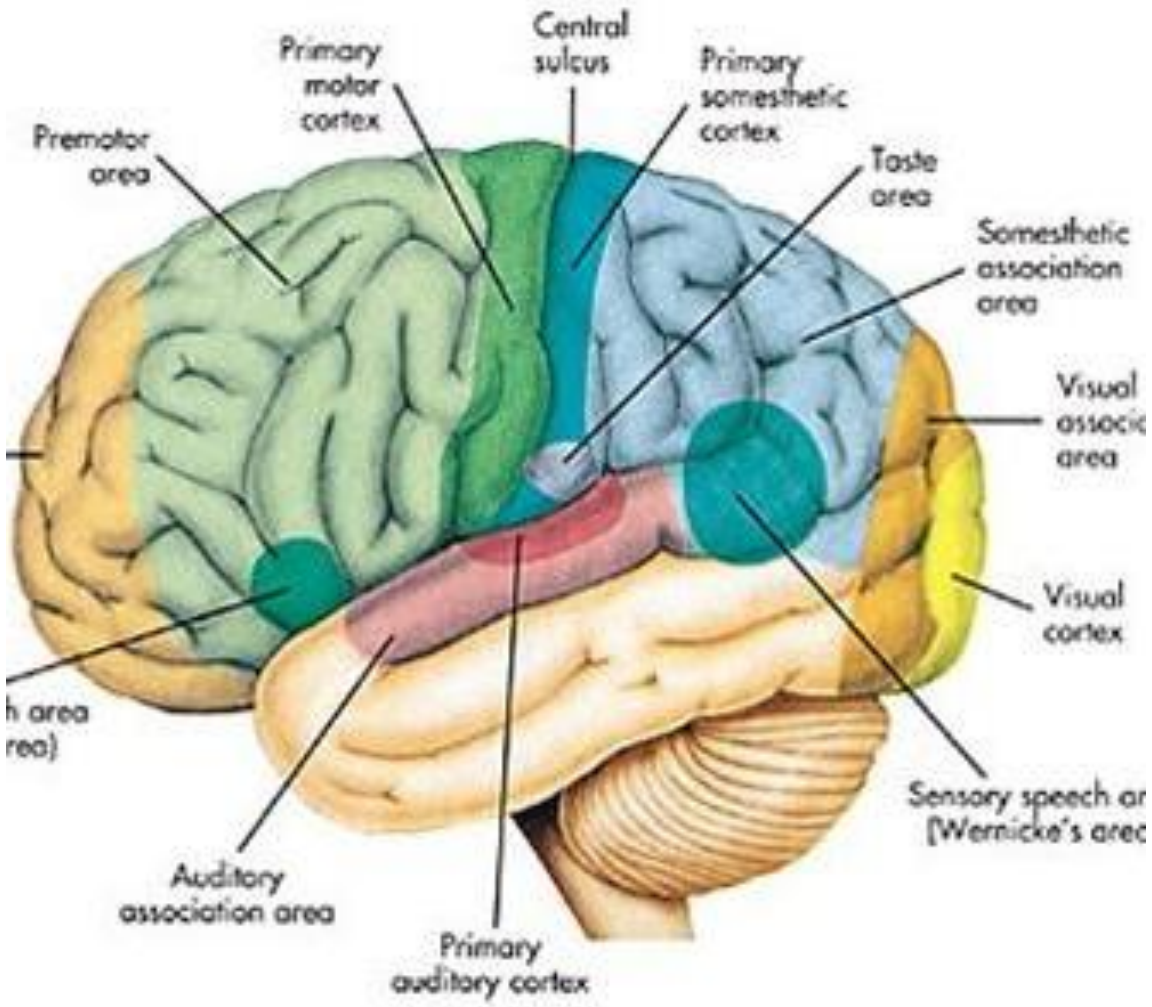
Central Auditory System

- Cochlear nerve enters pontomedullary junction
- Consists of cochlear nuclei at brainstem
  - Dorsal nucleus (all fibers ascend contralaterally)
  - Ventral nucleus (30% fibers go ipsilateral)

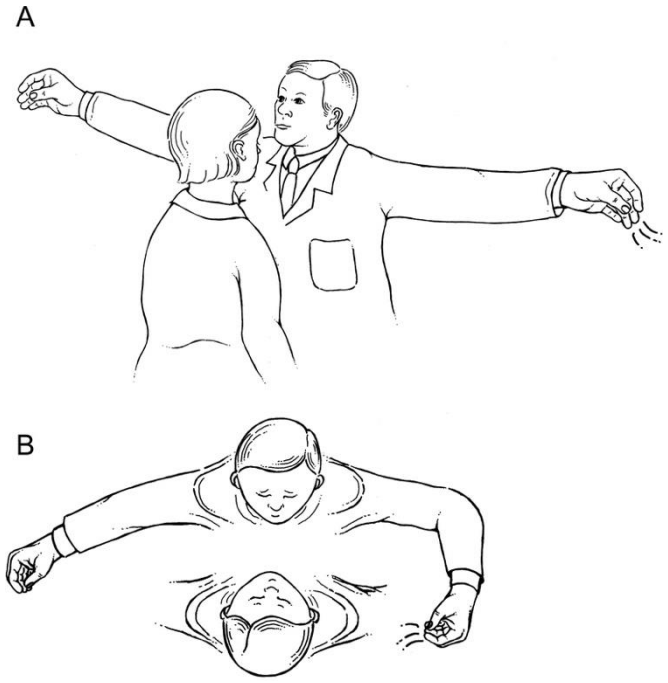


Auditory Cortex

- Hearing in noise
- Sound localization
- Temporal judgments of sound
- Decoding complex acoustic stimuli
- Music appreciation

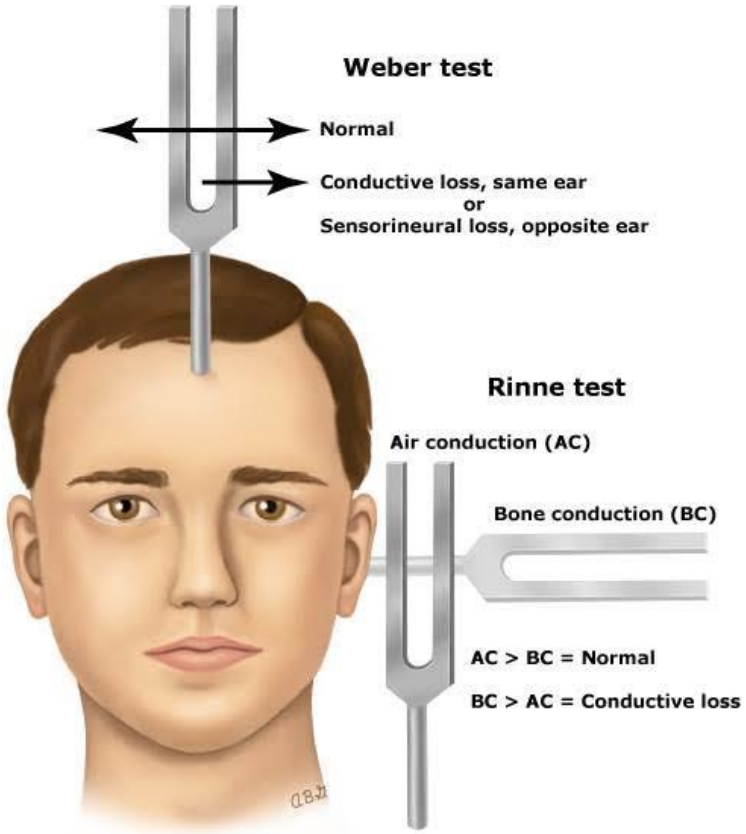


How to test hearing:



	No. of ears at specified CALFRAST threshold (%)	Mean auditory threshold, dB	Maximum auditory threshold, dB	Minimum auditory threshold, dB
CALFRAST-Faint 70	220 (49.8)	<21*	35	20
CALFRAST-Strong 70	130 (29.4)	29	65	20
CALFRAST-Strong 35	34 (7.7)	46	NR	30
CALFRAST-Strong 10	27 (6.1)	52	NR	30
CALFRAST-Strong 2	23 (5.2)	59	NR	40
CALFRAST-Strong NR	8 (1.8)	>80*	NR	70

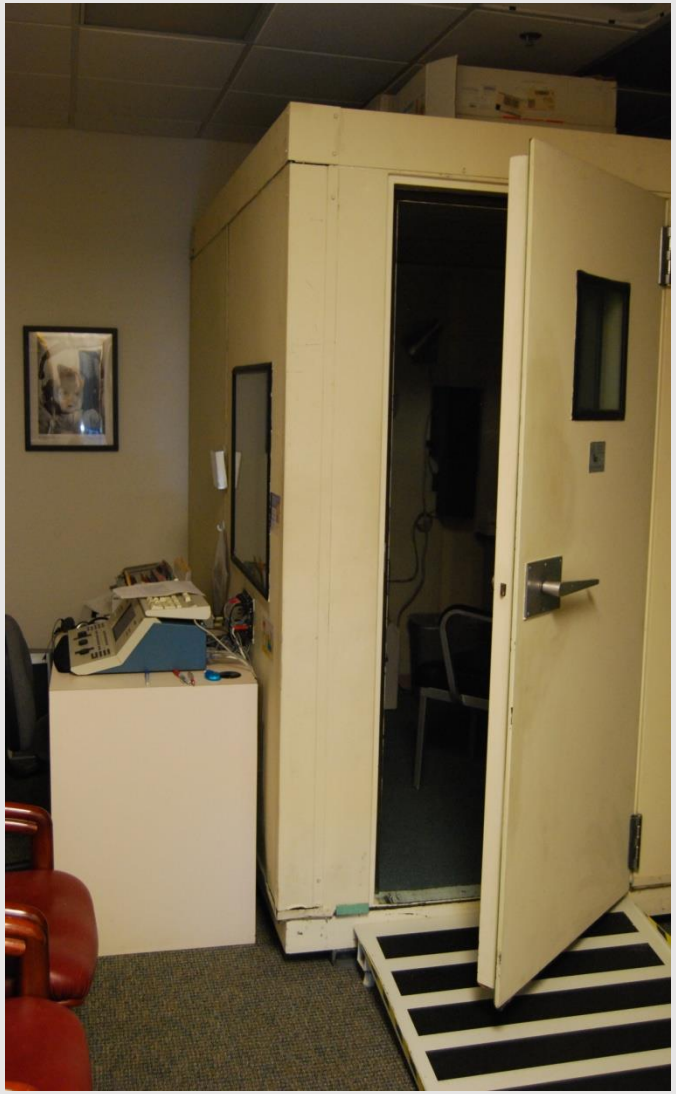
How to test hearing:





©KaWe

Audiology Booth and Testing





Air Conduction and Bone Conduction Testing



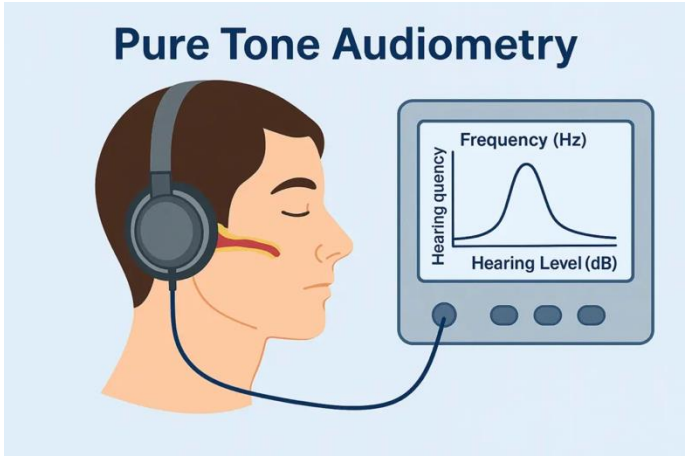
Supra-Aural Headphones



Insert Earphones



Pure Tone Testing & Word Recognition Testing

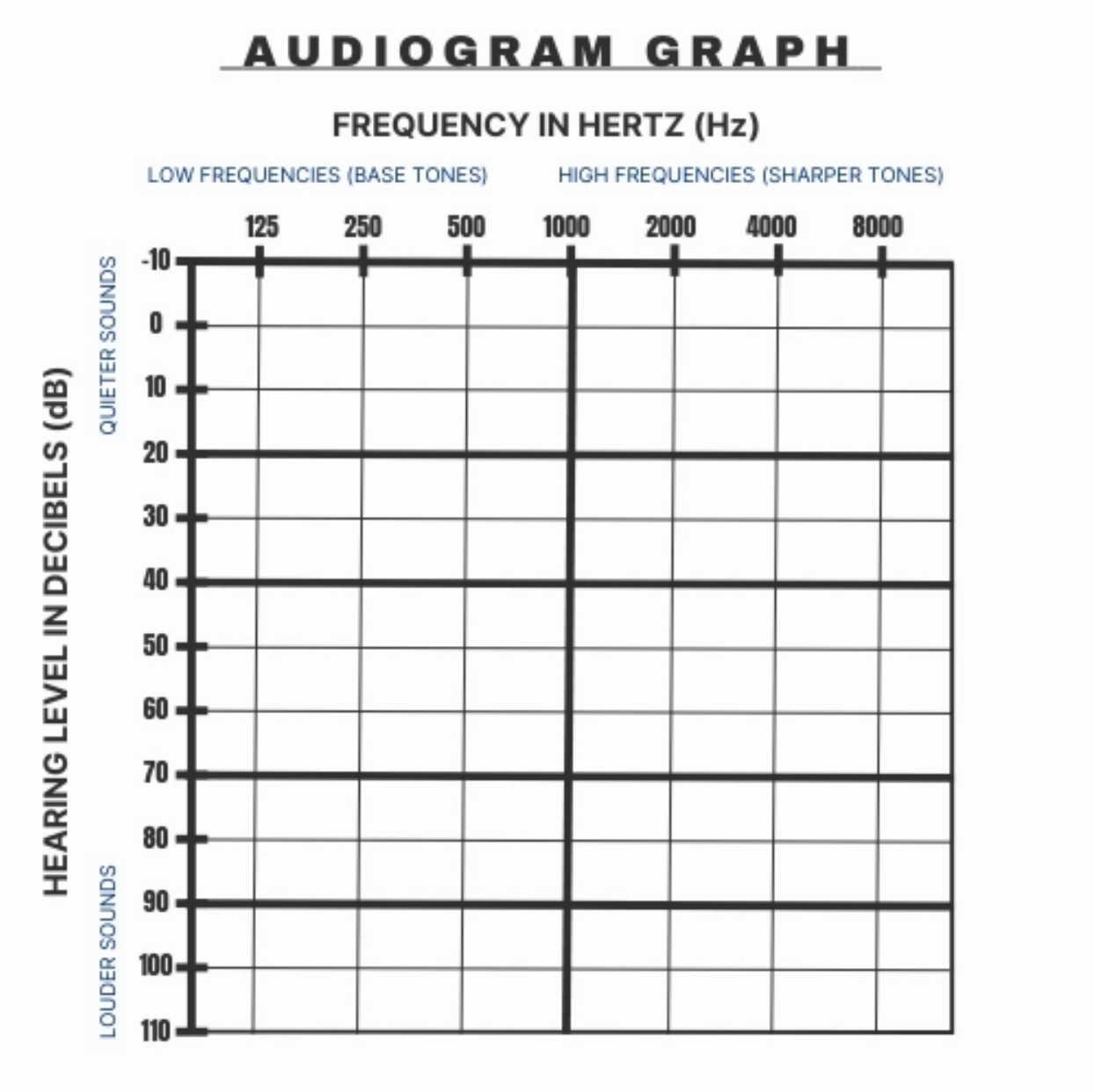


PBK List

Please	Hunt	Scab
Great	No	Lay
Sled	Box	Class
Pants	Are	Me
Rat	Teach	Dish
Bad	Slice	Neck
Pinch	Is	Beef
Such	Tree	Few
Bus	Smile	Use
Need	Bath	Did
Ways	Slip	Bit
Five	Ride	Pond
Mouth	End	Hot
Rag	Pink	Own
Put	Thank	Bead
Fed	Take	Shop
Fold	Cart	

Score	Descriptor
96-100%	Excellent
88-95%	Very Good
80-87%	Good
70-79%	Fair
50-69%	Poor
Less than 50%	Very Poor

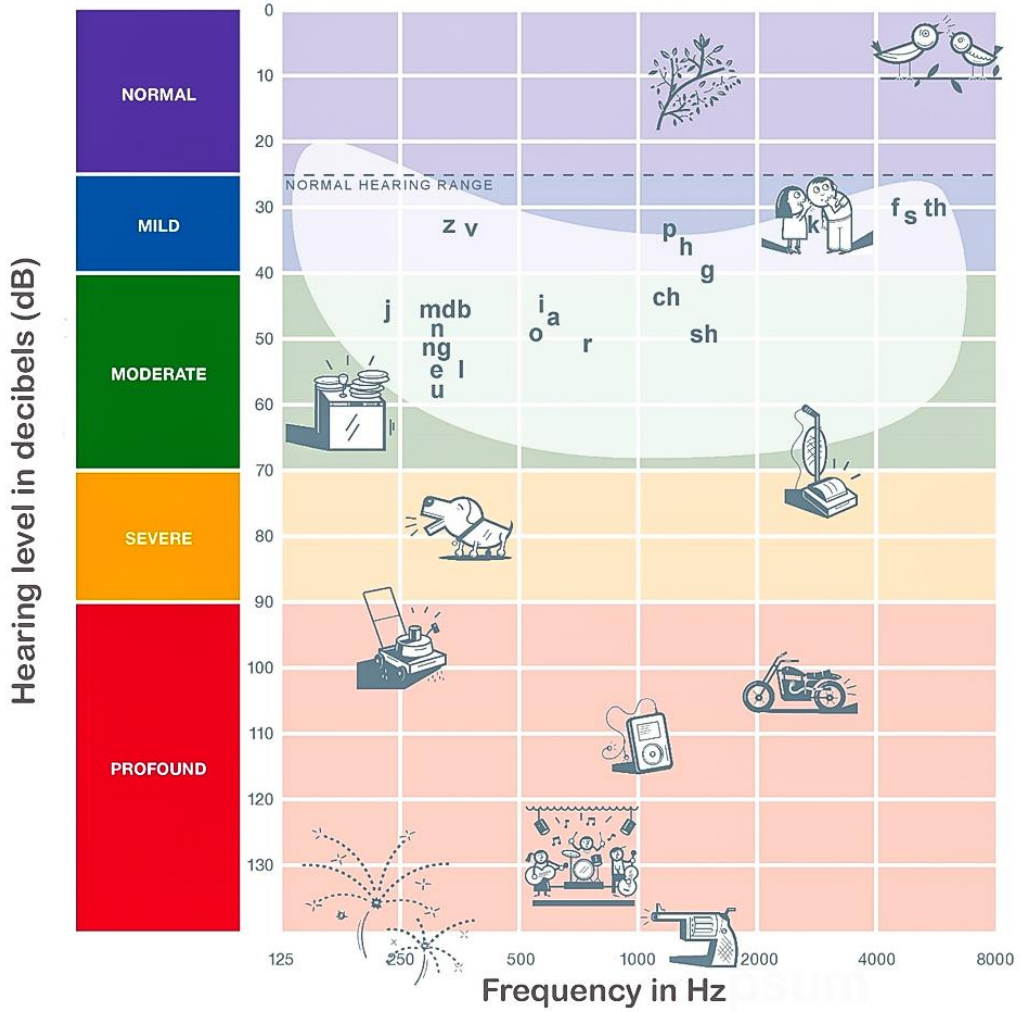
AUDIOGRAM


AUDIOGRAM GRAPH
FREQUENCY IN HERTZ (Hz)
LOW FREQUENCIES (BASE TONES)
HIGH FREQUENCIES (SHARPER TONES)
125
250
500
1000
2000
4000
8000
HEARING LEVEL IN DECIBELS (dB)
QUIETER SOUNDS
LOUDER SOUNDS
-10
0
10
20
30
40
50
60
70
80
90
100
110

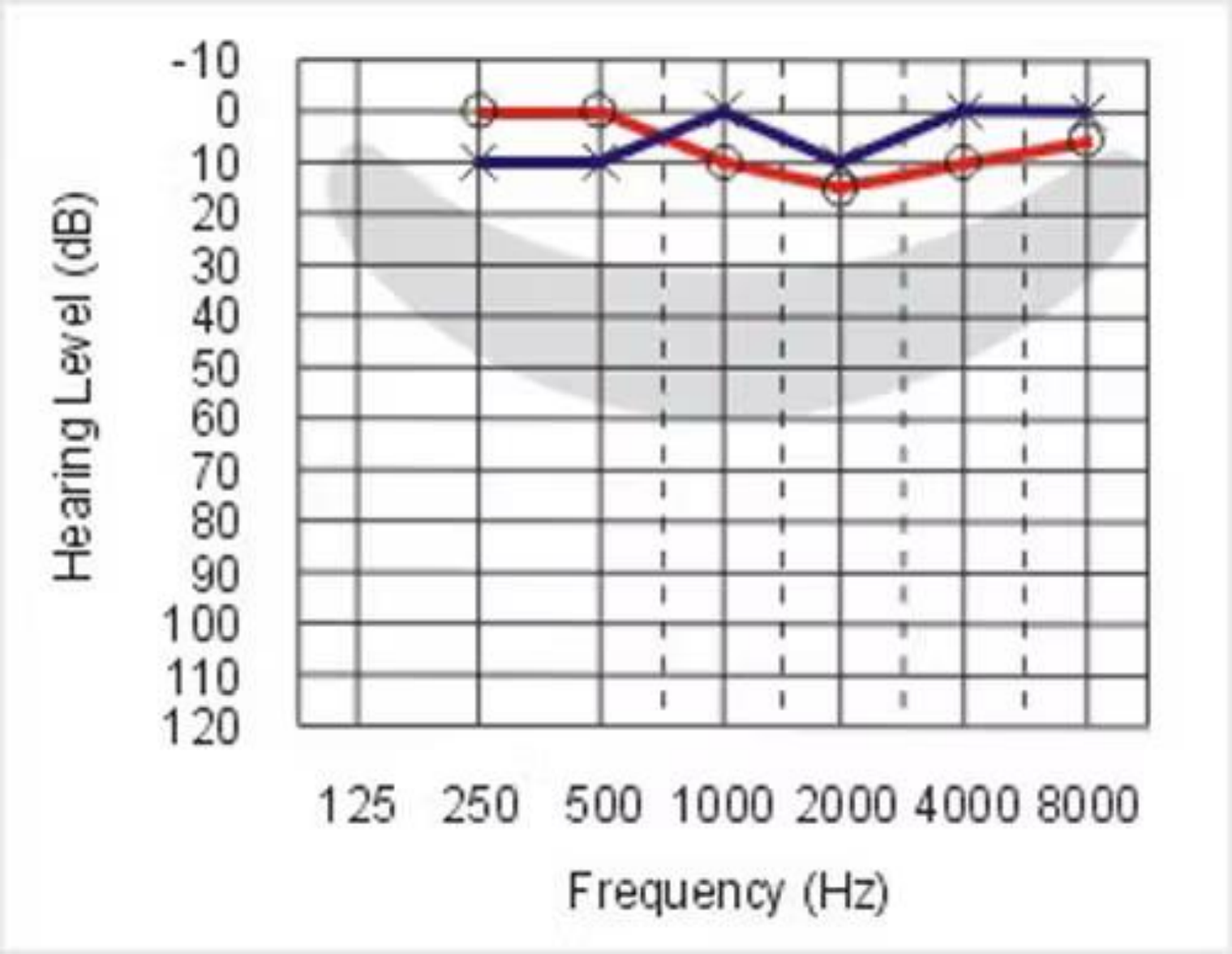
AUDIOGRAM- degrees of hearing loss



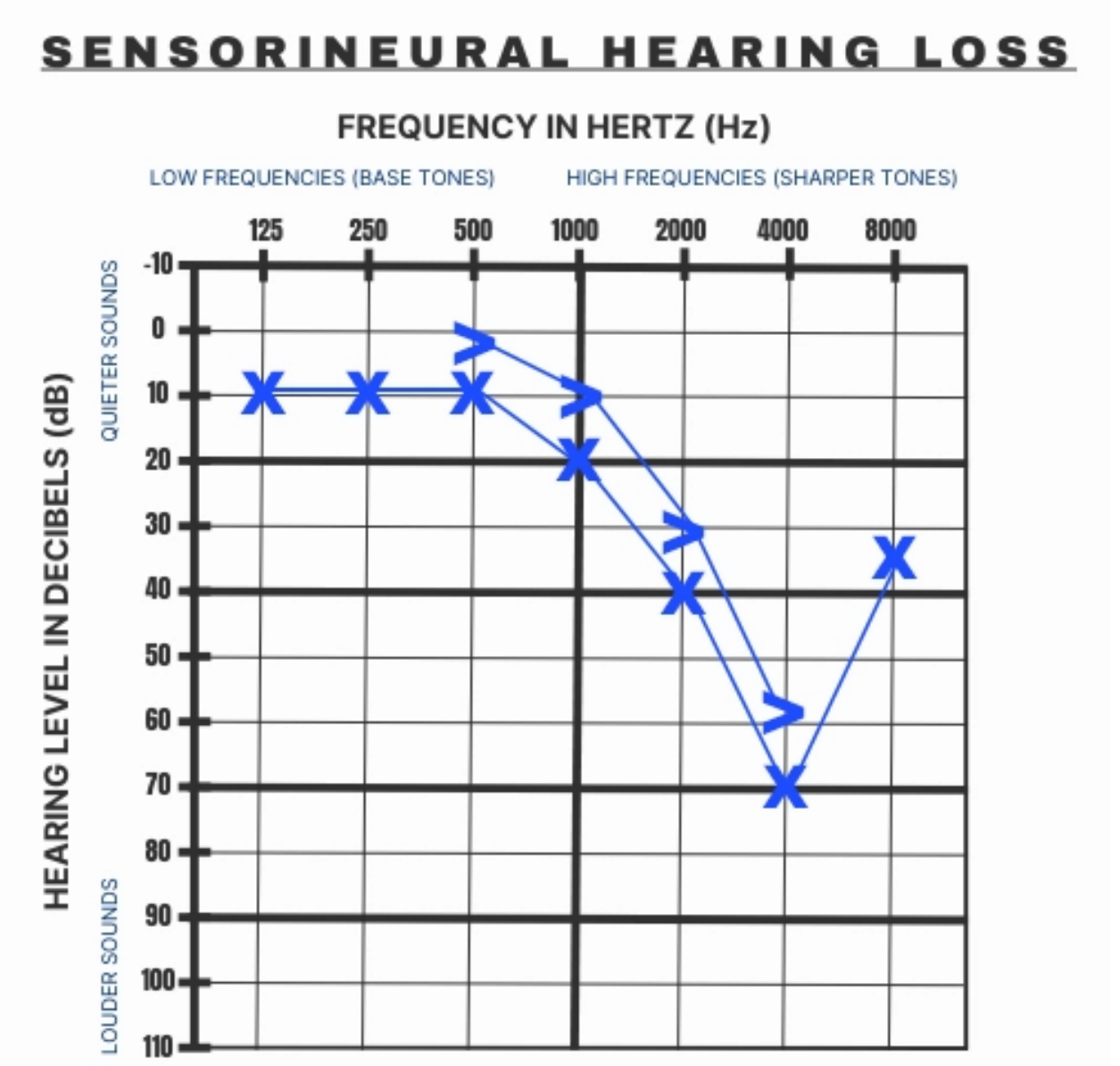
AUDIOGRAM


NORMAL
MILD
MODERATE
SEVERE
PROFOUND
Hearing level in decibels (dB)
0
10
20
30
40
50
60
70
80
90
100
110
120
130
NORMAL HEARING RANGE
z v
j
m d b
n
ng
e
l
u
i
a
o
r
p
h
g
ch
sh
f
s
th
k
125
250
500
1000
2000
4000
8000
Frequency in Hz

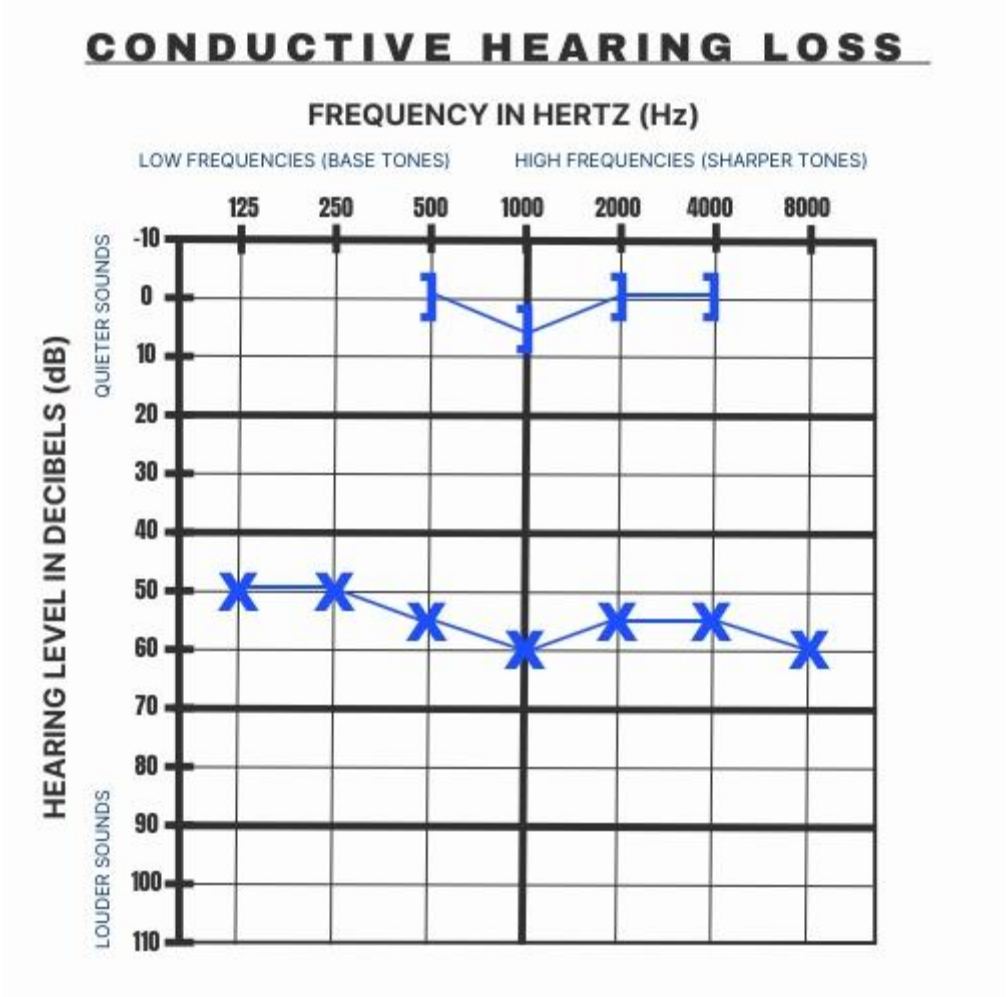
Normal hearing



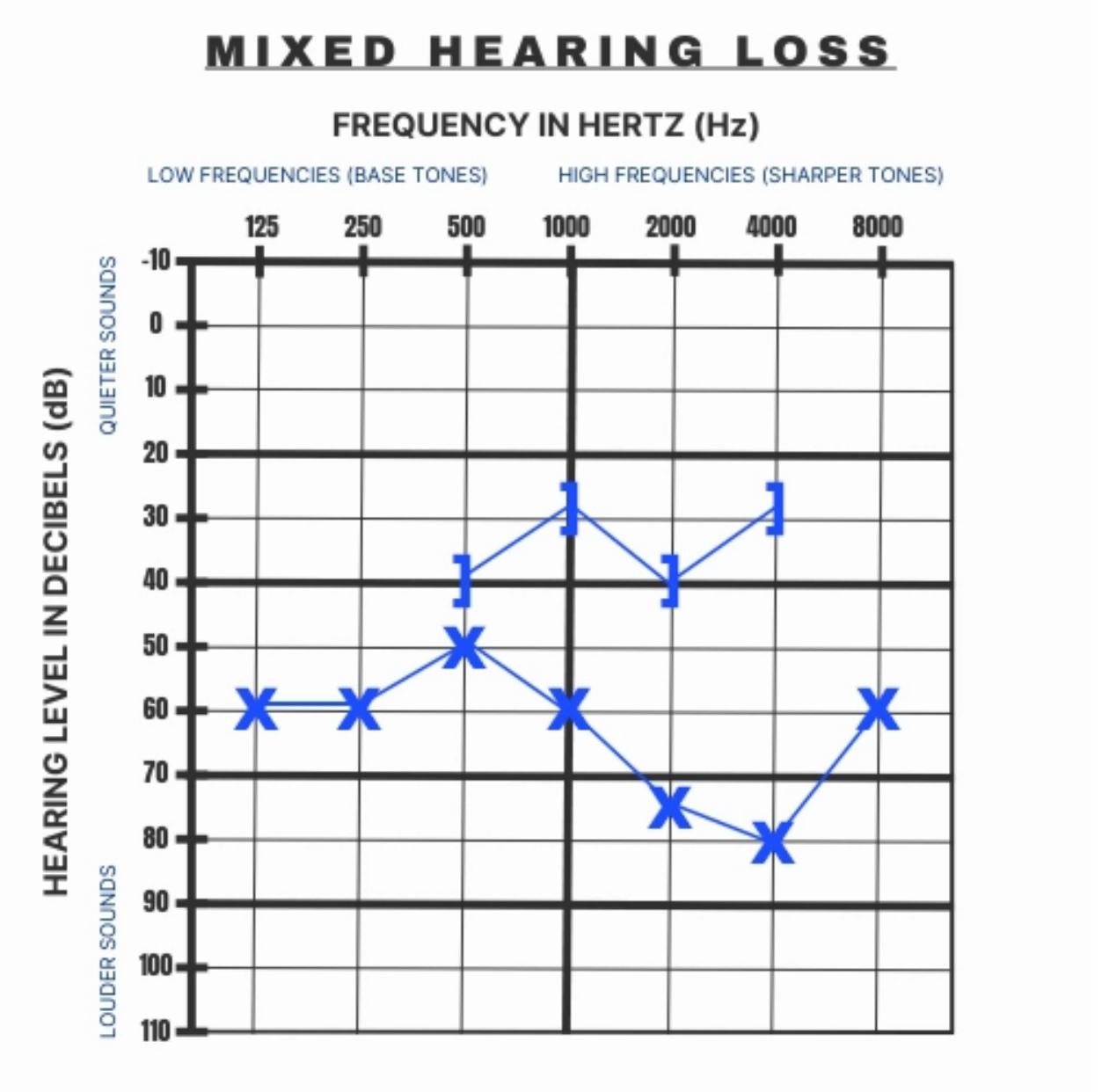
AUDIOGRAM- sensorineural hearing loss



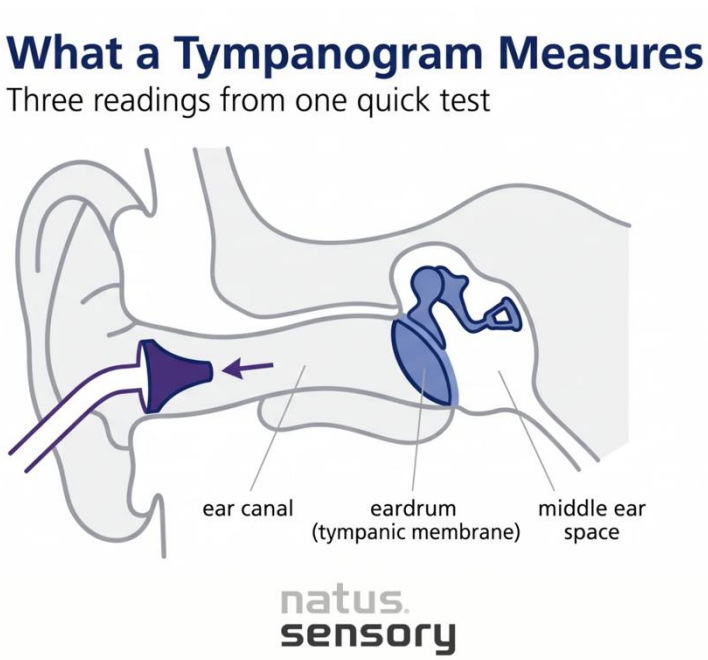
Conductive Hearing Loss



Mixed Hearing Loss



TYMPANOGRAM



1 Peak Pressure

Where the eardrum moves most freely

Reflects pressure inside the middle ear

2 Compliance

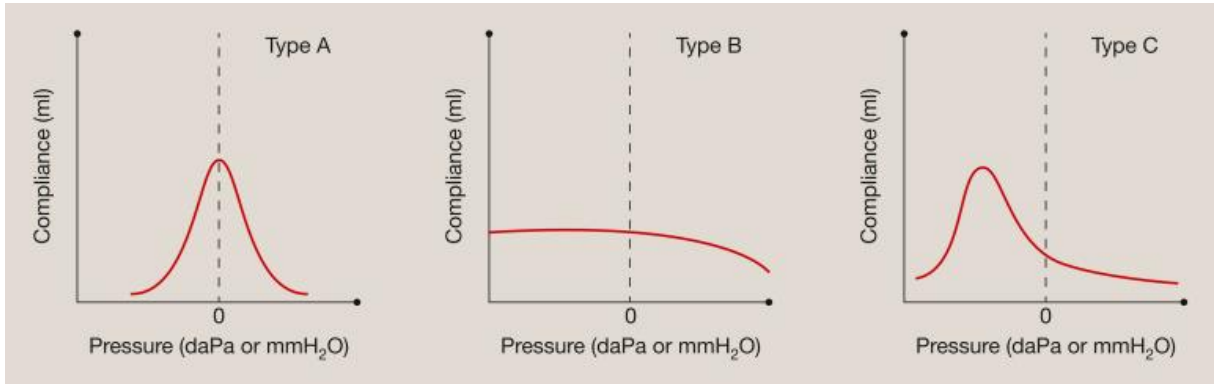
How easily the eardrum moves

A measure of middle ear mobility

3 Ear Canal Volume

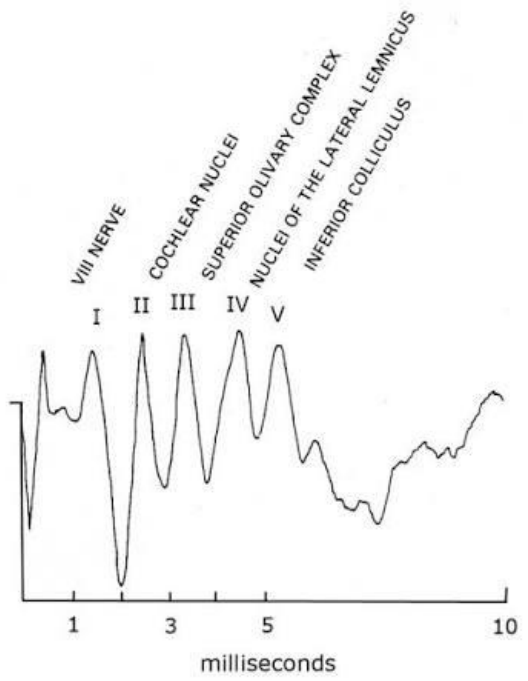
Air space between the probe and the eardrum

Helps tell a perforation apart from fluid

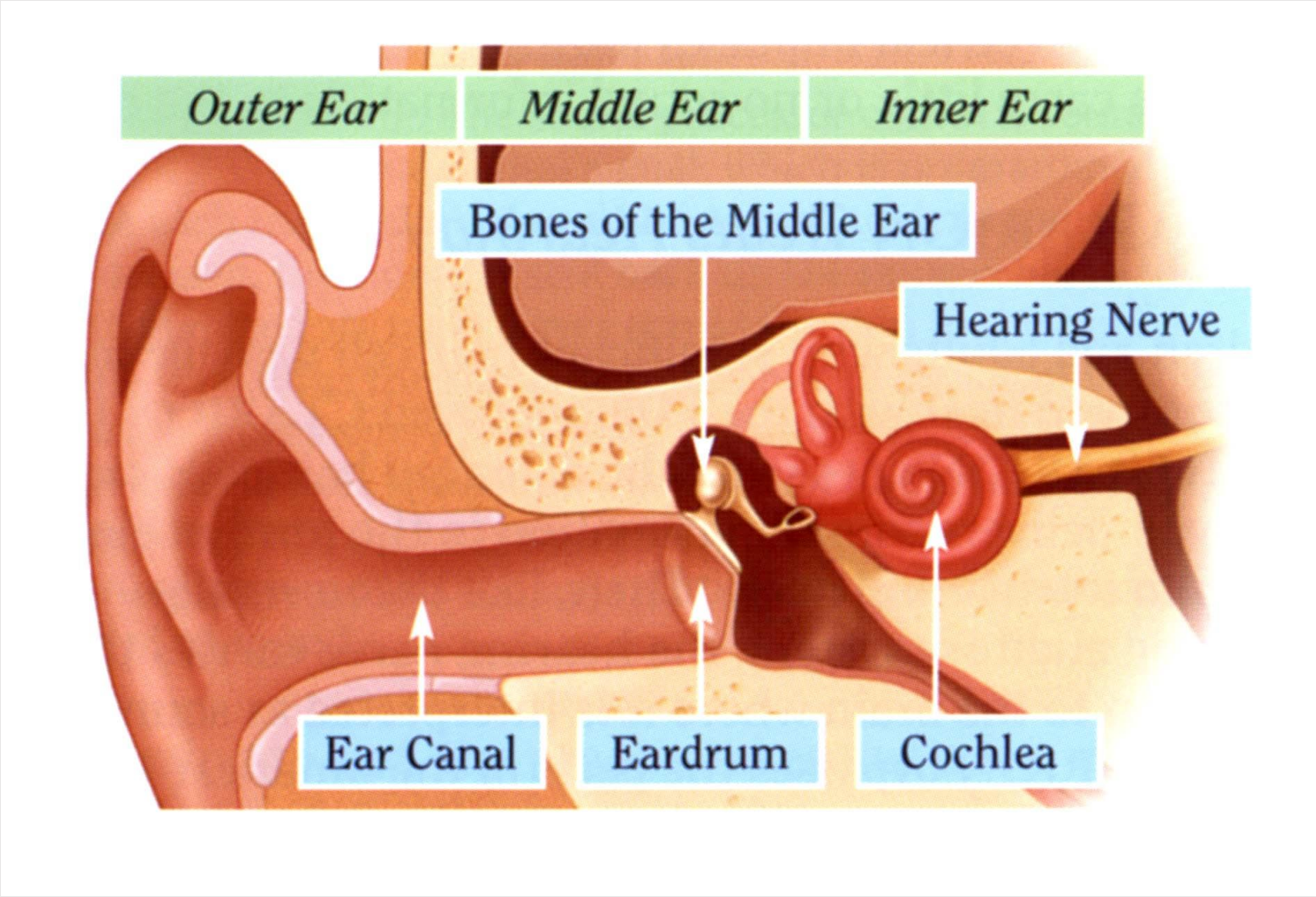


Auditory Brainstem Response Testing

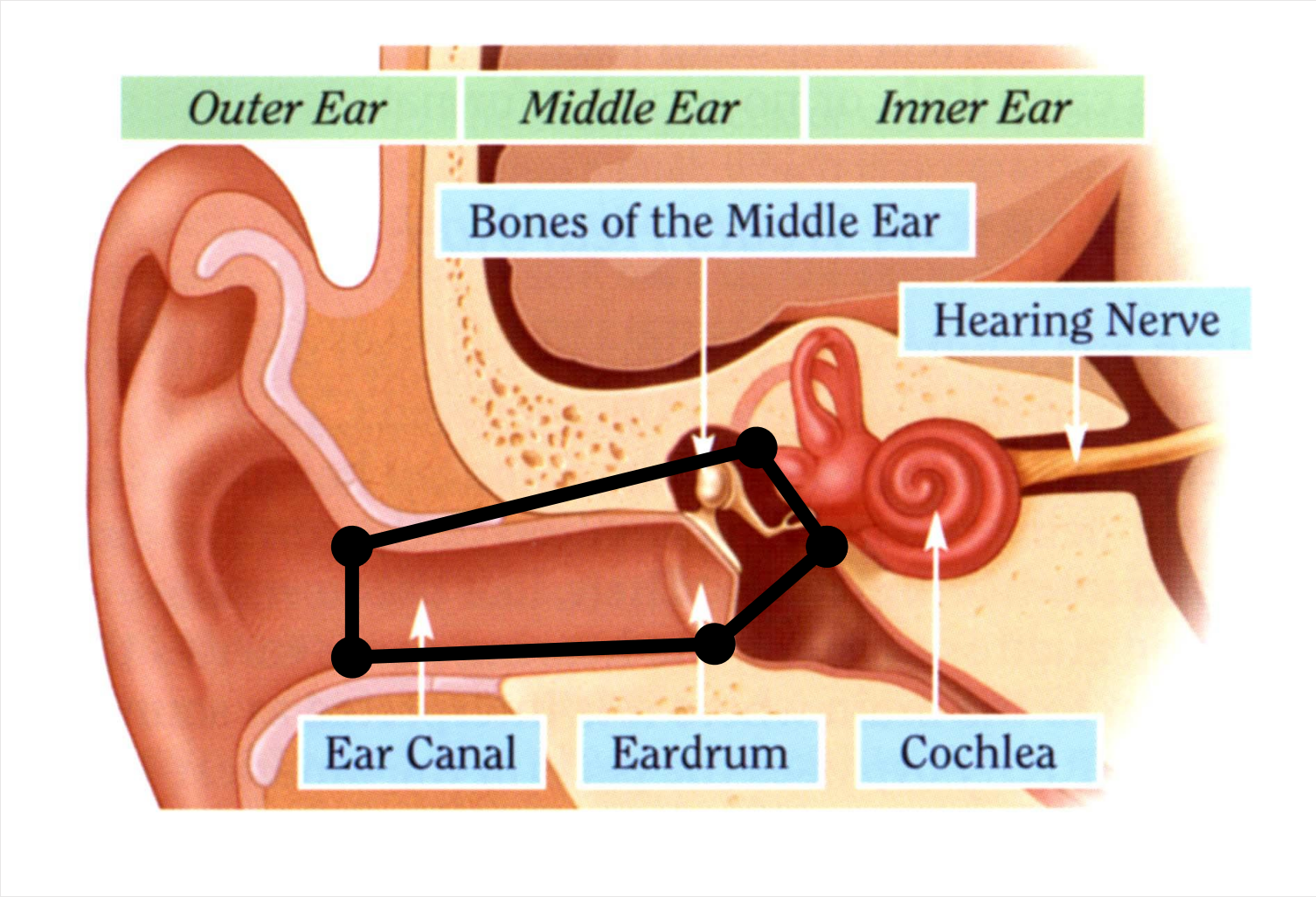




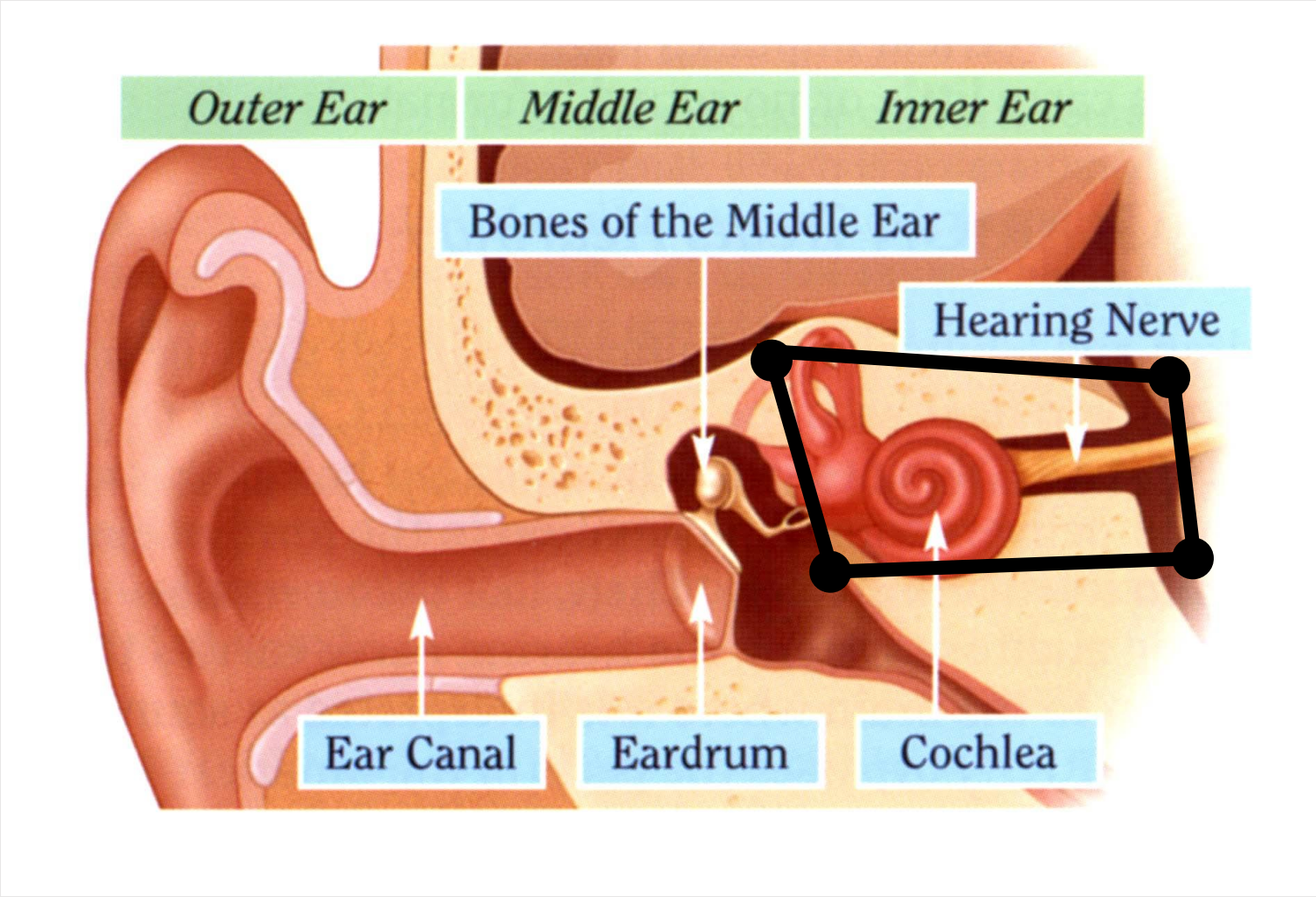
Types of hearing loss…



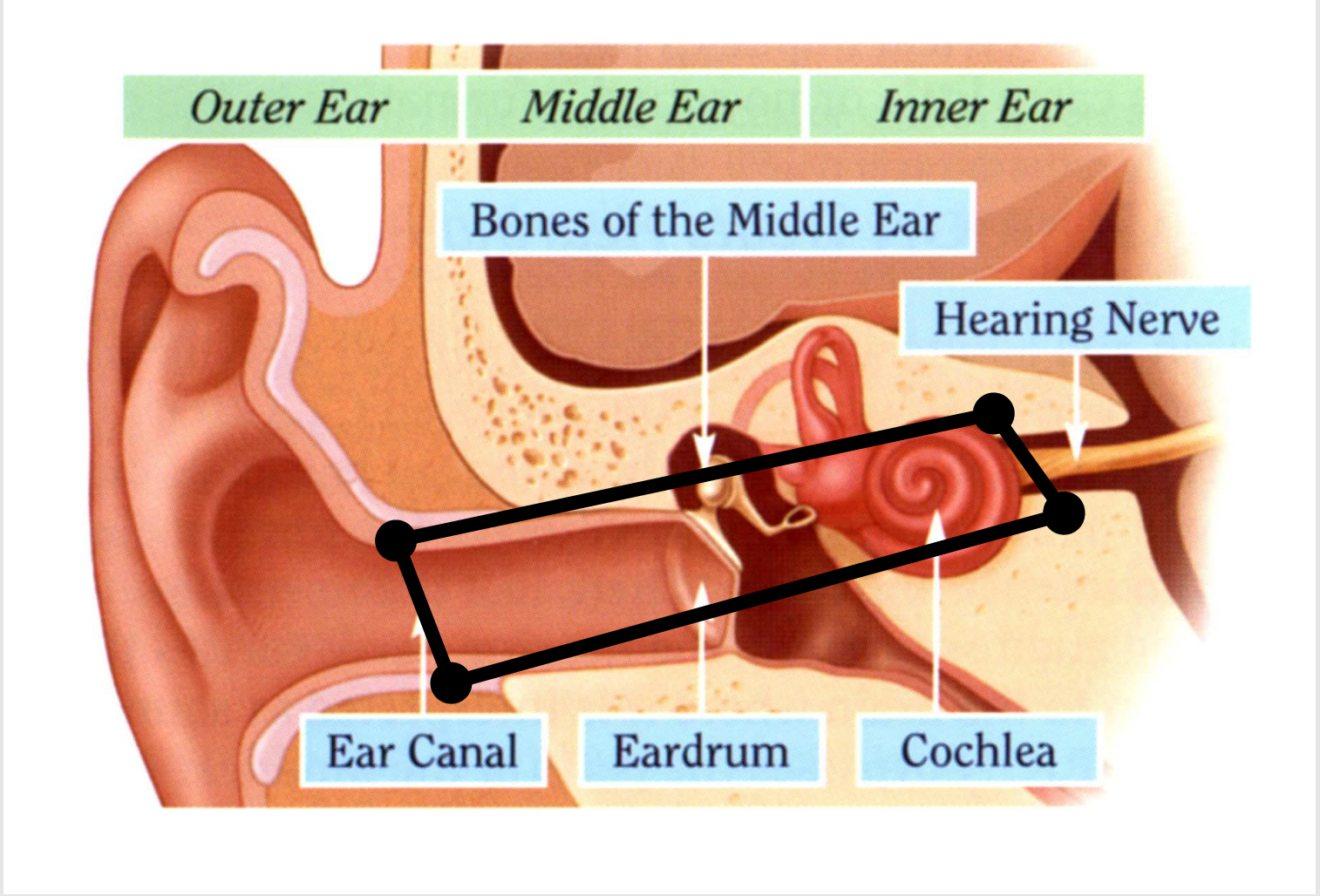
Conductive Hearing Loss



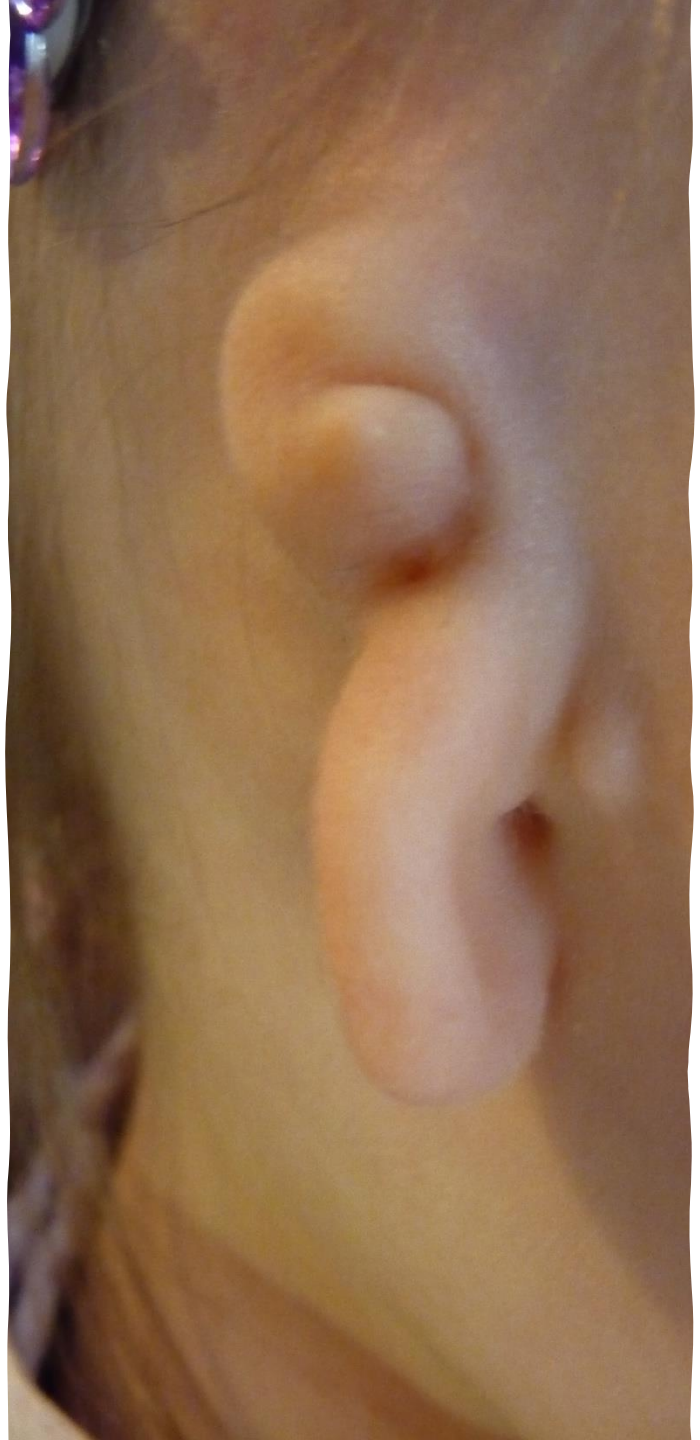
Sensorineural Hearing Loss



Mixed Hearing Loss



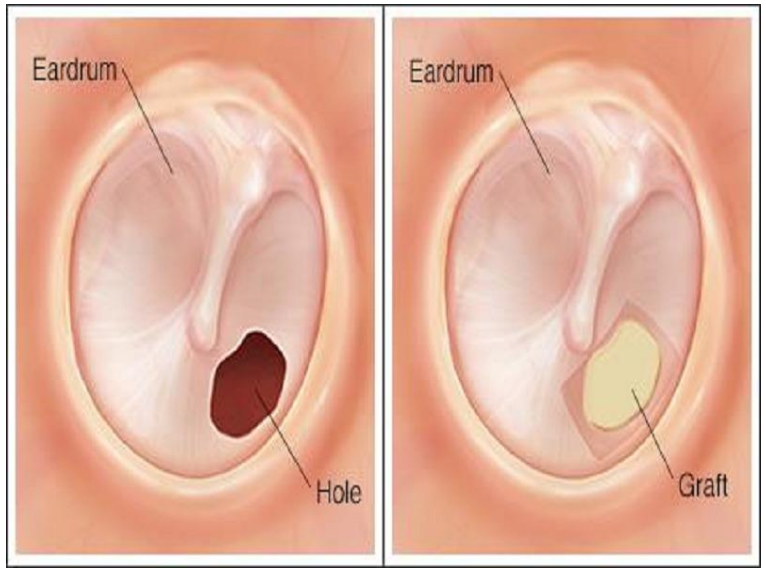
Causes for conductive hearing loss…

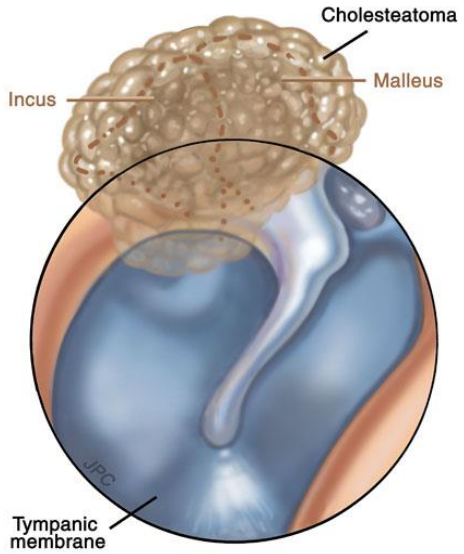


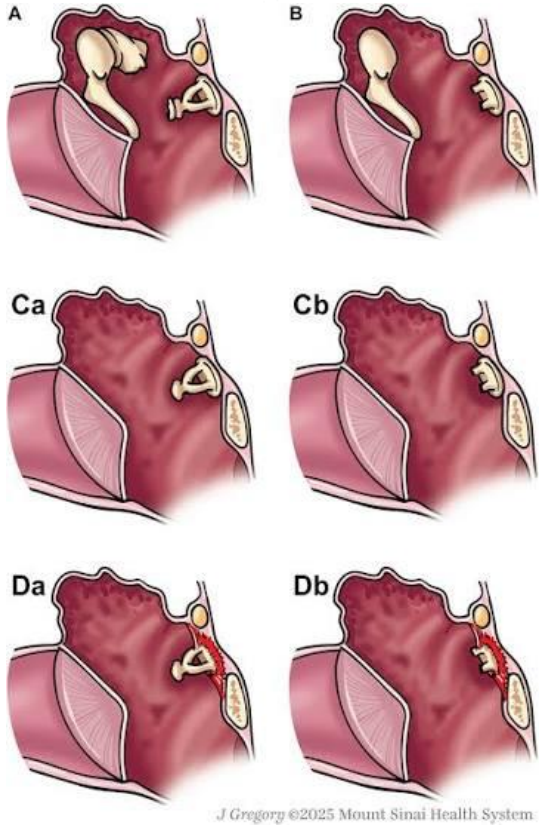




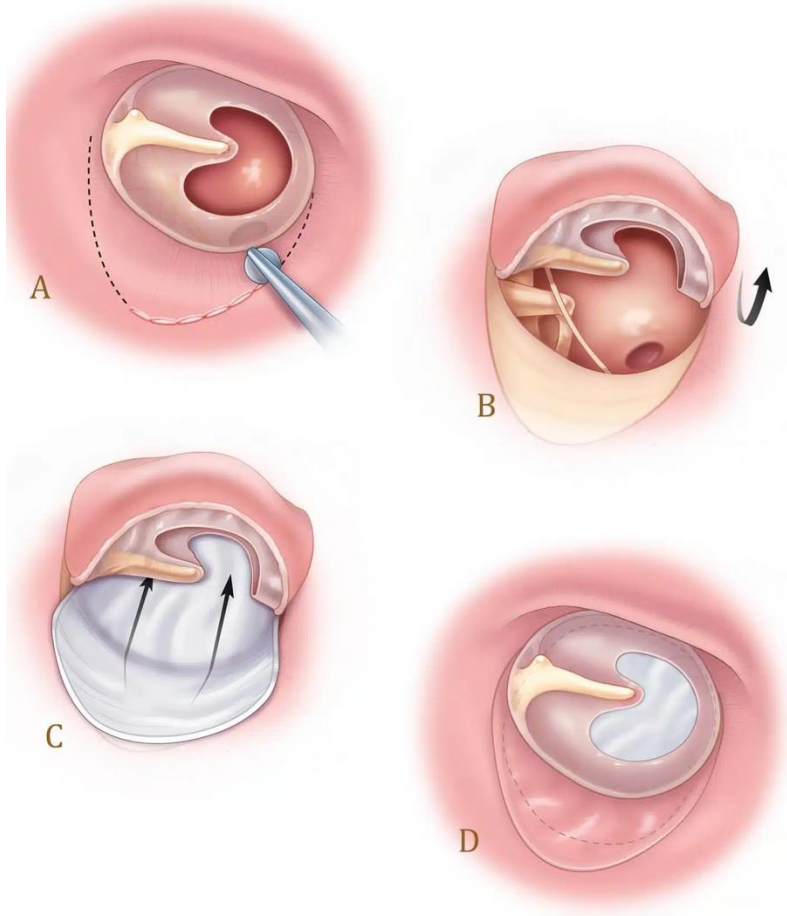
Causes of conductive hearing loss…

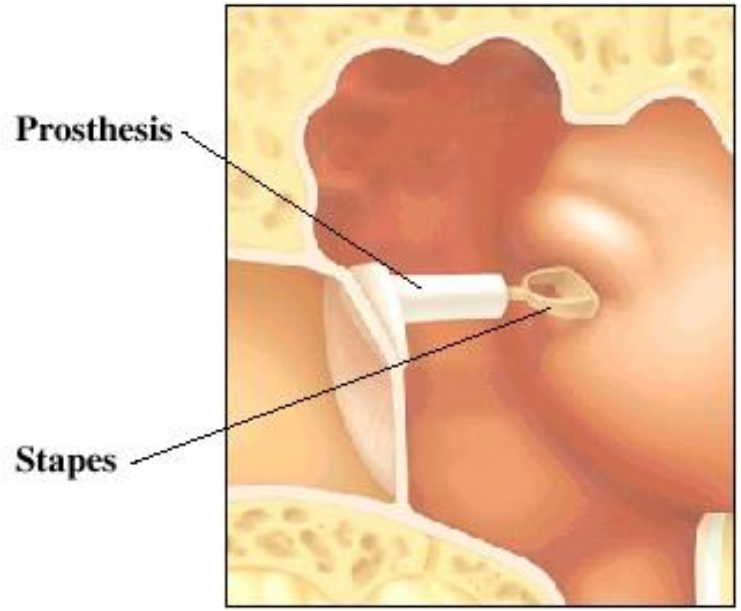


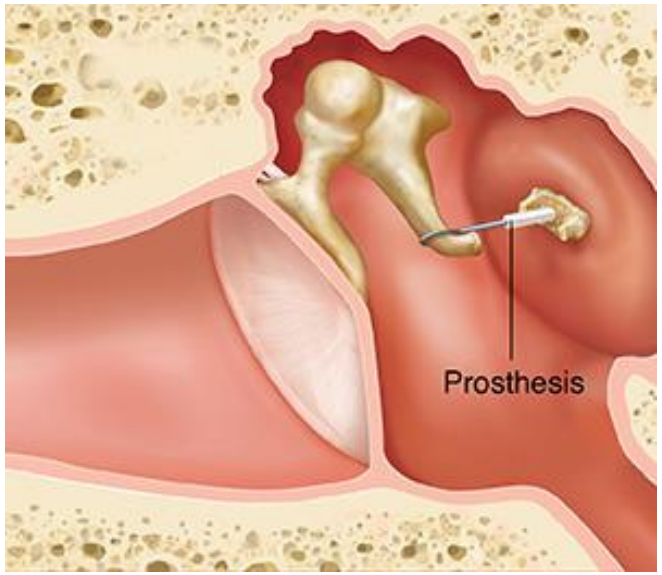




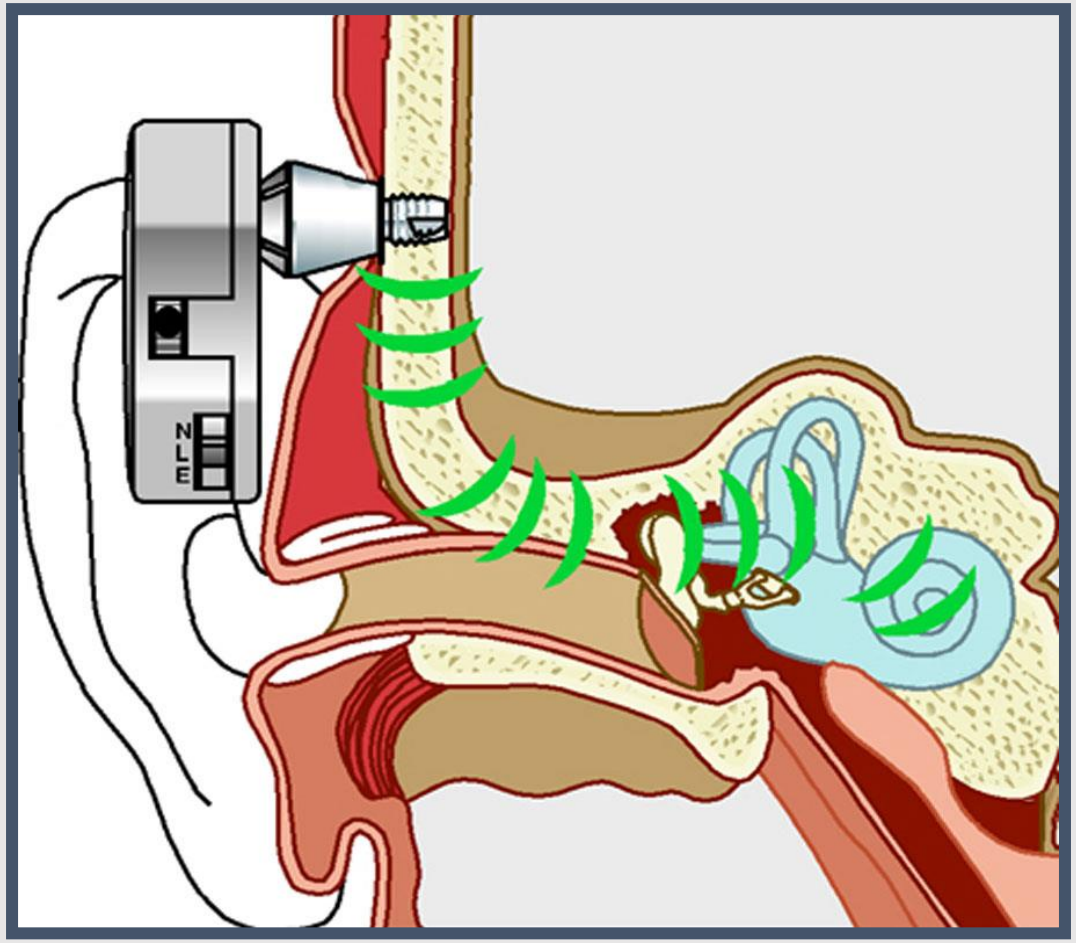
Ways to correct conductive hearing loss…







Bone conduction devices-percutaneous



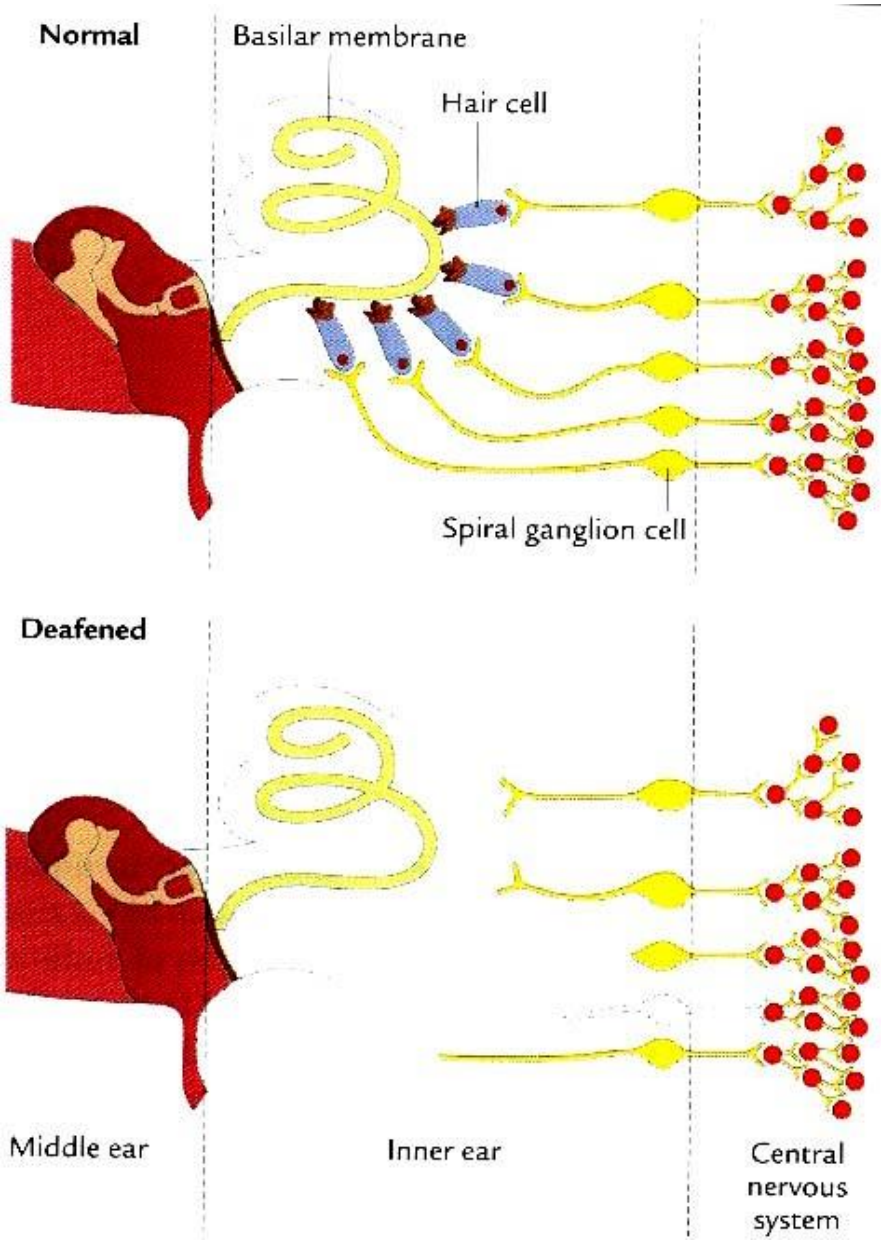


Bone conduction devices-transcutaneous



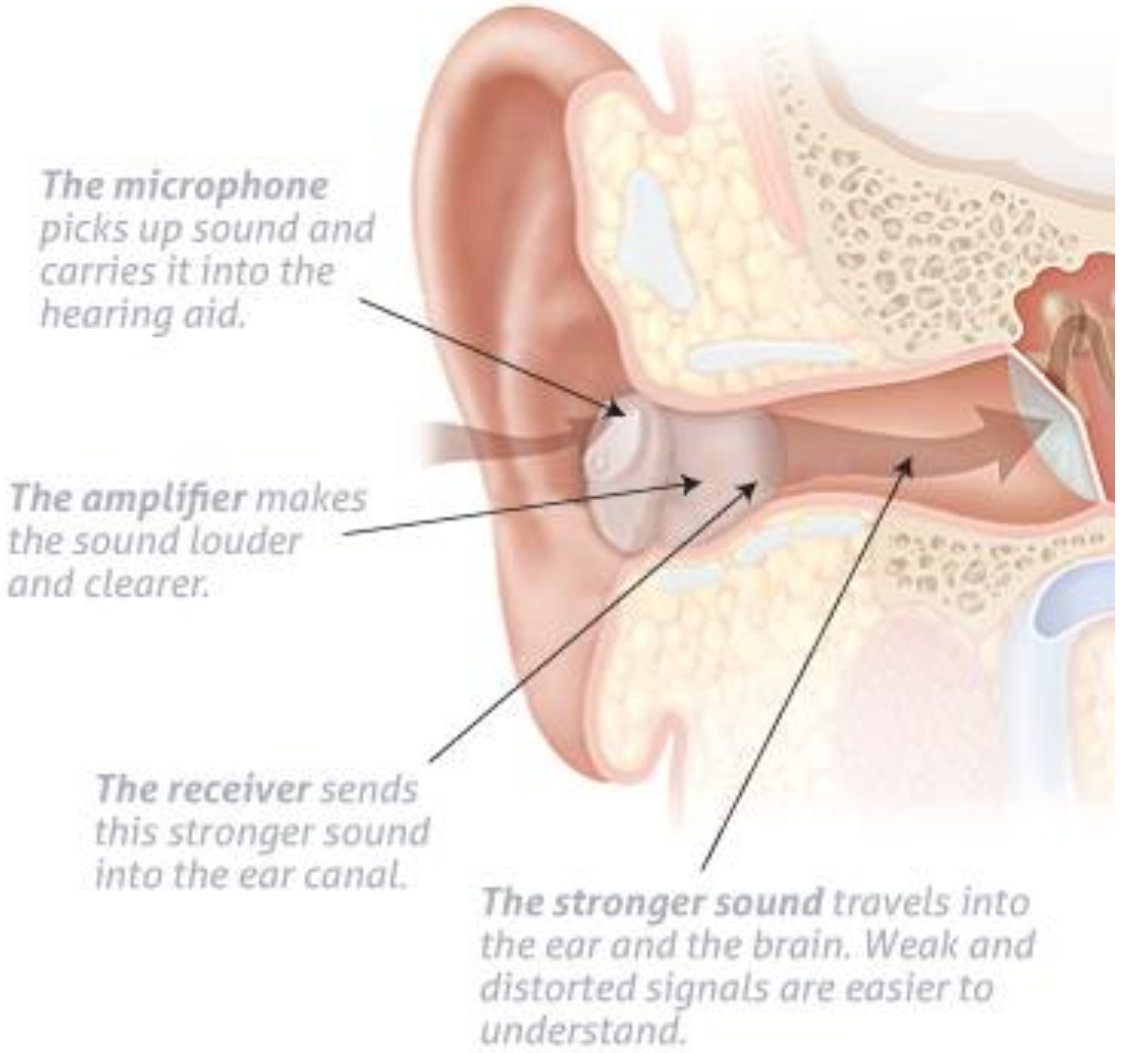


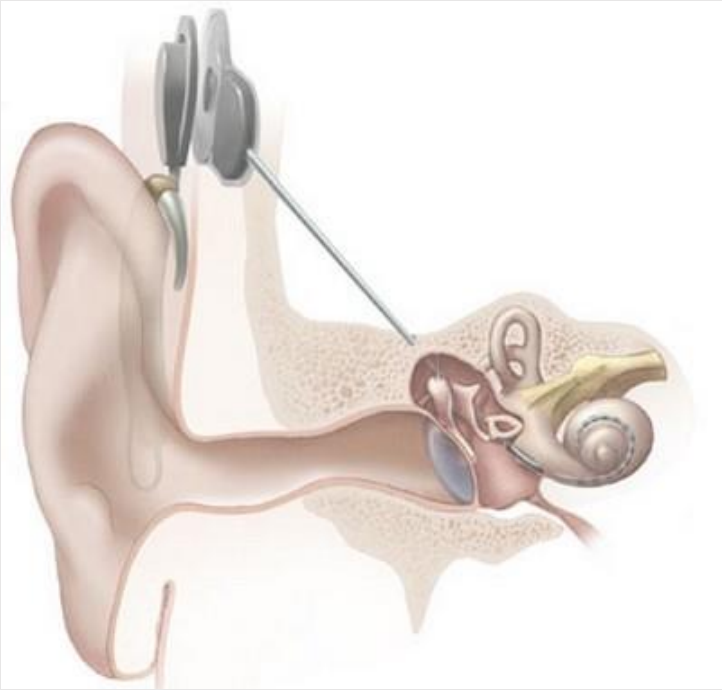
What happens in SNHL hearing loss?

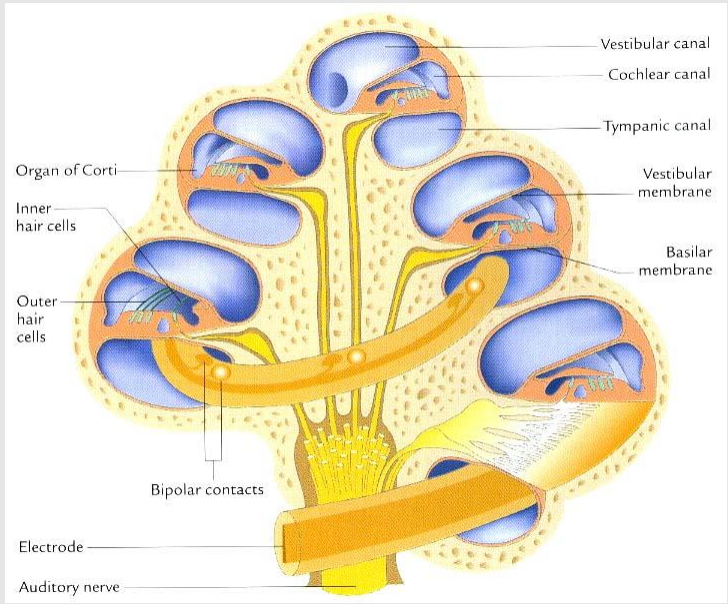


Ways to assist hearing…

Amplification





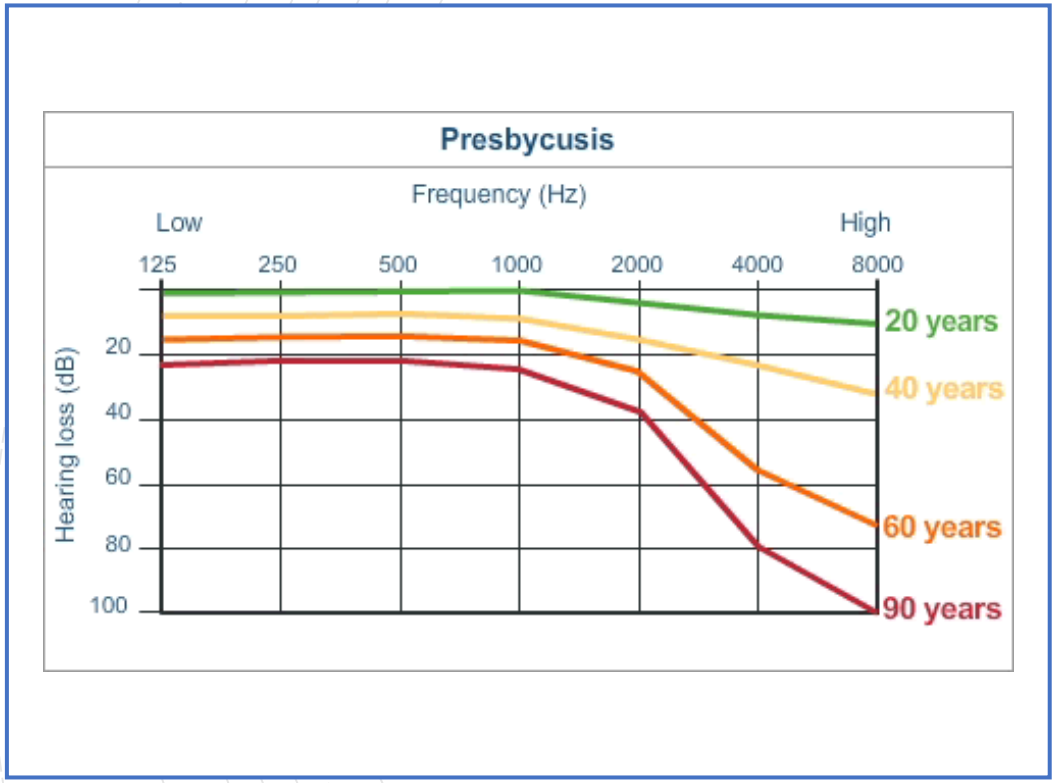






Cochlear Implantation

Presbycusis (age-related hearing loss)



- Auditory neural structures do not regenerate
- Loss of auditory sensitivity and decline in word recognition abilities
- Contributing factors: genetic predispositions, occupational noise, diet, CV disease, smoking, alcohol abuse, head trauma
- Symmetric sensorineural hearing loss

Noise-induced hearing loss

- Susceptibility to NIHL from sounds within our audible spectrum
- Damage to human hearing > 8 hrs of continuous sound 85dBA
- Most noise is intermittent and broad band
- OSHA 5-dB trade rule: reduction of ½ duration, permissible sound levels can increase by 5dB
- Most affected frequency band affected 3-6KHz
- Temporary vs. permanent threshold shifts
- Humans lack hair cell regeneration capability



Ototoxic hearing loss

- Therapeutic agents/chemical substances that cause functional impairment and cellular degeneration of inner ear tissues
- Common agents:
  - Aminoglycosides (OHC>IHC loss at basal turn)
    - Streptomycin, gentamicin, tobramycin, amikacin
  - Loop diuretics (stria vascularis; dose-related, reversible reduction of endocochlear potential)
    - furosemide
  - Salicylates (biochemical alterations, OHC motility affected, reversible)
    - aspirin
  - Cisplatin (loss of OHC; mechanism? Irreversible damage)
- Recommendations: reduce dosing frequency, avoid IV bolus administrations, monitor drug serum levels, pay attention to early symptomatology (HL, tinnitus)
- Vestibulotoxicity
  - Similar mechanisms of hair cell damage
  - Early signs: dizziness, vertigo, nystagmus, oscillopsia



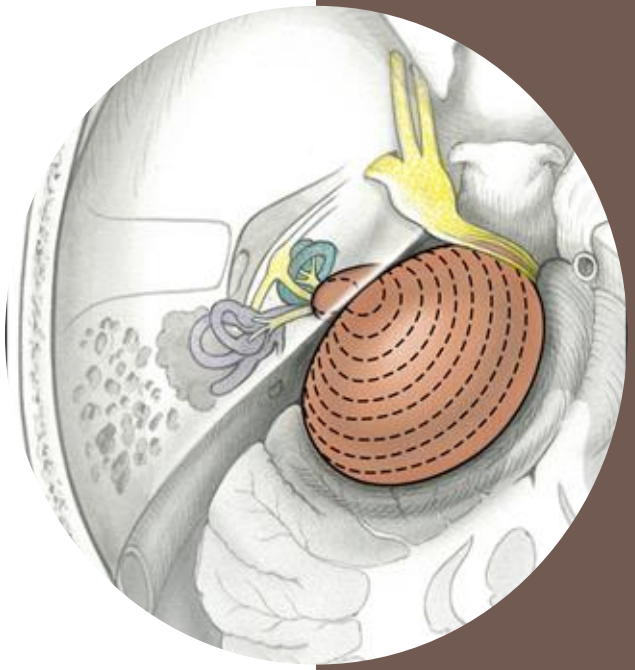
Sudden idiopathic SNHL

- Defined by loss of >30dB in at least 3 contiguous test frequencies within 3 days (rule of 3's)
- Atrophy of organ of Corti, stria vascularis, tectorial membrane
- Causes: viral, autoimmune, vasocclusive
- Treatment: oral steroids, intratympanic steroids

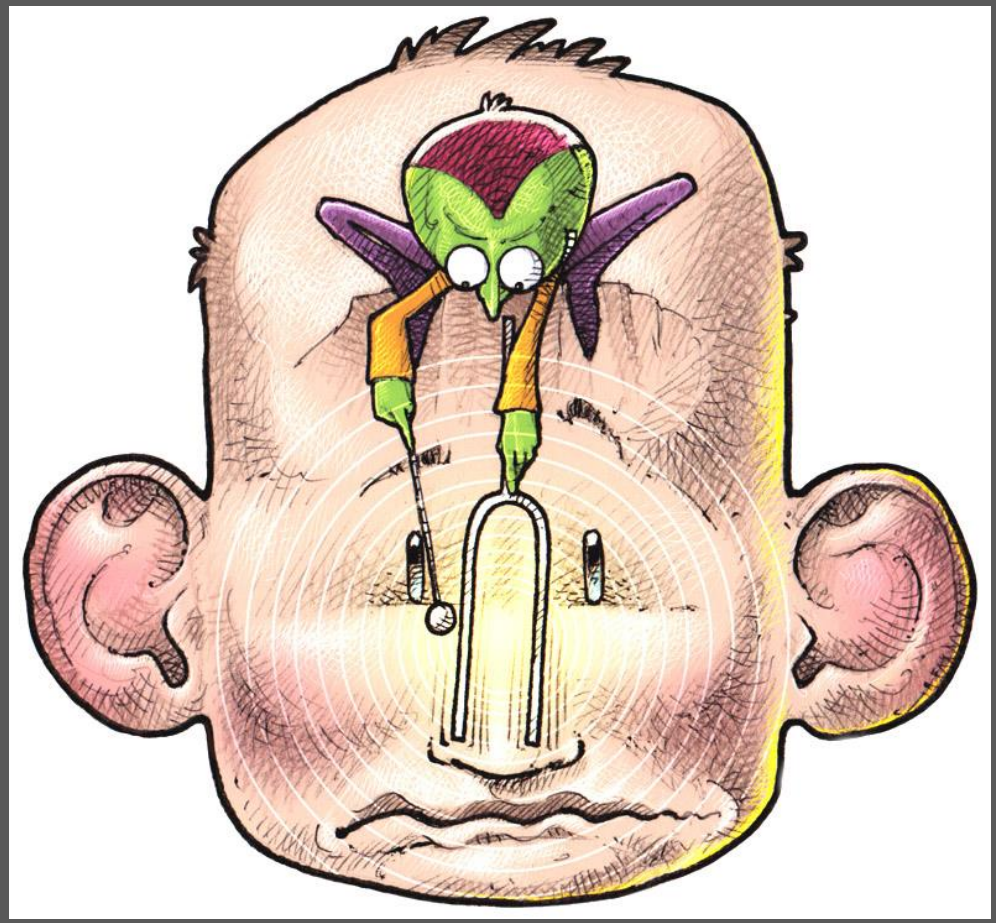


Asymmetric SNHL and Vestibular Schwannomas

- Asymmetric hearing loss is not common
- Defined as 10dB or worse over 2 test frequencies or > 15% difference in word recognition scores



TINNITUS



TINNITUS
Definition

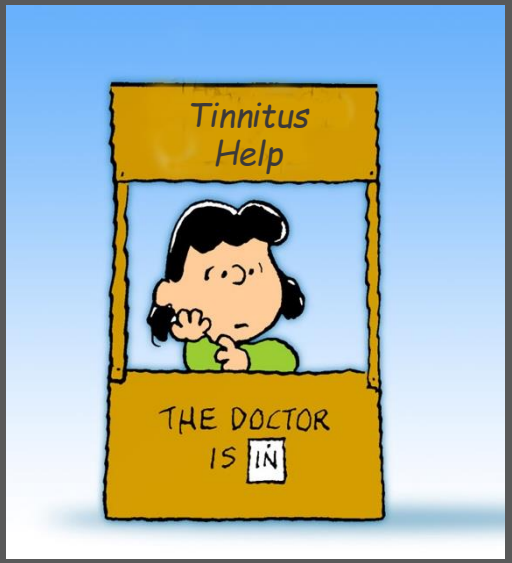
- Tinnitus: perception of noise in the absence of acoustic stimulus



TINNITUS

- Objective and subjective tinnitus

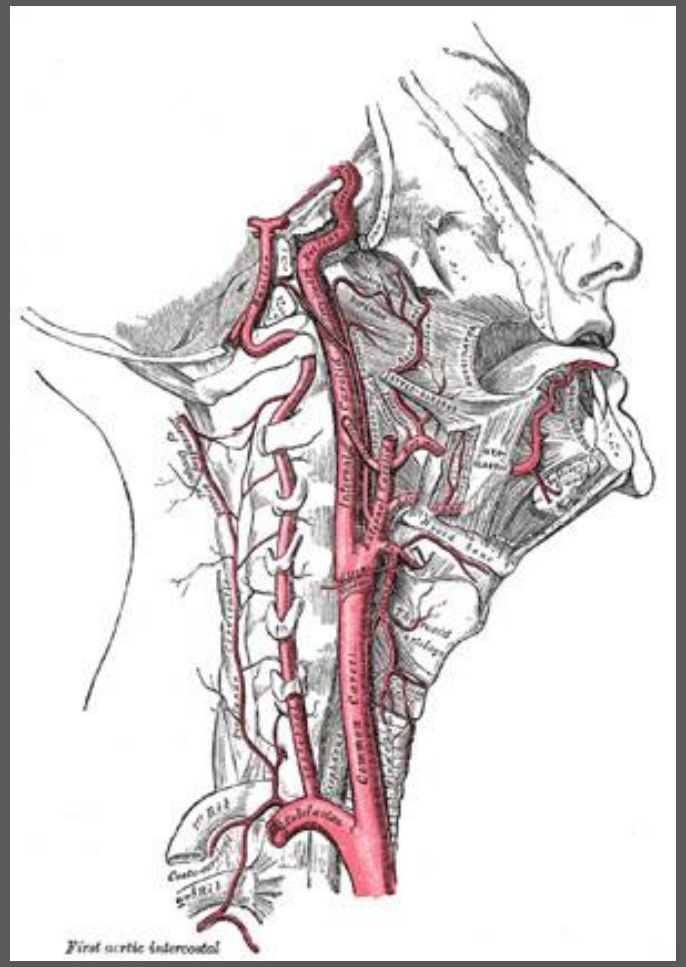
- Objective: that which is audible by physician or with pulsatile character
- Subjective: that which is audible by patient only

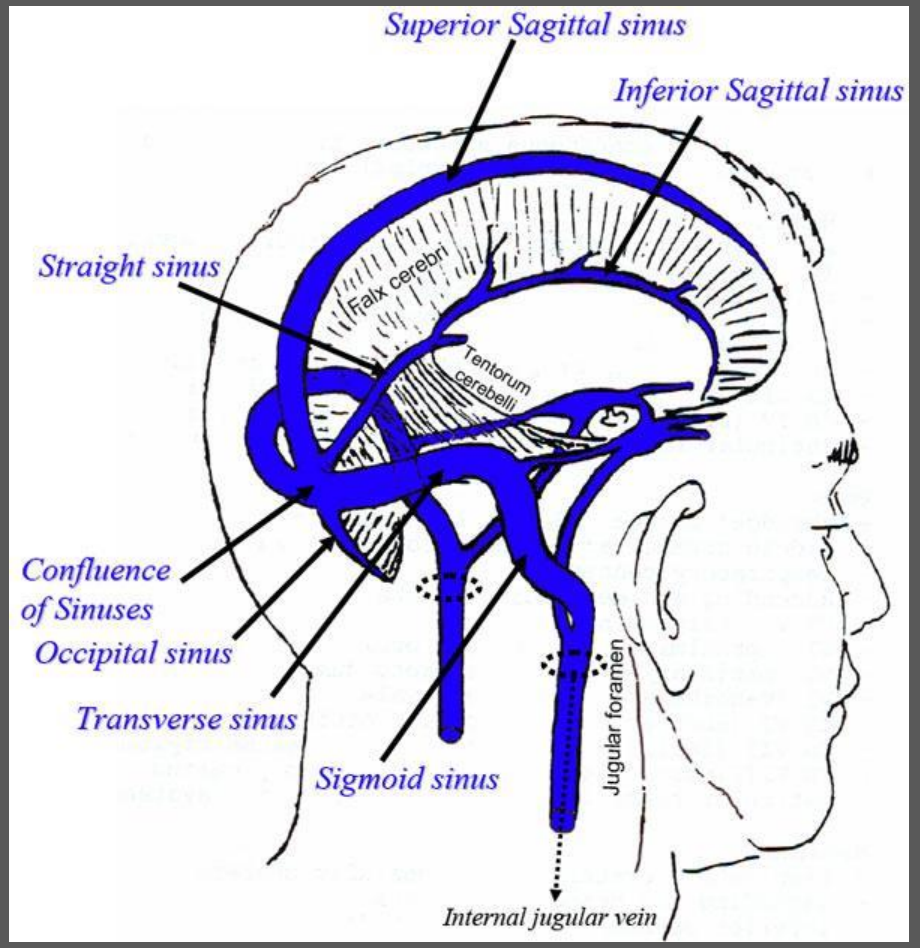


Objective Tinnitus

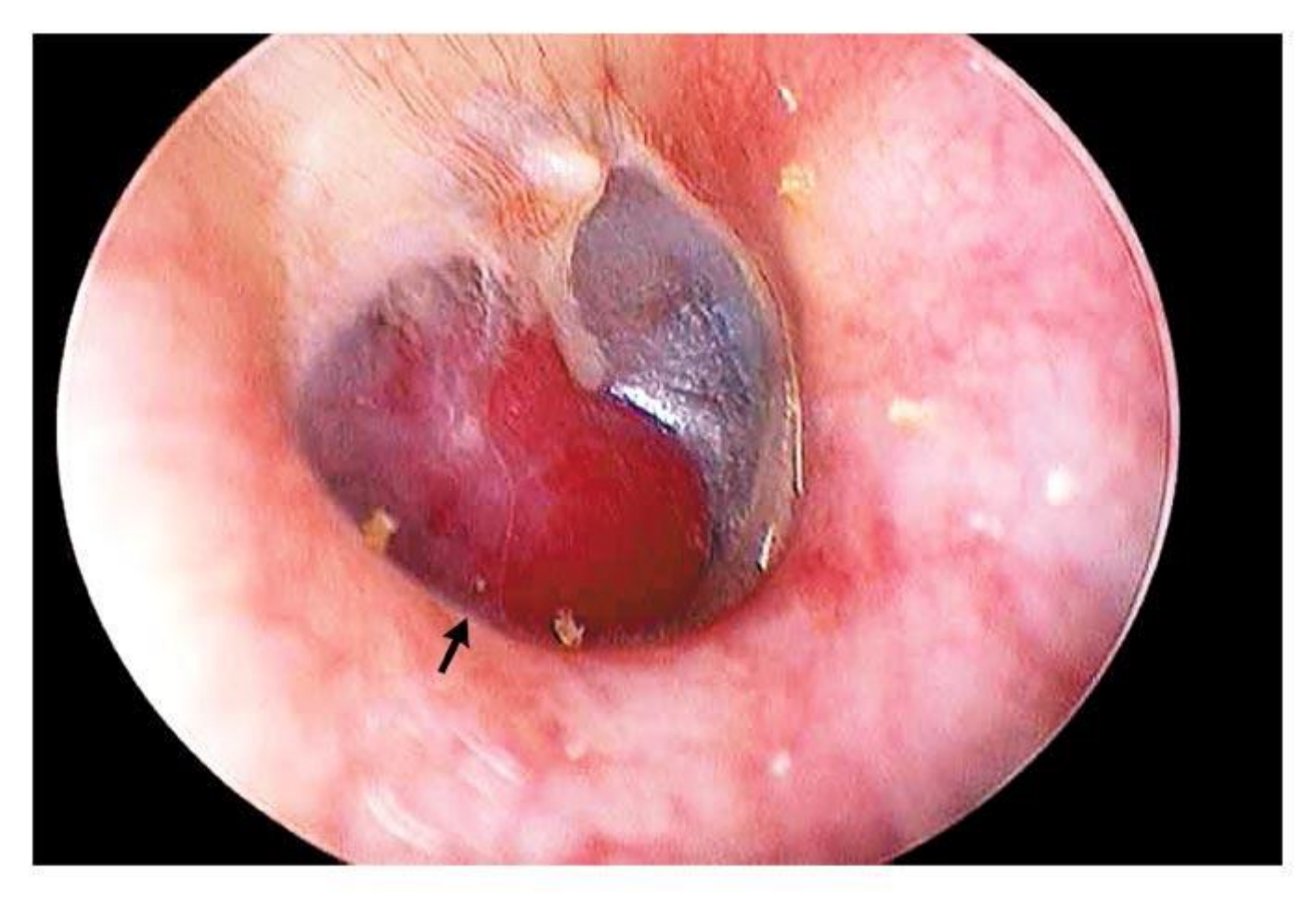
- Types of objective tinnitus
  - Vascular
  - Muscular
  - Eustachian tube dysfunction

Objective Tinnitus
Vascular





Objective Tinnitus
Vascular



Objective Tinnitus

Vascular

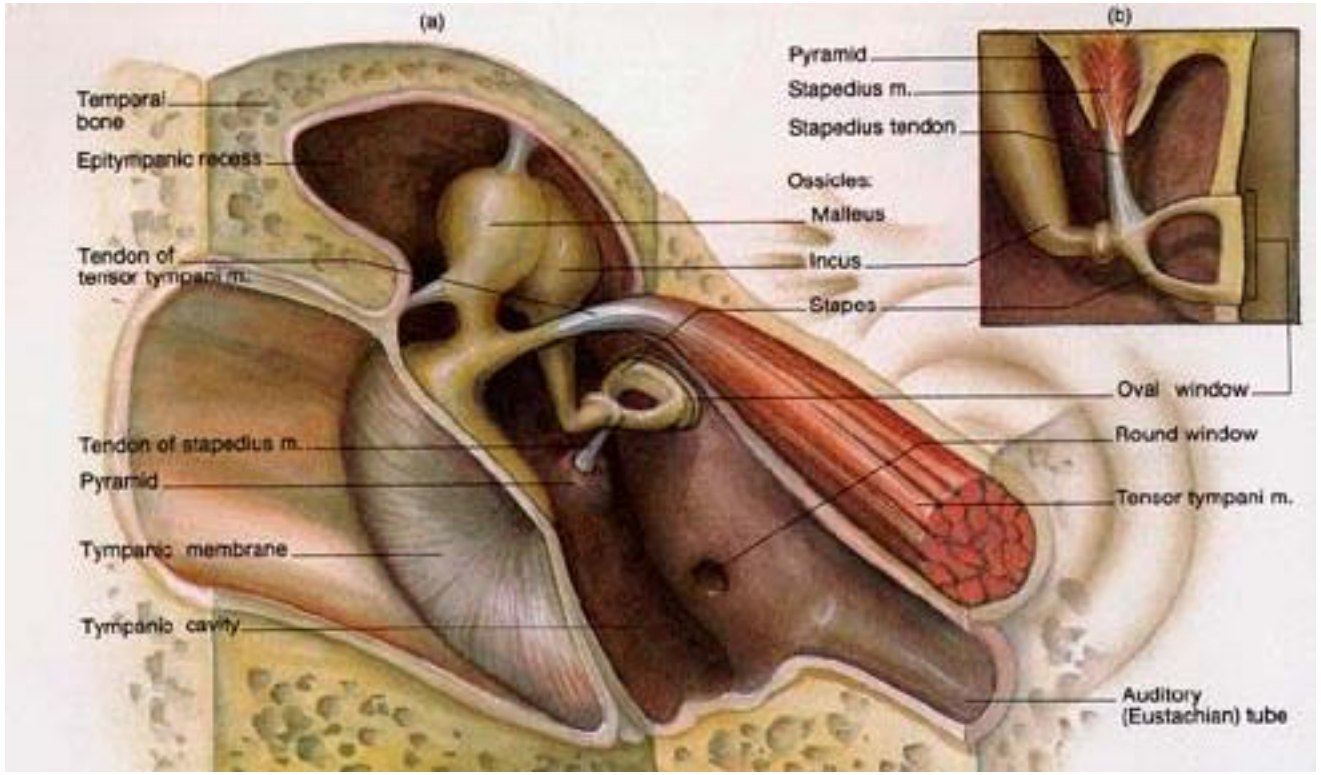
- Pulsatile sound
- Synchronous with heartbeat/pulse
- Tempo or rate increases with exercise
- Pathologies to consider:
  - Arteriovenous shunts (intracranial or periauricular, congenital or acquired)
  - Congenital AV malformations (direct connection between arterial and venous systems)
  - Arterial bruits (from carotid or vertebral A systems, stenoses)
  - Venous hums (related to HTN)
  - Idiopathic intracranial hypertention

Objective Tinnitus
Vascular

- Diagnosis: MRA/MRV; CT angiography; conventional angiography



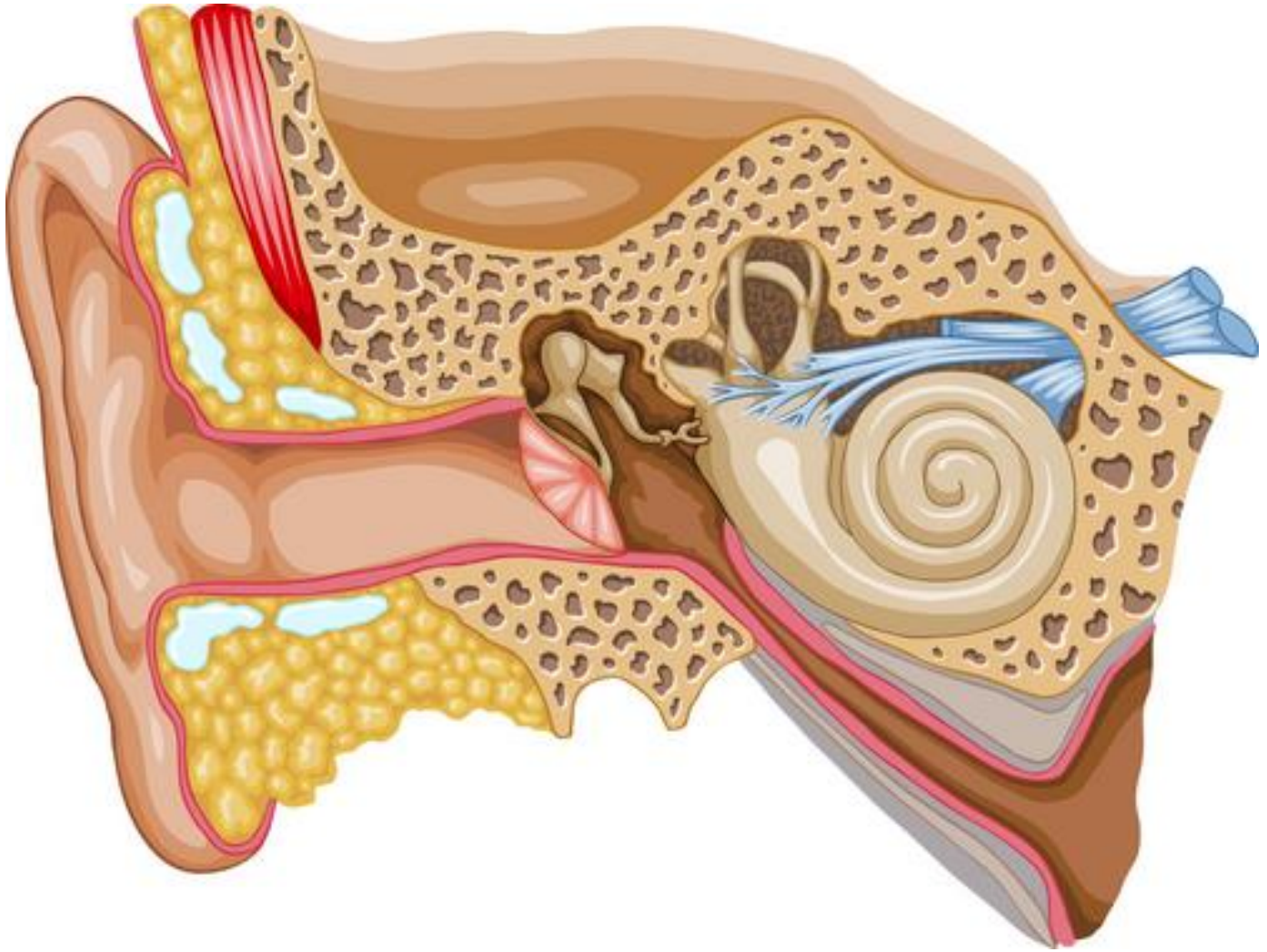
Objective Tinnitus Muscular



- Tensor tympani muscle/ stapedius muscle spasm

Objective Tinnitus Eustachian tube dysfunction

- Patulous Eustachian Tube
- Respiratory & phonatory autophony
- Aural fullness



Subjective Tinnitus

Subjective Tinnitus
Causes/ Theories

- Disordered hair cell theory: hair cell injury from hearing loss causes repetitive discharge in a way that central auditory system does not discriminate from actual sound

- Spontaneous neural activity within individual nerve fibers of the auditory nerve originate from neural injury or metabolic abnormality

- Hyperactivity of nuclei in ascending auditory pathway stimulates different areas of auditory cortex in absence of acoustic stimulus

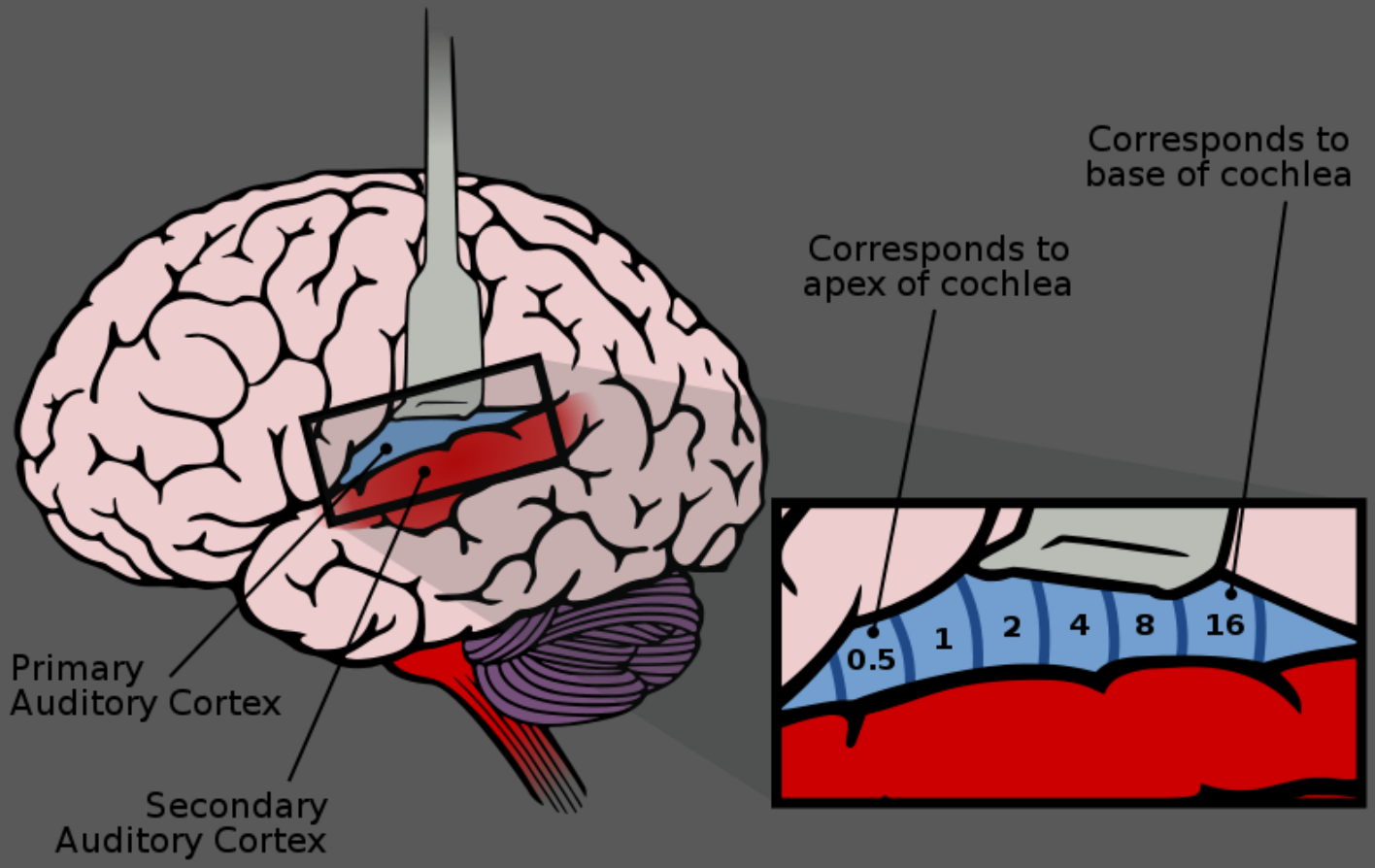
Subjective Tinnitus
Causes/ Theories

- **Cochlear damage reduces suppressive influence of CNS allowing increase in neural activity at higher centers in the auditory system

DEAFFERENTATION- LOSS OF INHIBITION HYPOTHESIS OF TINNITUS

Subjective Tinnitus

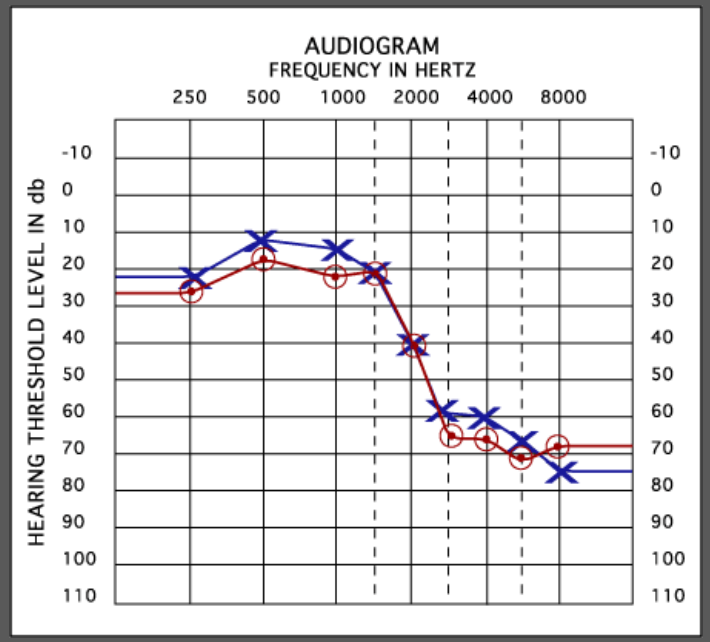
Causes/ Theories



Tinnitus

Common Presentation

- 75% with tinnitus have 30+ dB hearing loss from 3-8 KHz
- Pitch match of tinnitus correlates with frequency of hearing loss nadir















Tinnitus Treatment

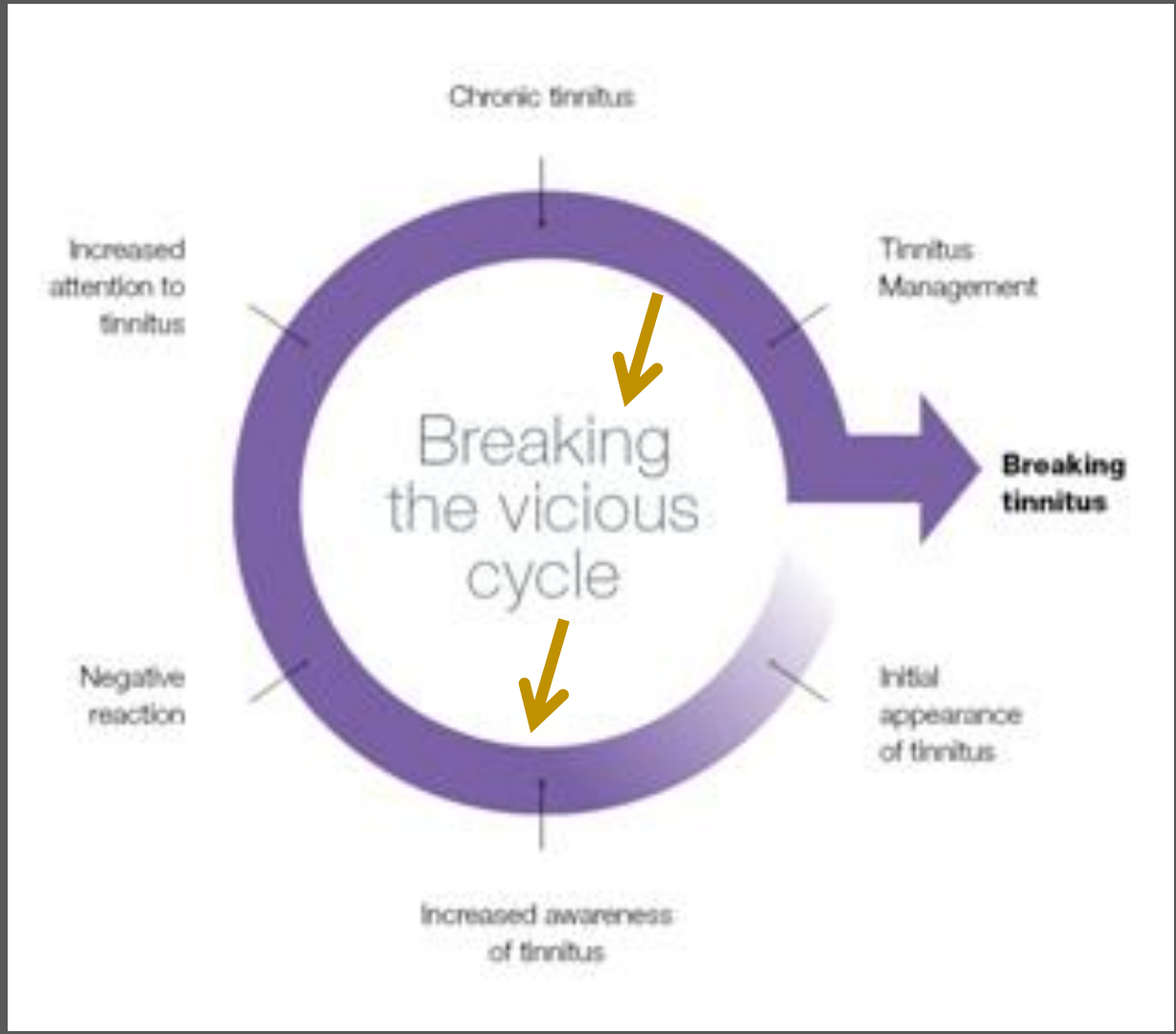
Tinnitus
Treatment

- General lifestyle and dietary modifications:
  - Reduce chocolate, caffeine, nicotine
  - Avoid noisy environments
  - Home masking techniques
  - Avoid ASA-containing medications
- Treatment should be directed to associated conditions (targeted-therapy)
  - TMJ (TMJ treatment)
  - Cervical muscular tension (muscle relaxant)
  - Depression/anxiety disorder (SSRI/anxiolytic)
  - Poor sleep patterns (melatonin)

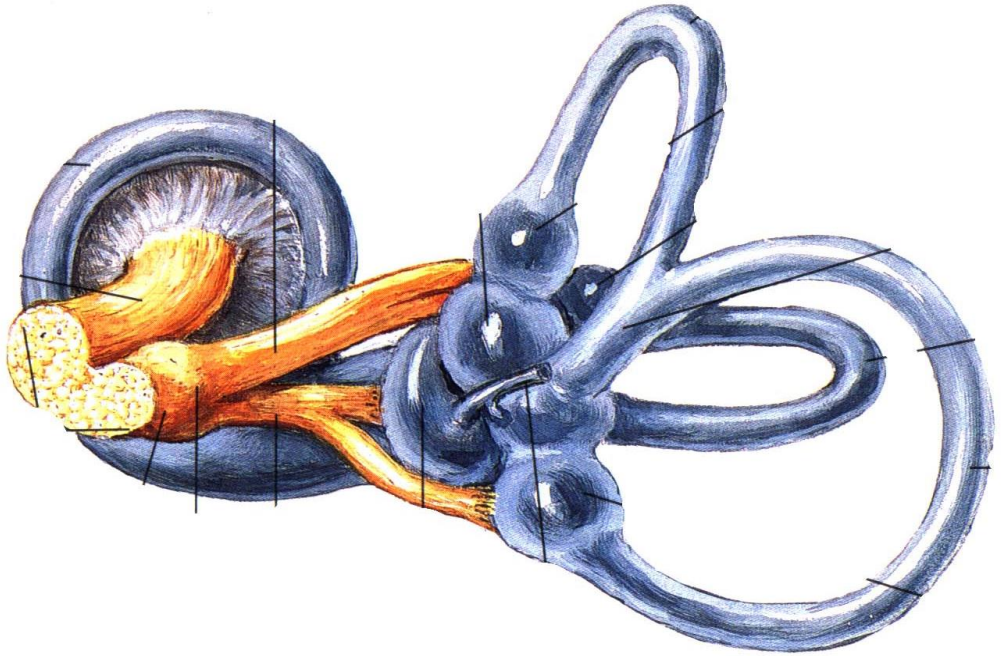
Tinnitus
Medical Treatment

- Anti-depressants
  - Heterocyclics (nortriptyline) /SSRI (sertraline)
- Anti-convulsants
  - Carbamazepine, gabapentin
- Tranquilizers
  - diazepam, clonazepam, oxazepam, alprazolam

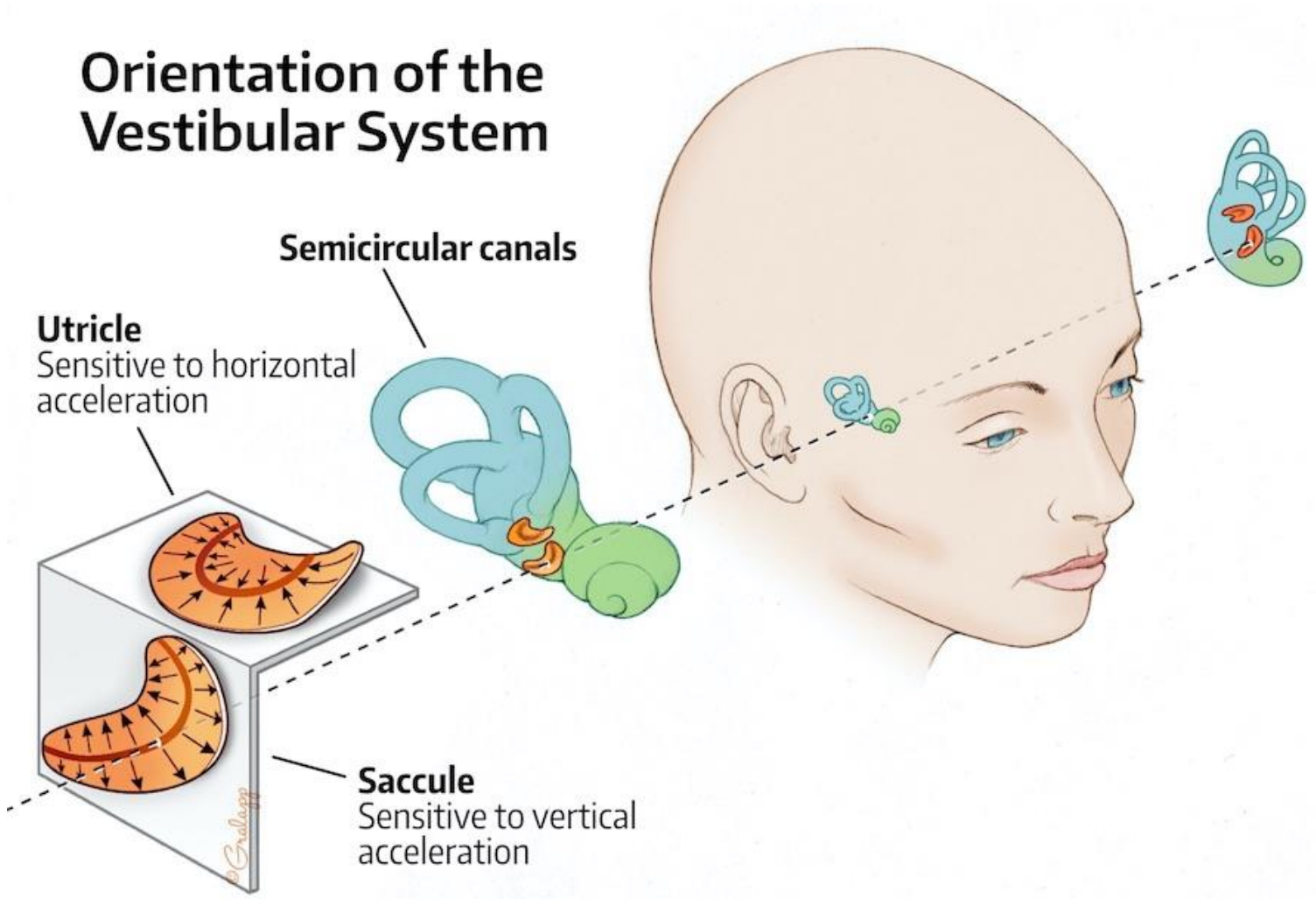
- *Rationale:*
  - *Enhance neural inhibition: GABA (inhibitory neurotransmitter) analogue such as gabapentin useful*
  - *Maintain serotonin levels (serotonin is a chemical modulator of neural sensory systems)*


Chronic tinnitus
Increased attention to tinnitus
Tinnitus Management
Breaking the vicious cycle
Breaking tinnitus
Negative reaction
Initial appearance of tinnitus
Increased awareness of tinnitus

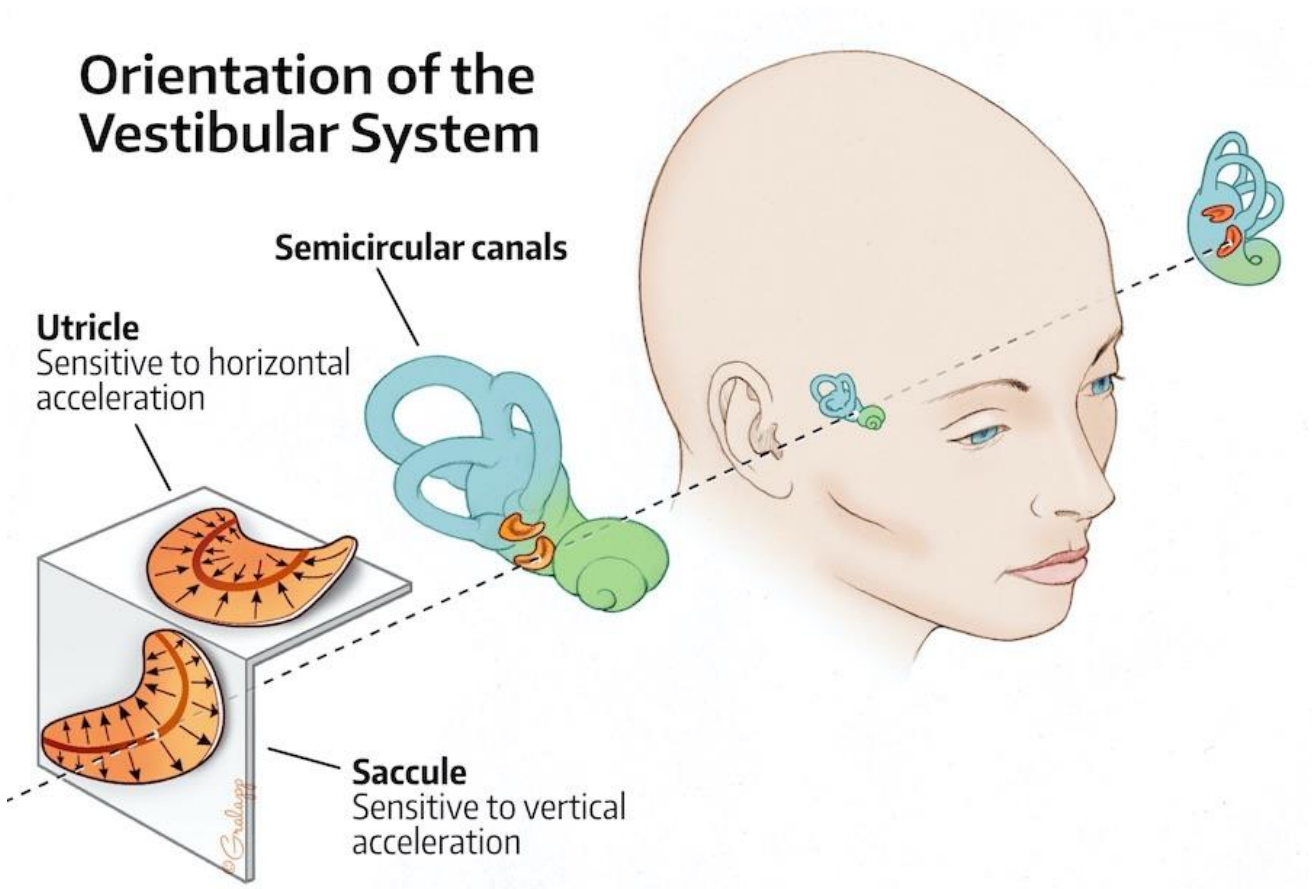
Vestibular System, Balance, Dizziness, & Vestibulopathies



Vestibular Endorgans

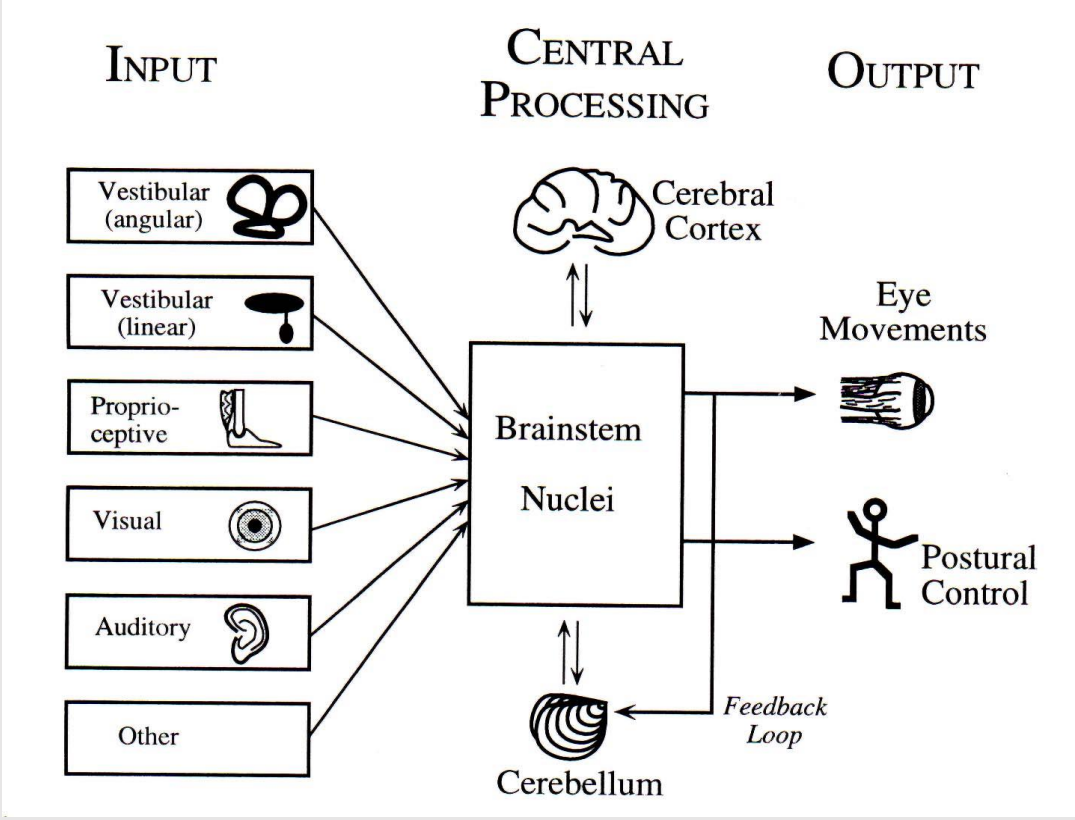


Vestibular Endorgans



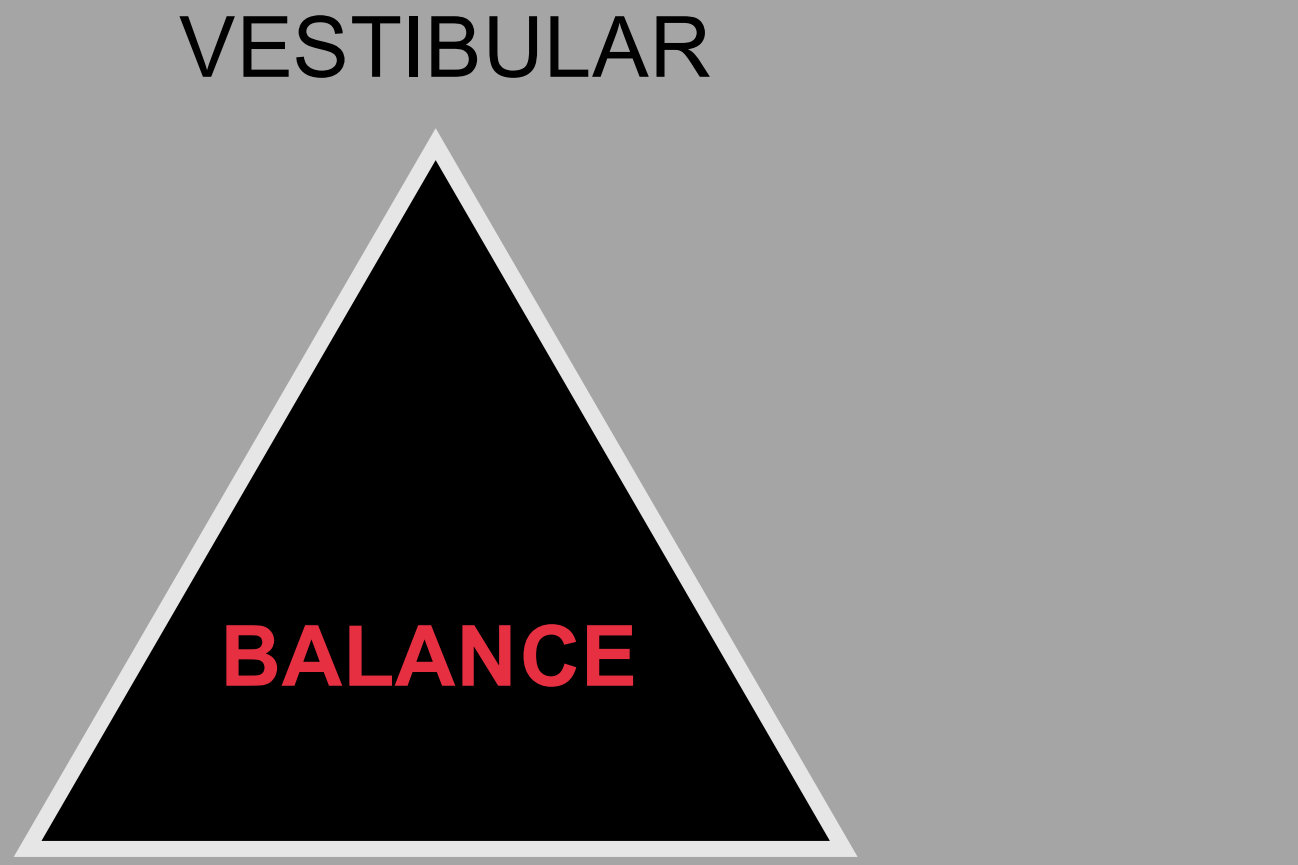


Maintaining Orientation and Balance

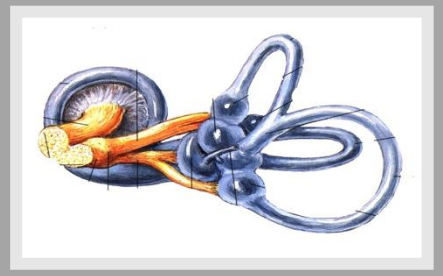




Maintaining Orientation and Balance



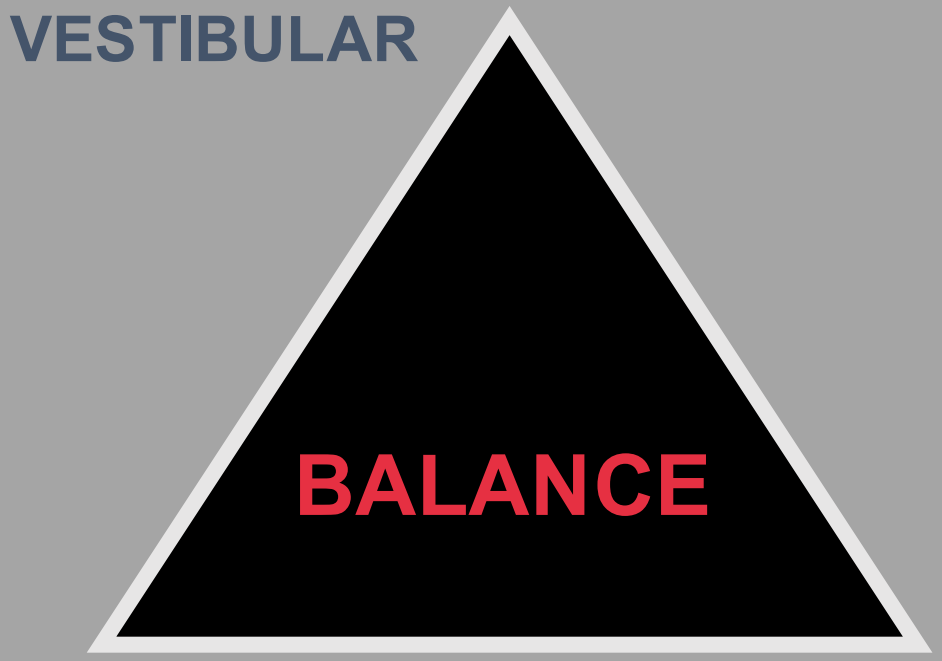
VESTIBULAR



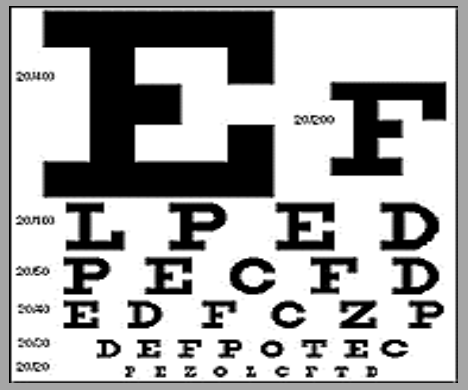
-Associated reflexes:

VESTIBULO-OCULAR REFLEX

VESTIBULO-SPINAL REFLEX







VESTIBULAR



PROPRIOCEPTION

VISION

- Visual acuity (clear vision)
- Oculomotor system
- Binocular vision

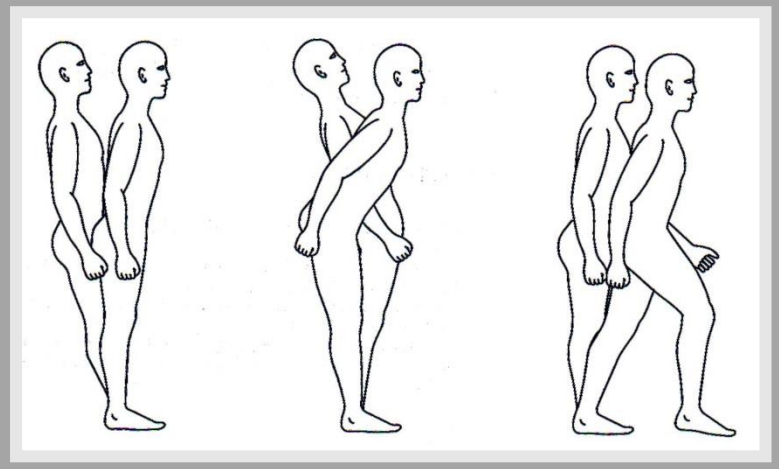
VESTIBULAR



VISION

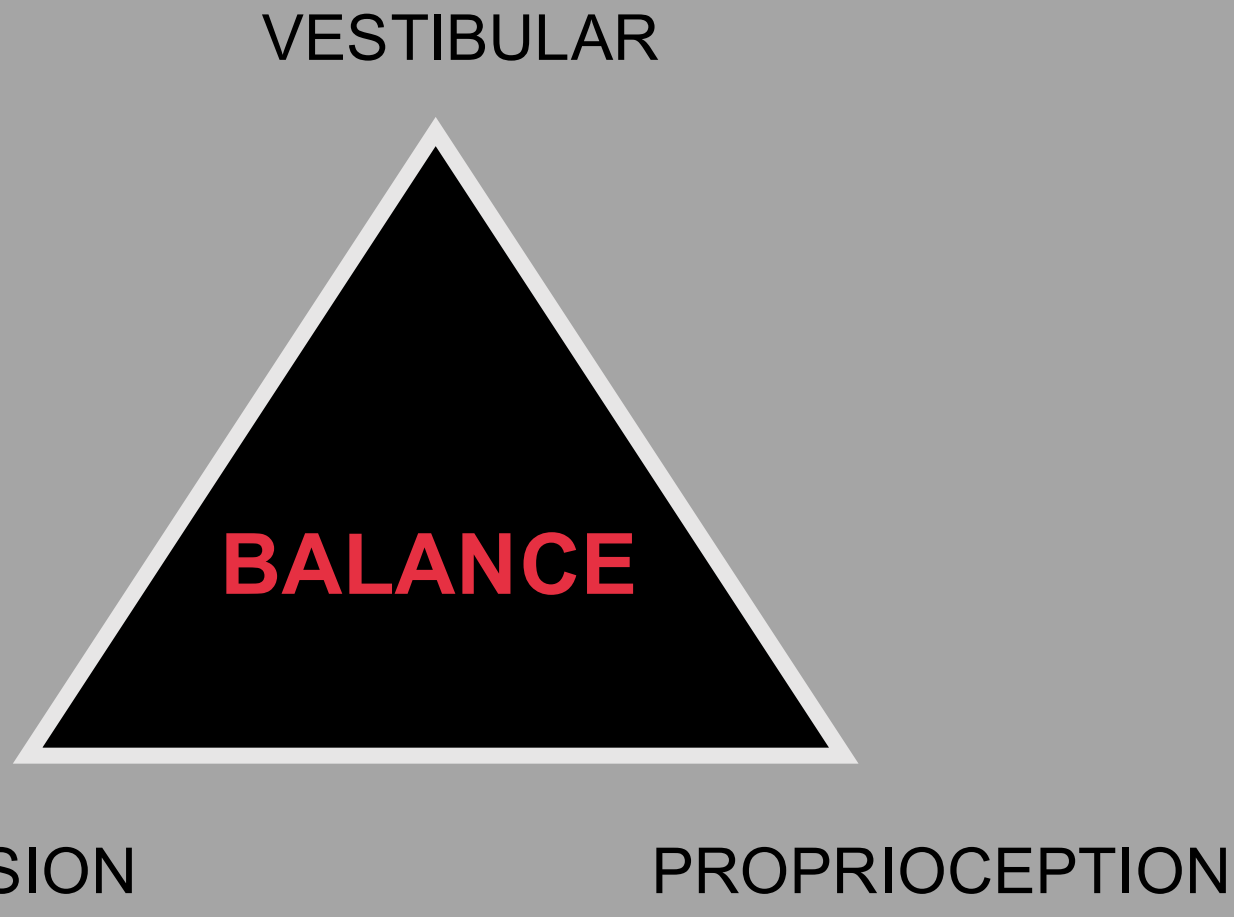
PROPRIOCEPTION

- Sensory input from major joints (hips, knees, ankles)
- Center of gravity shifts to remain upright



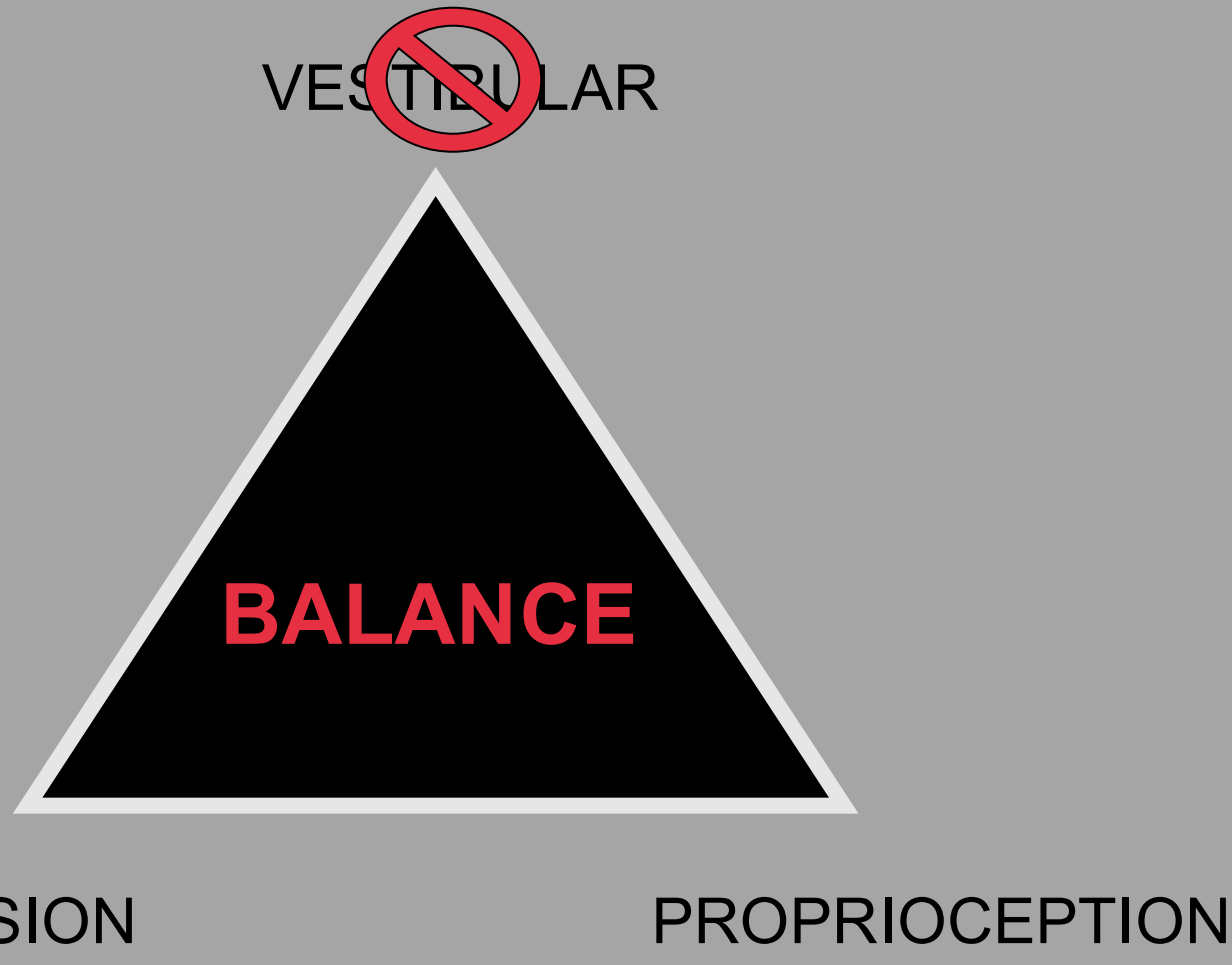
Maintaining Orientation and Balance

Sensory and Multisensory Conflict



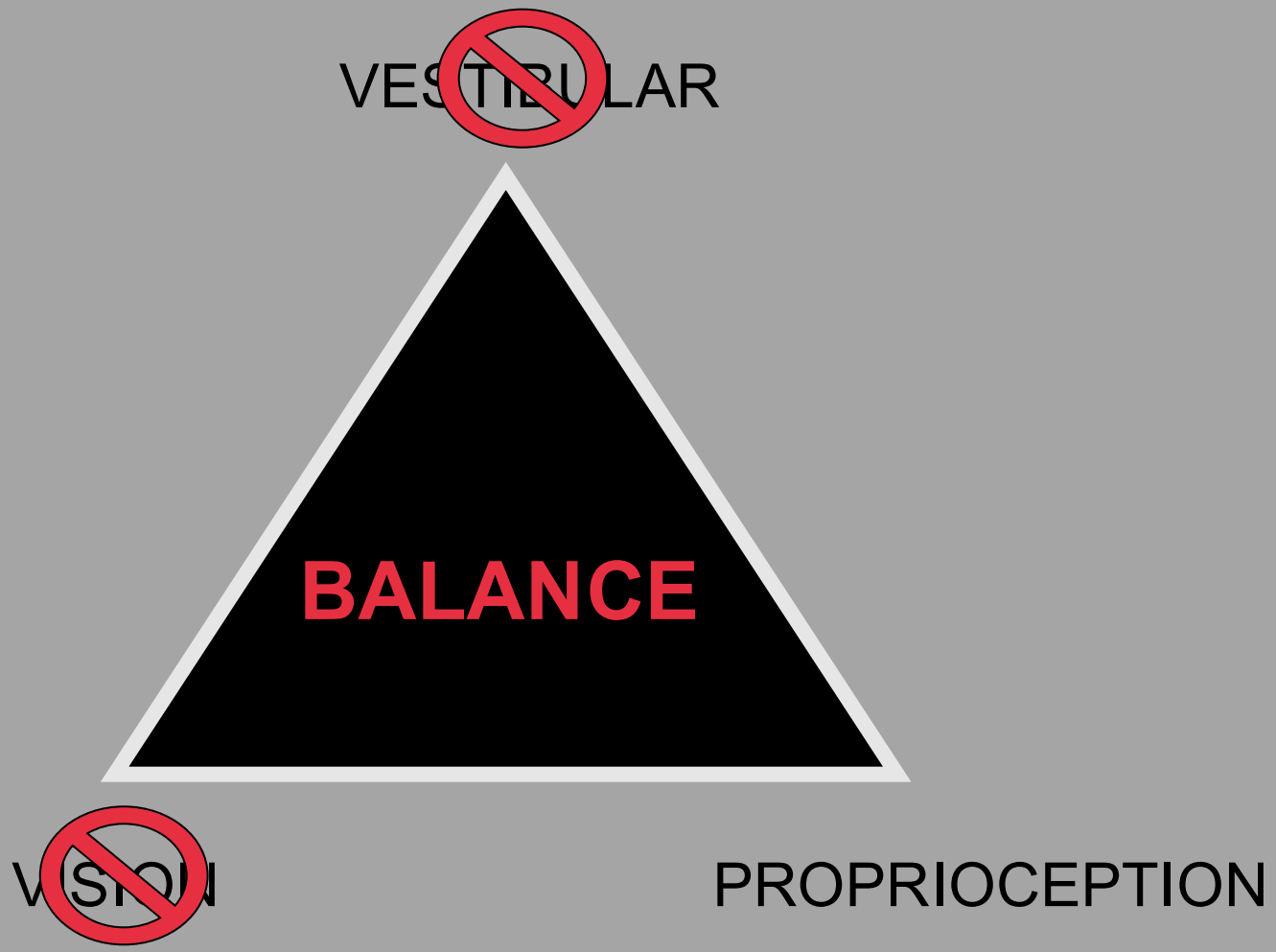
Maintaining Orientation and Balance

Sensory and Multisensory Conflict



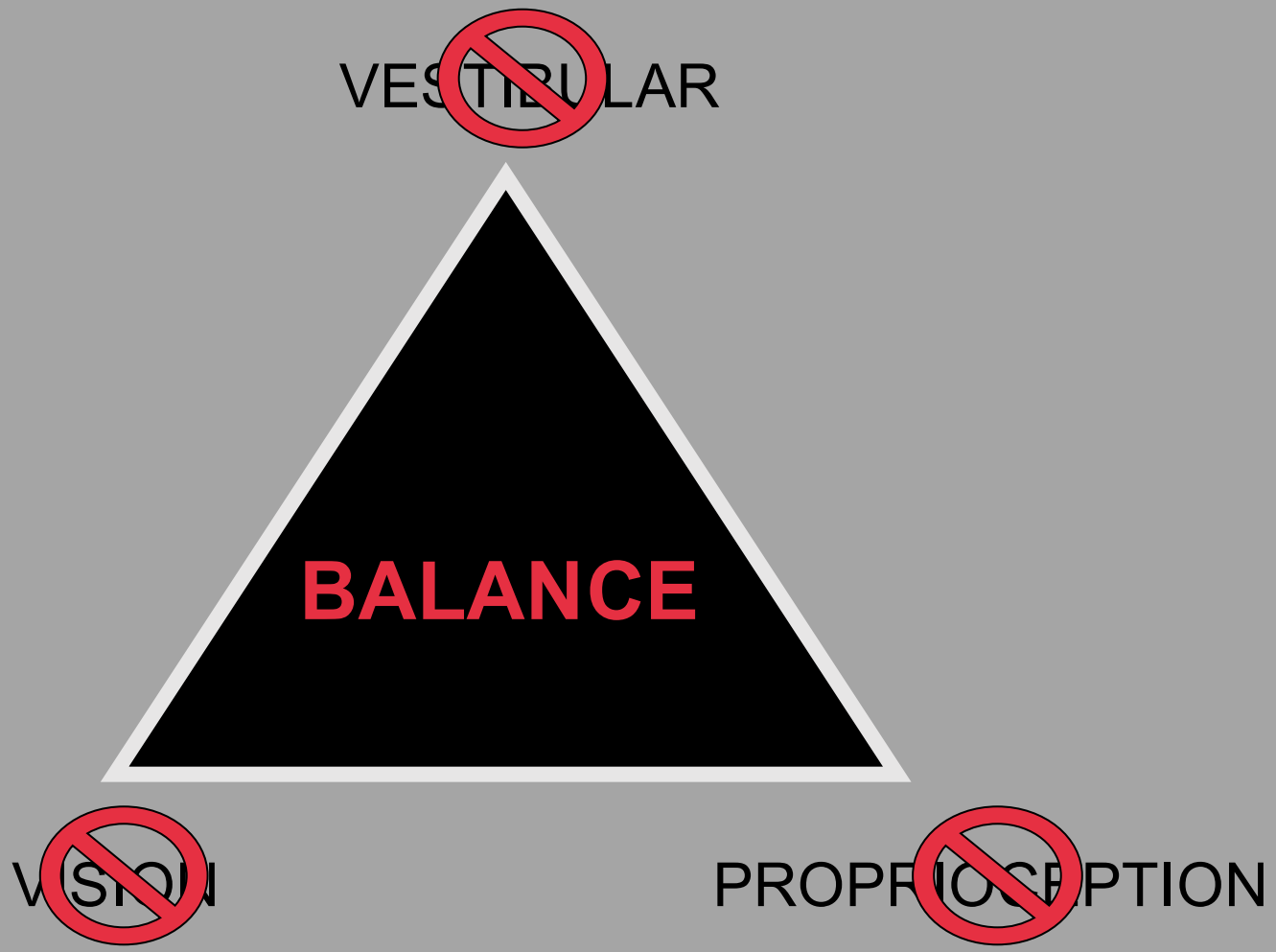
Maintaining Orientation and Balance

Sensory and Multisensory Conflict



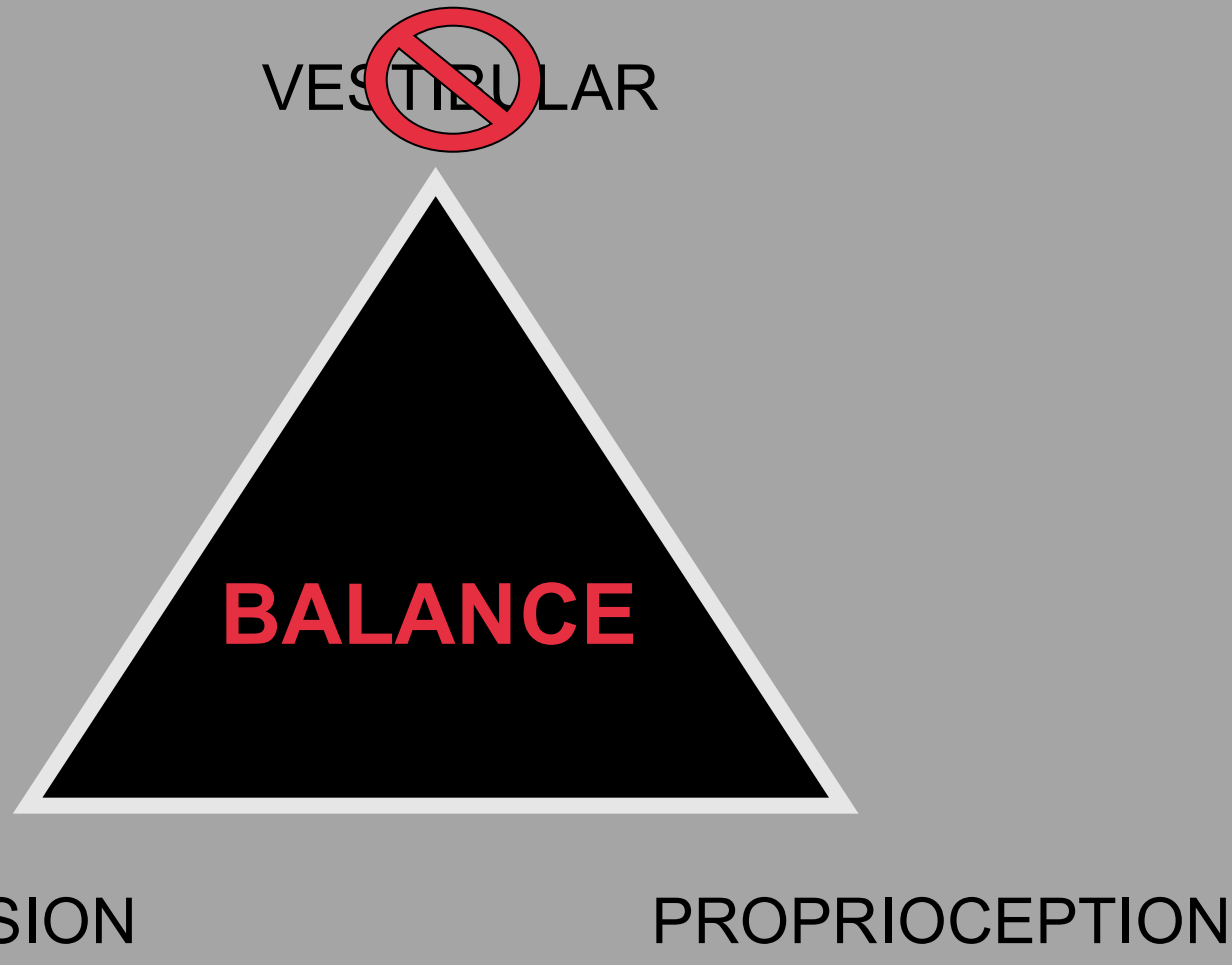
Maintaining Orientation and Balance

Sensory and Multisensory Conflict

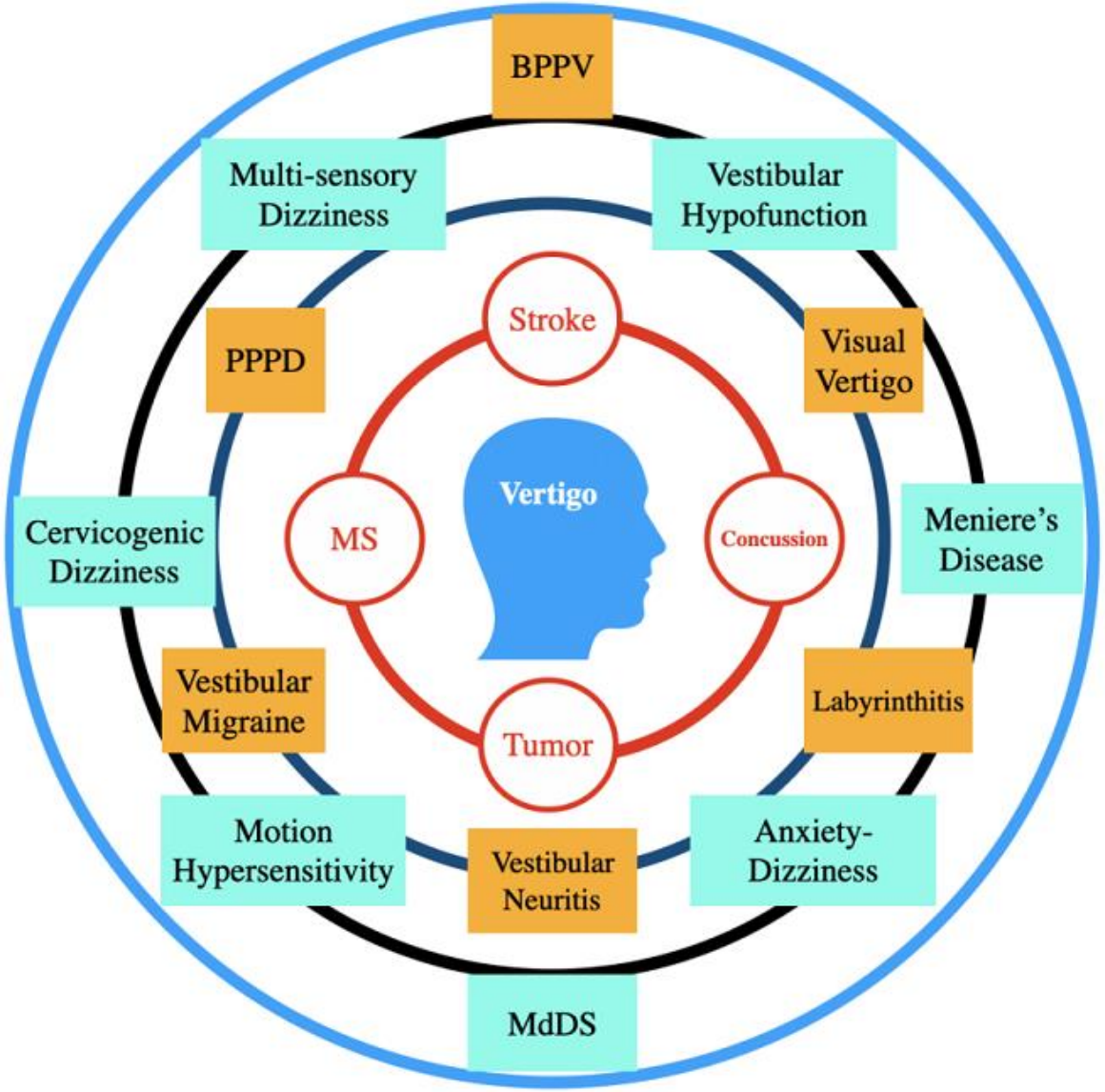


Maintaining Orientation and Balance

Sensory and Multisensory Conflict



Dizziness


BPPV
Multi-sensory Dizziness
Vestibular Hypofunction
PPPD
Stroke
Visual Vertigo
Cervicogenic Dizziness
MS
Vertigo
Concussion
Meniere's Disease
Vestibular Migraine
Tumor
Labyrinthitis
Motion Hypersensitivity
Vestibular Neuritis
Anxiety-Dizziness
MdDS

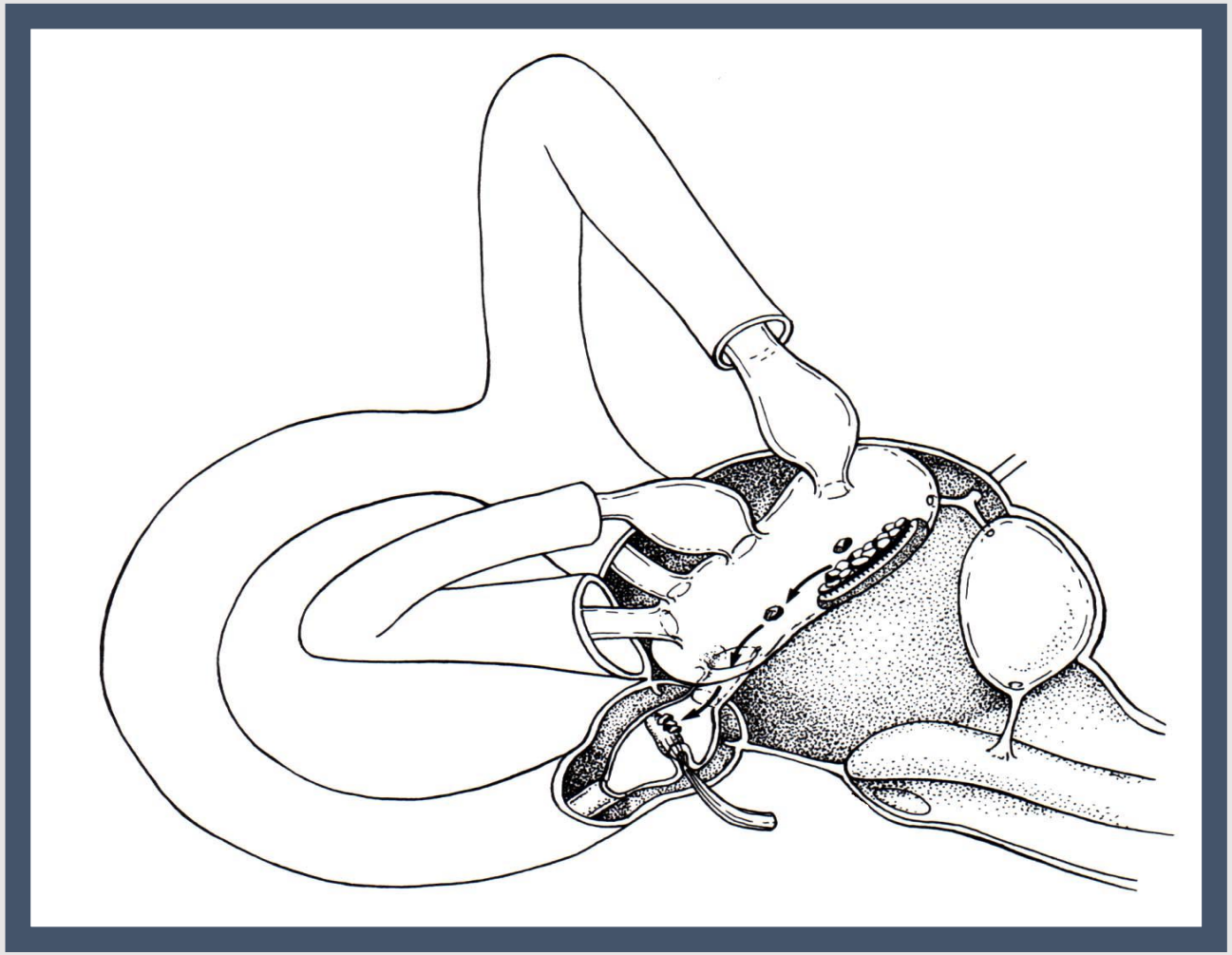
PERIPHERAL LABYRINTHINE DISORDERS

- Benign Paroxysmal Positional Vertigo
- Vestibular neuritis/ labyrinthitis
- Meniere’s disease
- Vestibulotoxicity

COMMON PERIPHERAL LABYRINTHINE DISORDERS

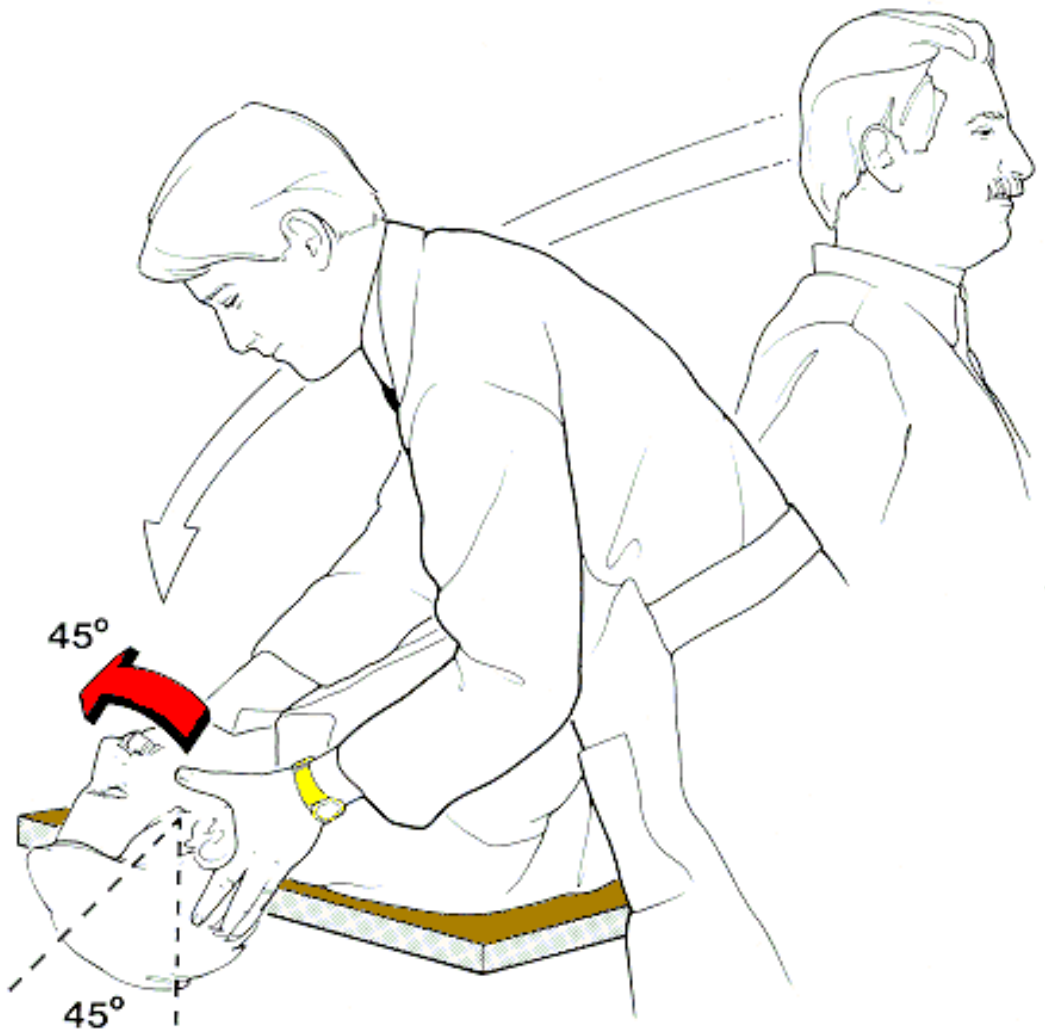
- Benign Paroxysmal Positional Vertigo (BPPV)
  - Cupulolithiasis/ Canalithiasis: Loose otoconial particles from posterior SCC from utricle
  - Positionally induced vertigo, lasting seconds
  - Provoking positions: turn over in bed, get in/out of bed, bending over, extend neck to look up and back
  - + Dix-Hallpike positioning test: torsional vertical positional nystagmus in plane of posterior SCC
  - Treatment: Canalith repositioning maneuver (Epley maneuver)

BENIGN PAROXYSMAL POSITIONAL VERTIGO

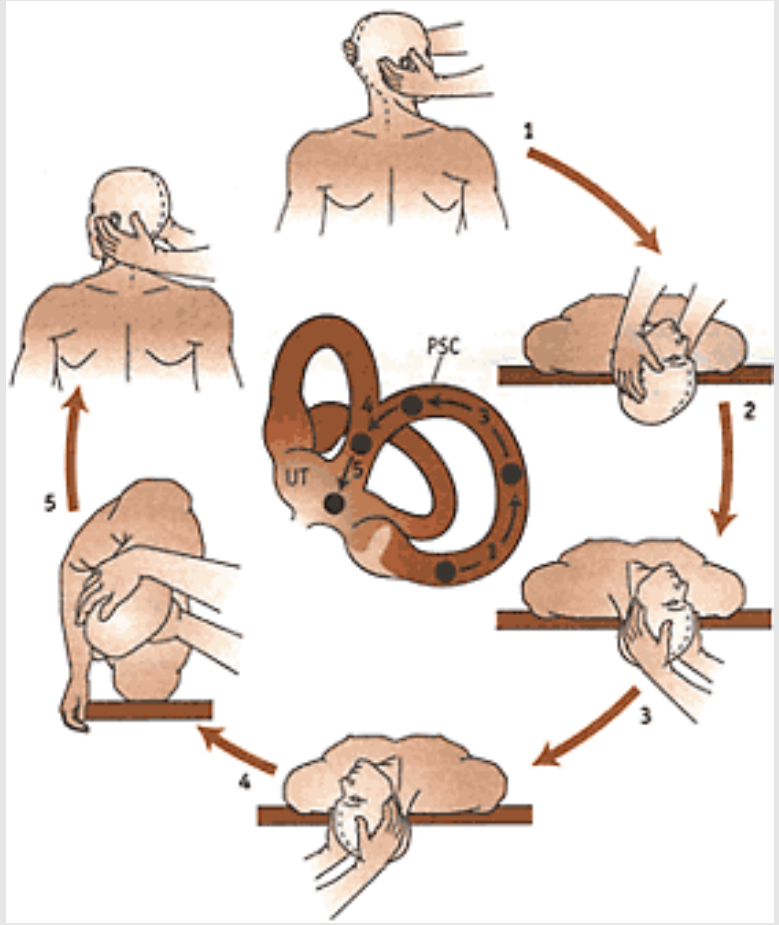


Dix-Hallpike test for BPPV

- Look for rotary, jerk nystagmus
- Geotropic (toward lowermost involved ear) rotational nystagmus



Epley Maneuver for BPPV



COMMON PERIPHERAL LABYRINTHINE DISORDERS

- Vestibular Neuritis
  - Inflammation of labyrinth or vestibular nerve from viral
  - Acute onset vertigo (peak at 24 hours), duration for days, and gradual improvement over weeks
  - 20-30% have recurrence
  - usually not associated hearing loss
  - Treatment: vestibular suppressants, antiemetics, corticosteorids

COMMON PERIPHERAL LABYRINTHINE DISORDERS

- Serous / suppurative labyrinthitis
  - Infection of the labyrinth
  - Entry of infection from middle ear or intracranial cavity (meningitis, internal auditory canal, cochlear aqueduct)
  - May have entry of inflammatory mediators / toxins / bacteria
  - Features hearing loss
  - Treatment: vestibular suppressants, antiemetics, corticosteorids + ANTIBIOTICS

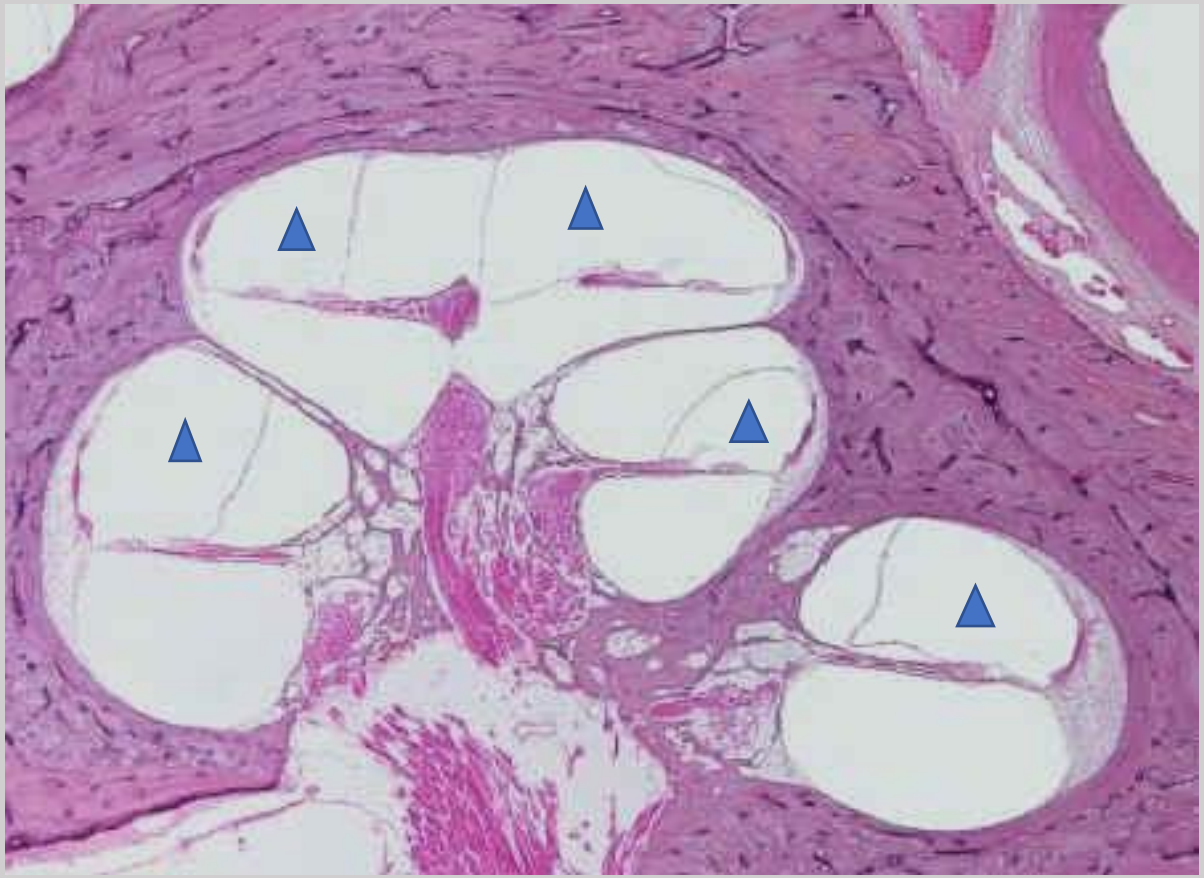
Meniere's disease

- Symptom complex described by Prosper Ménière in 1861
  - Spontaneous episodic vertigo
  - Tinnitus
  - Fluctuating hearing loss
  - Aural fullness

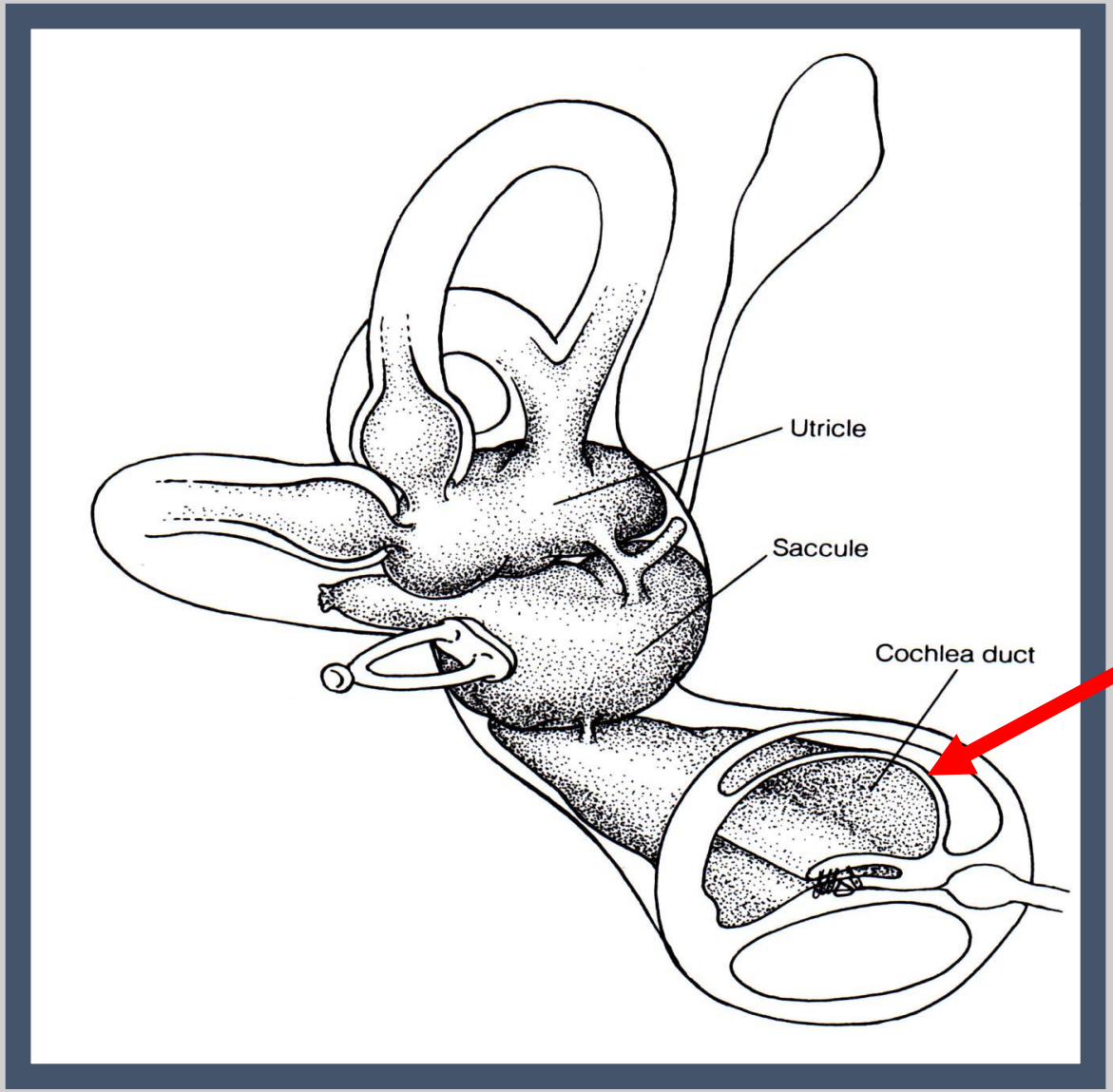


- Histopathologic correlate described independently by Hallpike & Cairns and Yamakawa (1938): endolymphatic hydrops

Endolymphatic Hydrops



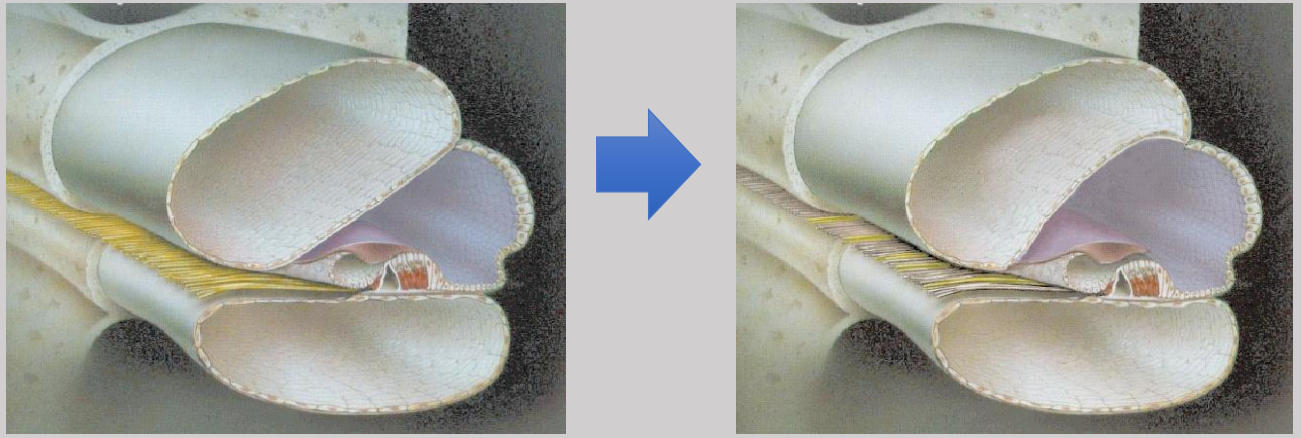
MENIERE'S DISEASE



DISTENTION OF ENDOLYMPHATIC COMPARTMENT

Endolymphatic Hydrops & Meniere's disease: a complex relationship

- Endolymph produced by stria vascularis and dark cells of the vestibular labyrinth
- Endolymphatic hydrops might result from overproduction or under-resorption of endolymph
- Local alteration of the biochemical environment & homeostasis of the labyrinth



Meniere's disease: Presentation (classic tetrad)

- Episodic vertigo (minutes to hrs)
- Fluctuating hearing loss (50-70%) (low frequency)
- Tinnitus (roaring, gets louder with attacks)
- Aural fullness (50-70%)

- Progression of Meniere's disease to contralateral ear (up to 50%)

Course of Meniere's Disease

- Spontaneous attacks of vertigo in clusters or on regular schedule
- Fluctuating hearing loss leading to progressive hearing loss
- Periods of remission for years followed by recurrence
- Burnt-out phase:
  - no more acute attacks of vertigo
  - chronic dysequilibrium
  - hearing stabilizes at 50dB flat audiometric configuration, 50% WRS

Medical Management

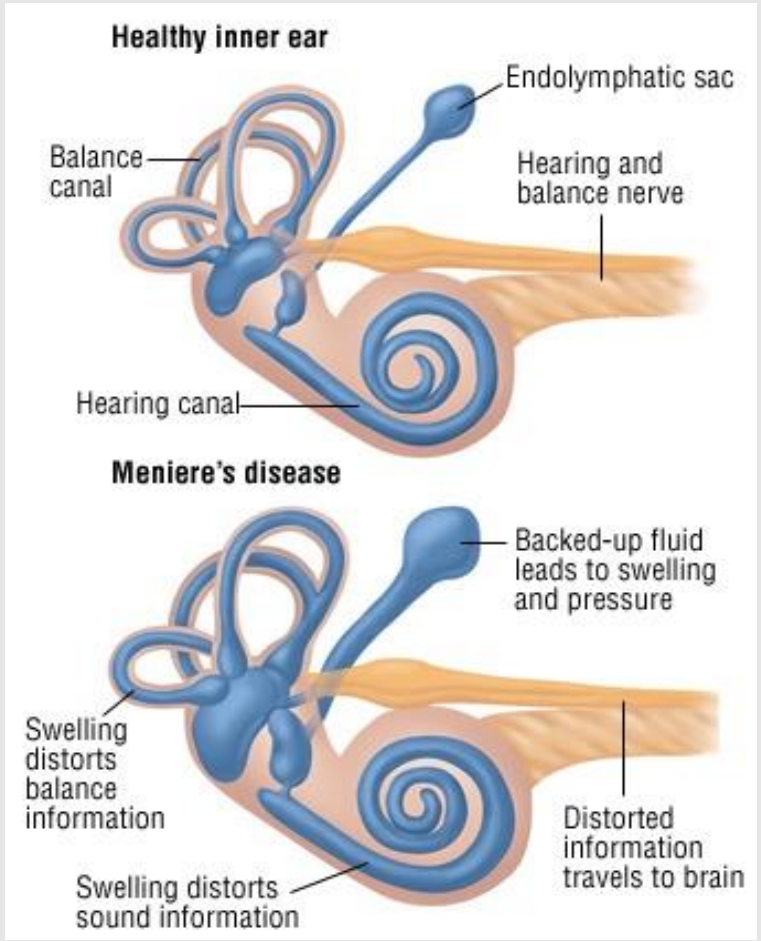
- Symptomatic treatment
  - Vestibular suppressants: meclizine, valium, clonazepam, *phenergan, ondansetron*
- Dietary modifications
  - Low sodium diet (< 2 grams/ day)
  - Reduce caffeine, alcohol intake
- General health maintenance: sleep, rest, control allergies, thyroid, DM control, avoid hormonal fluxes

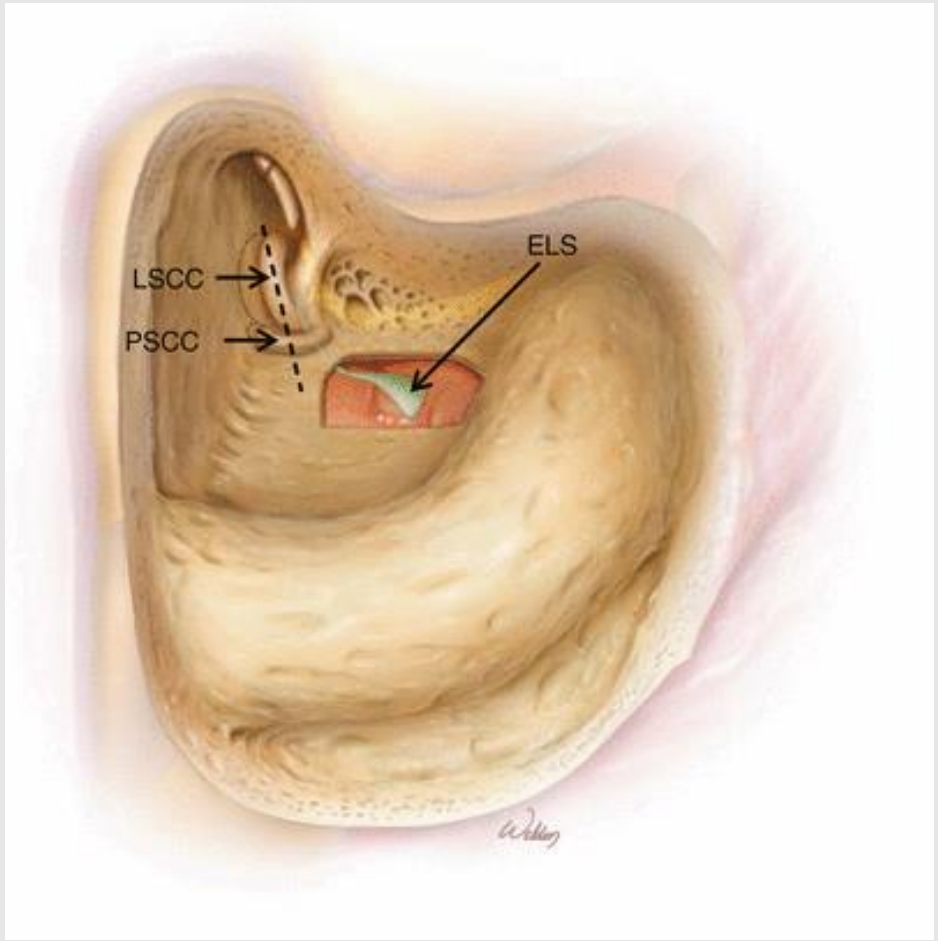
Medical Management

- Diuretics
  - Classes of diuretics used: thiazide (HCTZ), potassium sparing (Dyazide, Maxzide), carbonic anhydrase inhibitor (Diamox)\
- Vasodilators
  - Betahistine (Serc)
    - Oral analogue of histamine
    - Causes capillary dilation

Surgical Treatment

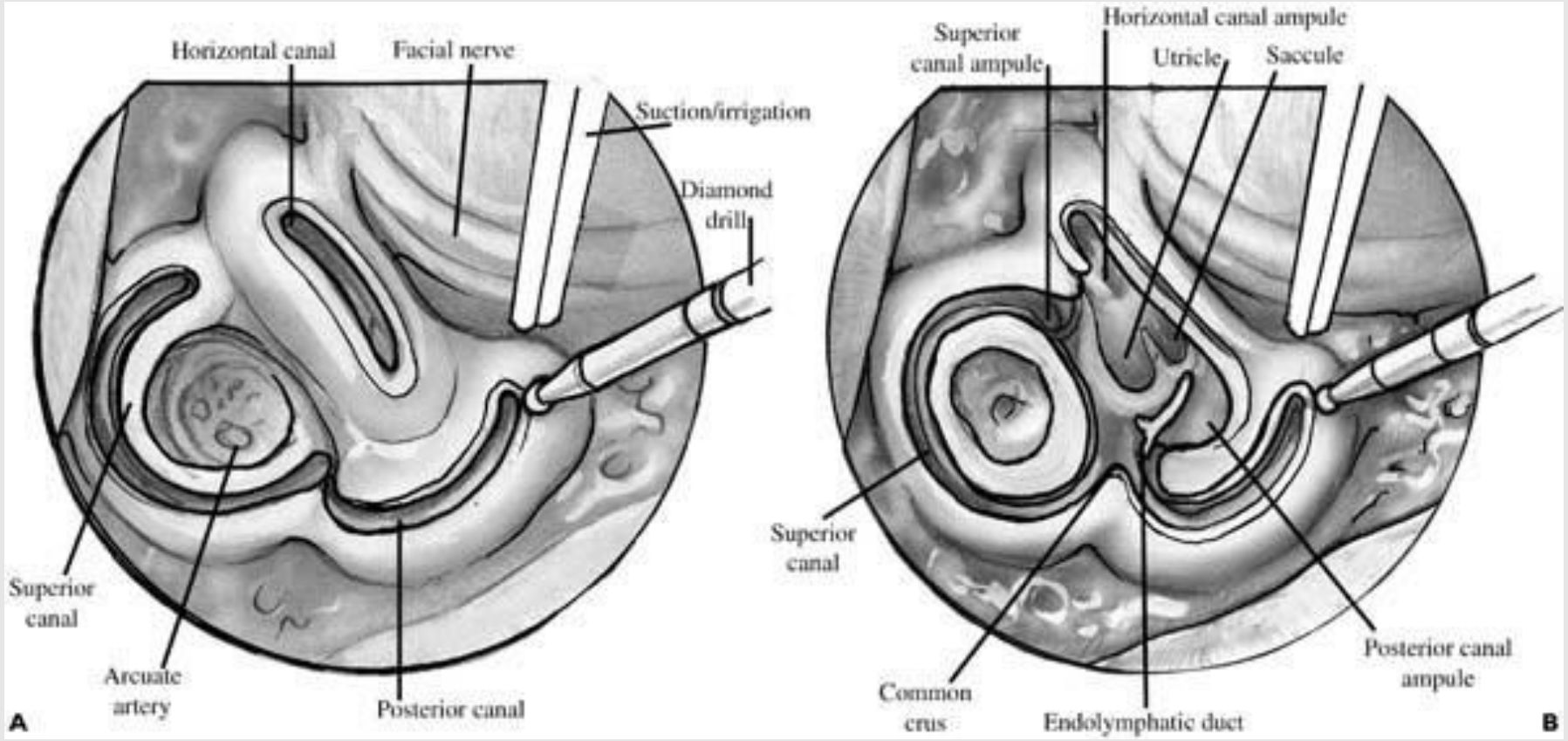
- Endolymphatic sac shunt / decompression





Surgical Treatment

- Labyrinthectomy



Surgical Treatment

- Vestibular Nerve Section

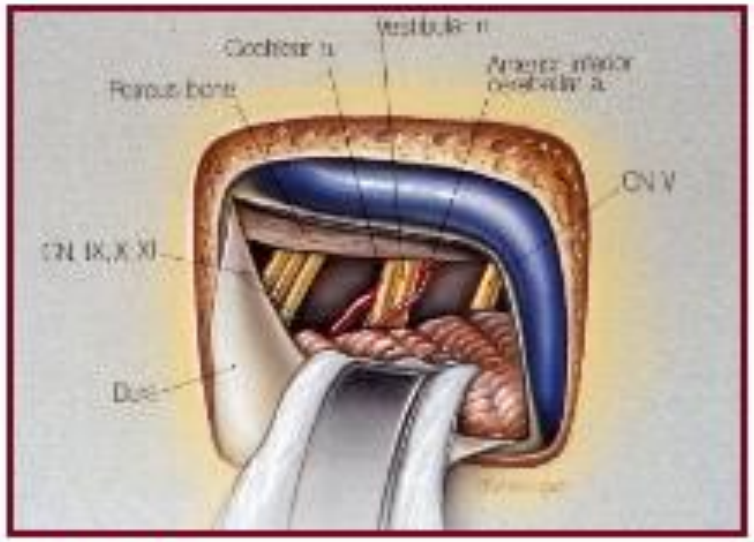


FIGURE 1
Vestibular Nerve Section. The vestibular and cochlear nerves are exposed through a retrosignoid craniotomy.



FIGURE 2
Vestibular Nerve Section. After exposure, the vestibular portion of the nerve is cut (sectioned). The facial nerve lies just in front.

Intratympanic Gentamicin



COMMON PERIPHERAL LABYRINTHINE DISORDERS

- Vestibulotoxicity
  - Drug-induced hair cell injury
  - Common agents: aminoglycosides (gentamicin), chemotherapeutic agents (cisplatin), salicylates
  - May lead to bilateral vestibular hypofunction
  - Symptoms: oscillopsia (visual bobbing), imbalance, high-frequency hearing loss

RECOVERY FROM VESTIBULAR INJURY

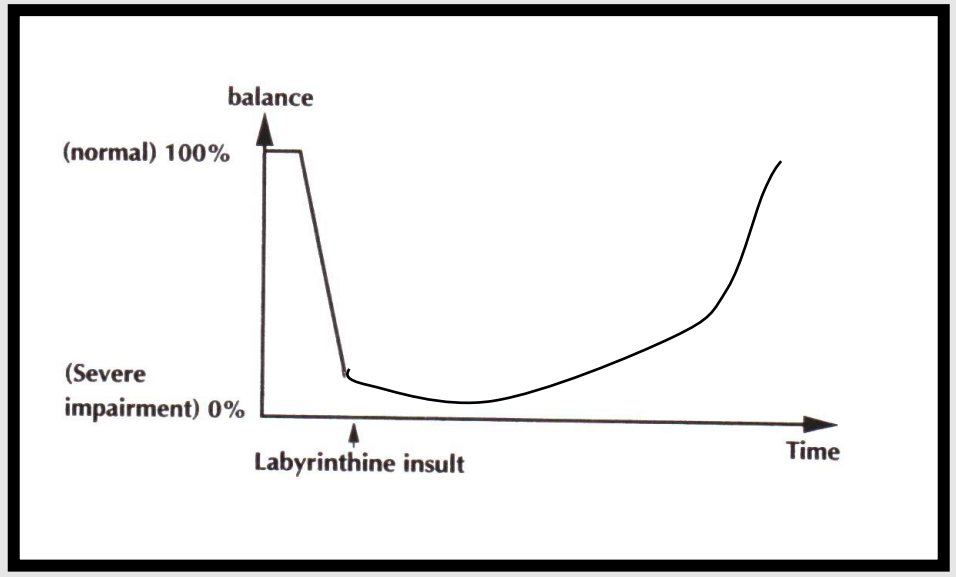
- Central nervous system has adaptive mechanisms to handle sensory conflicts

  (CENTRAL COMPENSATION)

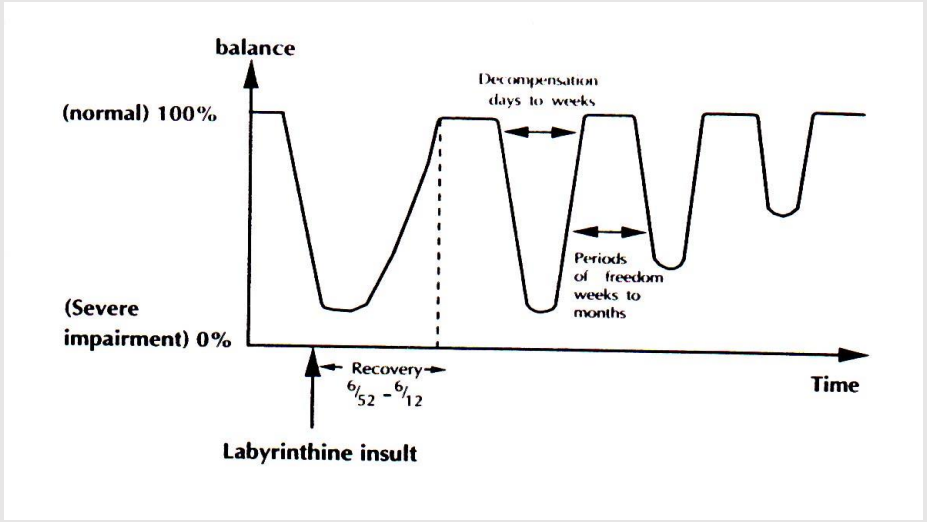
- Develop coping strategies
- Adaptive mechanisms will differ in each case; hence, recovery will be variable
- Factors: health of CNS, severity of insult

Recovery

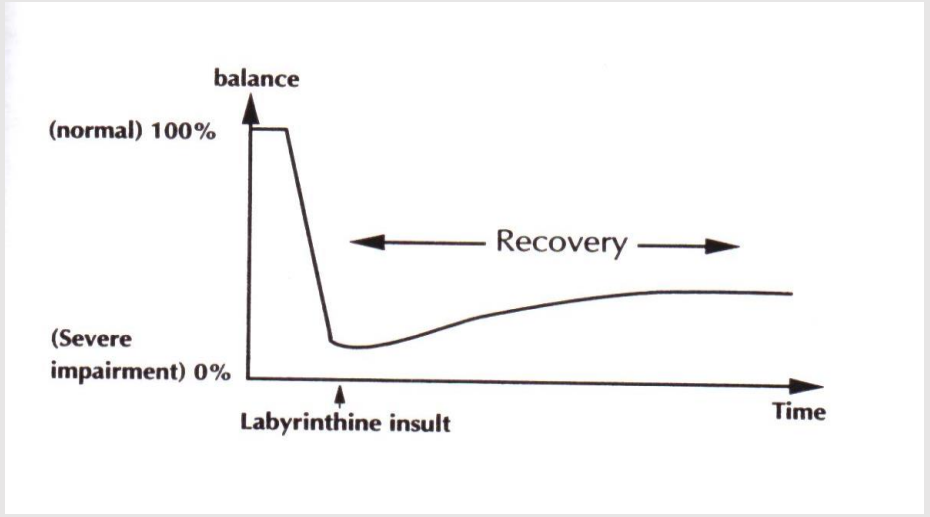
- Recovery through process of central compensation
- Consists of (2) phases:
  - Suppression of asymmetry in the acute phase
  - Re-balance neural activity (recalibration)
- CNS plasticity (ability to adapt to sensory input and maintain appropriate response)
- Rationale for vestibular rehabilitation
  - Habituation exercises



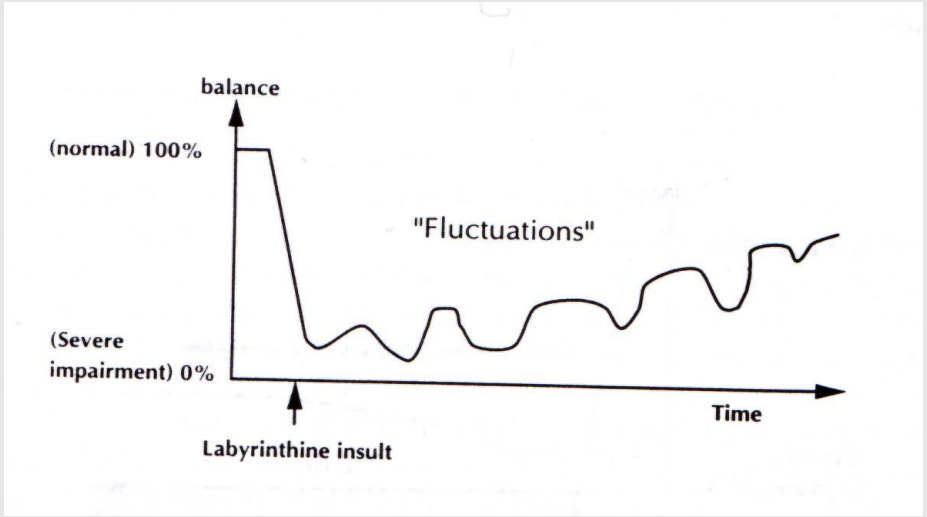
COMPENSATION PATTERNS



DECOMPENSATION



INCOMPLETE RECOVERY



FAILURE OF COMPENSATION

Central Compensation

- Compensation takes longer in older population
- Altered with medications
- Other neurological conditions
- Decompensation can still occur

VESTIBULAR REHABILITATION

- Balance exercises to maximize and enhance remaining vestibular function
- Exercises to improve gaze stability, balance, and gait
- Habituation exercises designed to extinguish pathologic responses to head motion
- Must have re-synthesis of visual, vestibular, proprioceptive information; changes on-going at cerebellum and vestibular nuclei level

Vestibular Rehabilitation

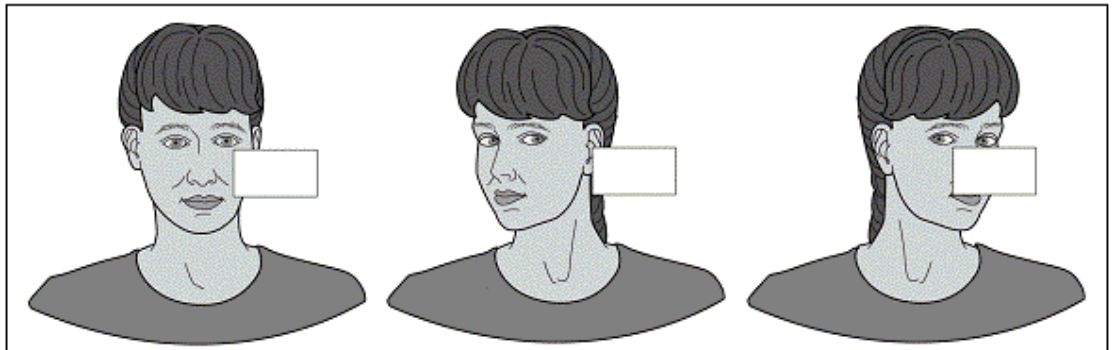


Figure 2A: Look straight ahead.

Figure 2B: Turn your head 45 degrees towards the right.

Figure 2C: Turn your head 45 degrees towards the left.

Note: Business card should be positioned at eye level.

(c) T.C.Hain, 2002

Vestibular Rehabilitation







Vestibular Rehabilitation





CONCLUSIONS

- Understanding the basics of hearing and balance function contributes to a better understanding of the pathophysiology of hearing and balance disorders to better treat clinical disorders
- Results in more effective and rational treatments

matthew.ng@unlv.edu